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DTK

# DTK 26.10
# 兼容性手册

DTK-261000-003-0915-918

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## 版本更新

- DTK-24.04：

  更新操作系统与内核兼容性列表。

- DTK-23.04：

  更新数学库兼容性列表，新增通讯组件兼容性列表。

- DTK-22.10：

  更新操作系统与内核兼容性列表，补充支持的其他操作系统。

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# 1 操作系统和内核兼容性列表

| 系统 | 版本 | 内核 |
| --- | --- | --- |
| CentOS | 7.6 | 3.10.0-957.el7.x86_64 |
| CentOS | 8.5 | 4.18.0-348.el8.x86_64 |
| Rocky | 8.6 | 4.18.0-348.el8.x86_64 |
| Rocky | 9.2 | 4.18.0-348.el8.x86_64 |
| Ubuntu | 20.04.1 | 5.4.0-42-generic |
| Ubuntu | 22.04 | 5.15.0-25-generic |
| Ubuntu | 24.04 | 6.8.0-110-generic |
| NFS | 3.2 | 3.10.0-957.nfs.5.x86_64 |
| NFS | 4.0 | 4.19.113-14.1.nfs4.x86_64 |
| NFSCNS-4.0 | 4.0 | 4.19.113-18.237.nfs4.x86_64 |
| UOS | 1021e | 4.19.90-2109.1.0.0108.up2.uel20.x86_64 |
| UOS | 1070e | 4.19.90-2409.6.0.0297.101.uel20.x86_64 |
| UOS | 1070a | 4.19.0-91.82.190.uelc20.x86_64 |
| Kylin | v10 SP3 | 4.19.90-89.11.v2401.ky10.x86_64 |
| Kylin | v11 | 6.6.0-32.1.v2505.ky11.x86_64 |
| Anolis | 8.4 | 4.19.91-23.4.an8.x86_64 |
| Anolis | 8.6 | 4.19.91-26.an8.x86_64 |
| Anolis | 8.10 | 5.10.134-18.an8.x86_64 |
| openEuler | 22.03 | 5.10.0-60.18.0.50.oe2203.x86_64 |
| openEuler | 24.03 | 6.6.0-132.0.0.111.oe2403sp3.x86_64 |
| BCLinux | 8.2 | 4.19.0-240.23.11.el8_2.bclinux.x86_64 |
| BCLinux | 21.10u4 | 5.10.0-200.0.0.157.206.bclinux.21.10U4.x86_64 |
| Alinux | 3.0 | 5.10.134-17.1.al8.x86_64 |
| Alinux | 3.2304 | 5.10.134-16.3.al8.x86_64 |
| RedHat | 7.9 | 3.10.0-1160.el7.x86_64 |

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| | | |
|---|---|---|
| RedHat | 8.7 | 4.18.0-425.3.1.el8.x86_64 |
| SugonOS | 8.9 | 5.10.134-17.1.3.sga8.x86_64 |
| TencentOS | 3.3 | 5.4.241-24.0017.23 |
| TencentOS | 4.4 | 6.6.98-40.2.tl4.x86_64 |
| TencentOS | 3.1 | 5.4.119-19.0009.59 |
| Opencloud | 8.8 | 5.4.119-20.0009.31 |
| Opencloud | 9.4 | 6.6.114-43.oc9.x86_64 |
| Debian | 11 | 5.10.0-39-amd64 |
| Debian | 12 | 6.1.0-25-amd64 |
| ScaleOS | 26 | 6.6.0-132.0.0.112.scale26.x86_64 |

# 2 异构程序兼容性

基于 DTK-21.10、DTK-21.10.1、DTK-22.04、DTK-22.04.1、DTK-22.04.2、DTK-22.10、DTK-22.10.1、DTK-23.04、DTK-23.04.1、DTK-23.10、DTK-23.10.1、DTK-24.04、DTK-24.04.1、DTK-24.04.2、DTK-24.04.3、DTK-25.04、DTK-25.04.1、DTK-25.04.2、DTK-25.04.3、DTK-25.04.4、DTK-26.04、DTK-26.04.1 编译的异构可执行程序和异构共享库可以直接在 DTK-26.10 上使用。

# 3 CUDA API 兼容性列表

## 3.1 Device Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaChooseDevice | hipChooseDevice | |
| cudaDeviceGetAttribute | hipDeviceGetAttribute | |
| cudaDeviceGetByPCIBusId | hipDeviceGetByPCIBusId | |
| cudaDeviceGetCacheConfig | hipDeviceGetCacheConfig | |
| cudaDeviceGetLimit | hipDeviceGetLimit | |

| | | |
|---|---|---|
| cudaDeviceGetNvSciSyncAttributes | | 10.2 |
| cudaDeviceGetPCIBusId | hipDeviceGetPCIBusId | |
| cudaDeviceGetSharedMemConfig | hipDeviceGetSharedMemConfig | |
| cudaDeviceGetStreamPriorityRange | hipDeviceGetStreamPriorityRange | |
| cudaDeviceReset | hipDeviceReset | |
| cudaDeviceSetCacheConfig | hipDeviceSetCacheConfig | |
| cudaDeviceSetLimit | | |
| cudaDeviceSetSharedMemConfig | hipDeviceSetSharedMemConfig | |
| cudaDeviceSynchronize | hipDeviceSynchronize | |
| cudaGetDevice | hipGetDevice | |
| cudaGetDeviceCount | hipGetDeviceCount | |
| cudaGetDeviceFlags | | |
| cudaGetDeviceProperties | hipGetDeviceProperties | |
| cudaIpcCloseMemHandle | hipIpcCloseMemHandle | |
| cudaIpcGetEventHandle | | |
| cudaIpcGetMemHandle | hipIpcGetMemHandle | |
| cudaIpcOpenEventHandle | | |
| cudaIpcOpenMemHandle | hipIpcOpenMemHandle | |

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| | | |
|---|---|---|
| cudaSetDevice | hipSetDevice | |
| cudaSetDeviceFlags | hipSetDeviceFlags | |
| cudaSetValidDevices | | |
| cudaDeviceGetP2PAttribute | | 8.0 |

## 3.2 Error Handling

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaGetErrorName | hipGetErrorName | |
| cudaGetErrorString | hipGetErrorString | |
| cudaGetLastError | hipGetLastError | |
| cudaPeekAtLastError | hipPeekAtLastError | |

## 3.3 Stream Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaStreamAddCallback | hipStreamAddCallback | |
| cudaStreamAttachMemAsync | | |
| cudaStreamBeginCapture | | 10.0 |
| cudaStreamEndCapture | | 10.0 |
| cudaStreamIsCapturing | | 10.0 |
| cudaStreamGetCaptureInfo | | 10.1 |
| cudaStreamCreate | hipStreamCreate | |
| cudaStreamCreateWithFlags | hipStreamCreateWithFlags | |
| cudaStreamCreateWithPriority | hipStreamCreateWithPriority | |
| cudaStreamDestroy | hipStreamDestroy | |
| cudaStreamGetFlags | hipStreamGetFlags | |

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| | | |
|---|---|---|
| cudaStreamGetPriority | hipStreamGetPriority | |
| cudaStreamQuery | hipStreamQuery | |
| cudaStreamSynchronize | hipStreamSynchronize | |
| cudaStreamWaitEvent | hipStreamWaitEvent | |
| cudaThreadExchangeStreamCaptureMode | | 10.1 |

## 3.4 Event Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaEventCreate | hipEventCreate | |
| cudaEventCreateWithFlags | hipEventCreateWithFlags | |
| cudaEventDestroy | hipEventDestroy | |
| cudaEventElapsedTime | hipEventElapsedTime | |
| cudaEventQuery | hipEventQuery | |
| cudaEventRecord | hipEventRecord | |
| cudaEventSynchronize | hipEventSynchronize | |

## 3.5 External Resource Interoperability

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaSignalExternalSemaphoresAsync | | 10.0 |
| cudaWaitExternalSemaphoresAsync | | 10.0 |
| cudaImportExternalMemory | | 10.0 |
| cudaExternalMemoryGetMappedBuffer | | 10.0 |
| cudaExternalMemoryGetMappedMipmappedArray | | 10.0 |
| cudaDestroyExternalMemory | | 10.0 |
| cudaImportExternalSemaphore | | 10.0 |
| cudaDestroyExternalSemaphore | | 10.0 |

## 3.6 Execution Control

| CUDA | HIP | CUDA version* |
|---|---|---|

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| | | |
|---|---|---|
| cudaFuncGetAttributes | hipFuncGetAttributes | |
| cudaFuncSetAttribute | | 9.0 |
| cudaFuncSetCacheConfig | hipFuncSetCacheConfig | |
| cudaFuncSetSharedMemConfig | | |
| cudaGetParameterBuffer | | |
| cudaGetParameterBufferV2 | | |
| cudaLaunchKernel | hipLaunchKernel | |
| cudaSetDoubleForDevice | | |
| cudaSetDoubleForHost | | |
| cudaLaunchCooperativeKernel | | 9.0 |
| cudaLaunchCooperativeKernel MultiDevice | | 9.0 |
| cudaLaunchHostFunc | | 10.0 |

## 3.7 Occupancy

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaOccupancyMaxActiveBlocksPer Multiprocessor | | |
| cudaOccupancyMaxActiveBlocksPer MultiprocessorWithFlags | | |

## 3.8 Execution Control [DEPRECATED since 7.0]

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaConfigureCall | hipConfigureCall | |
| cudaLaunch | hipLaunchByPtr | |
| cudaSetupArgument | hipSetupArgument | |

## 3.9 Memory Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaArrayGetInfo | | |
| cudaFree | hipFree | |
| cudaFreeArray | hipFreeArray | |
| cudaFreeHost | hipHostFree | |
| cudaFreeMipmappedArray | | |
| cudaGetMipmappedArrayLevel | | |
| cudaGetSymbolAddress | hipGetSymbolAddress | |
| cudaGetSymbolSize | hipGetSymbolSize | |
| cudaHostAlloc | hipHostMalloc | |
| cudaHostGetDevicePointer | hipHostGetDevicePointer | |
| cudaHostGetFlags | hipHostGetFlags | |
| cudaHostRegister | hipHostRegister | |
| cudaHostUnregister | hipHostUnregister | |
| cudaMalloc | hipMalloc | |
| cudaMalloc3D | hipMalloc3D | |
| cudaMalloc3DArray | hipMalloc3DArray | |
| cudaMallocArray | hipMallocArray | |
| cudaMallocHost | hipHostMalloc | |
| cudaMallocManaged | hipMallocManaged | |
| cudaMallocMipmappedArray | | |
| cudaMallocPitch | | |
| cudaMemGetInfo | hipMemGetInfo | |
| cudaMemPrefetchAsync | | 8.0 |
| cudaMemcpy | hipMemcpy | |
| cudaMemcpy2D | hipMemcpy2D | |

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| | | |
|---|---|---|
| cudaMemcpy2DArrayToArray | | |
| cudaMemcpy2DAsync | hipMemcpy2DAsync | |
| cudaMemcpy2DFromArray | | |
| cudaMemcpy2DFromArrayAsync | | |
| cudaMemcpy2DToArray | hipMemcpy2DToArray | |
| cudaMemcpy2DToArrayAsync | | |
| cudaMemcpy3D | hipMemcpy3D | |
| cudaMemcpy3DAsync | | |
| cudaMemcpy3DPeer | | |
| cudaMemcpy3DPeerAsync | | |
| cudaMemcpyArrayToArray | | |
| cudaMemcpyAsync | hipMemcpyAsync | |
| cudaMemcpyFromArray | hipMemcpyFromArray | |
| cudaMemcpyFromArrayAsync | | |
| cudaMemcpyFromSymbol | hipMemcpyFromSymbol | |
| cudaMemcpyFromSymbolAsync | hipMemcpyFromSymbolAsync | |
| cudaMemcpyPeer | hipMemcpyPeer | |
| cudaMemcpyPeerAsync | hipMemcpyPeerAsync | |
| cudaMemcpyToArray | hipMemcpyToArray | |
| cudaMemcpyToArrayAsync | | |
| cudaMemcpyToSymbol | hipMemcpyToSymbol | |
| cudaMemcpyToSymbolAsync | hipMemcpyToSymbolAsync | |
| cudaMemset | hipMemset | |
| cudaMemset2D | hipMemset2D | |
| cudaMemset2DAsync | hipMemset2DAsync | |
| cudaMemset3D | hipMemset3D | |
| cudaMemset3DAsync | hipMemset3DAsync | |
| cudaMemsetAsync | hipMemsetAsync | |

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| | | |
|---|---|---|
| make_cudaExtent | make_hipExtent | |
| make_cudaPitchedPtr | | |
| make_cudaPos | make_hipPos | |

## 3.10 Unified Addressing

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaPointerGetAttributes | hipPointerGetAttributes | |

## 3.11 Peer Device Memory Access

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaDeviceCanAccessPeer | hipDeviceCanAccessPeer | |
| cudaDeviceDisablePeerAccess | hipDeviceDisablePeerAccess | |
| cudaDeviceEnablePeerAccess | hipDeviceEnablePeerAccess | |

## 3.12 Version Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaDriverGetVersion | hipDriverGetVersion | |
| cudaRuntimeGetVersion | hipRuntimeGetVersion | |

## 3.13 Graph Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| cudaGraphAddChildGraphNode | | 10.0 |
| cudaGraphAddDependencies | | 10.0 |
| cudaGraphAddEmptyNode | | 10.0 |
| cudaGraphAddHostNode | | 10.0 |
| cudaGraphAddKernelNode | | 10.0 |
| cudaGraphAddMemcpyNode | | 10.0 |
| cudaGraphAddMemsetNode | | 10.0 |
| cudaGraphChildGraphNodeGetGraph | | 10.0 |
| cudaGraphClone | | 10.0 |
| cudaGraphCreate | | 10.0 |

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| | | |
|---|---|---|
| cudaGraphDestroy | | 10.0 |
| cudaGraphDestroyNode | | 10.0 |
| cudaGraphExecDestroy | | 10.0 |
| cudaGraphGetEdges | | 10.0 |
| cudaGraphGetNodes | | 10.0 |
| cudaGraphGetRootNodes | | 10.0 |
| cudaGraphHostNodeGetParams | | 10.0 |
| cudaGraphHostNodeSetParams | | 10.0 |
| cudaGraphInstantiate | | 10.0 |
| cudaGraphExecKernelNodeSetParams | | 10.1 |
| cudaGraphExecMemcpyNodeSetParams | | 10.2 |
| cudaGraphExecMemsetNodeSetParams | | 10.2 |
| cudaGraphExecHostNodeSetParams | | 10.2 |
| cudaGraphExecUpdate | | 10.2 |
| cudaGraphKernelNodeGetParams | | 10.0 |
| cudaGraphKernelNodeSetParams | | 10.0 |
| cudaGraphLaunch | | 10.0 |
| cudaGraphMemcpyNodeGetParams | | 10.0 |
| cudaGraphMemcpyNodeSetParams | | 10.0 |
| cudaGraphMemsetNodeGetParams | | 10.0 |
| cudaGraphMemsetNodeSetParams | | 10.0 |
| cudaGraphNodeFindInClone | | 10.0 |
| cudaGraphNodeGetDependencies | | 10.0 |
| cudaGraphNodeGetDependentNodes | | 10.0 |
| cudaGraphNodeGetType | | 10.0 |
| cudaGraphRemoveDependencies | | 10.0 |

## 3.14 Profiler Control

| CUDA | HIP | CUDA version* |
|---|---|---|

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| | | |
|---|---|---|
| cudaProfilerInitialize | | |
| cudaProfilerStart | hipProfilerStart | |
| cudaProfilerStop | hipProfilerStop | |

## 3.15 Device Library 支持列表

数学函数支持列表

| | | | |
|---|---|---|---|
| printf | umul64hi | __float2half_rn | __ull2double_rz |
| __popc | mul64hi | __half2float | __ull2float_rd |
| __popcll | __umulhi | __float2int_rd | __ull2float_rn |
| __clz | mulhi | __float2int_rn | __ull2float_ru |
| __clzll | mulhi | __float2int_ru | __ull2float_rz |
| __ffs | mulhi | __float2int_rz | float2int |
| __ffsll | __urhadd | __float2ll_rd | float2uint |
| __ffs | __usad | __float2ll_rn | int2float |
| __ffsll | __lane_id | __float2ll_ru | uint2float |
| __brev | __mbcnt_lo | __float2ll_rz | _clock64 |
| __brevll | __mbcnt_hi | __float2uint_rd | _clock |
| __lastbit_u32_u64 | __hip_ds_bpermute | __float2uint_rn | lock64 |
| __bitextract_u32 | __hip_ds_bpermutef | __float2uint_ru | lock |
| __bitextract_u64 | __hip_ds_permute | __float2uint_rz | __named_sync |
| __bitinsert_u32 | __hip_ds_permutef | __float2ull_rd | __clock64 |
| __bitinsert_u64 | __hip_ds_swizzle_N | __float2ull_rn | __clock |
| __byte_perm | __hip_ds_swizzlef_N | __float2ull_ru | clock64 |

| | | | |
|---|---|---|---|
| __hadd | __hip_move_dpp_N | __float2ull_rz | clock |
| __mul24 | __shfl | __float_as_int | __named_sync |
| __mul64hi | __shfl_up | float_as_int | __all |
| __mulhi | __shfl_down | __float_as_uint | __any |
| __rhadd | __shfl_xor | float_as_uint | __ballot |
| __sad | __double2float_rd | __hiloint2double | __ballot64 |
| __uhadd | __double2float_rn | __int2double_rn | __lanemask_gt |
| __umul24 | __double2float_ru | __int2float_rd | __lanemask_lt |
| __umul64hi | __double2float_rz | __int2float_rn | __lanemask_eq |
| __umulhi | __double2hiint | __int2float_ru | __local_to_generic |
| __urhadd | __double2loint | __int2float_rz | __get_dynamicgroupbaseptr |
| __usad | __double2int_rd | __int_as_float | __get_dynamicgroupbaseptr |
| __byte_perm | __double2int_rn | int_as_float | __amdgcn_get_dynamicgroupbaseptr |
| __hadd | __double2int_ru | __ll2double_rd | __threadfence |
| __mul24 | __double2int_rz | __ll2double_rn | __threadfence_block |
| mul24 | __double2ll_rd | __ll2double_ru | __threadfence_system |
| __mul64hi | __double2ll_rn | __ll2double_rz | abort |
| mul64hi | __double2ll_ru | __ll2float_rd | _wassert |
| __mulhi | __double2ll_rz | __ll2float_rn | __assert_fail |
| mulhi | __double2uint_r | __ll2float_ru | __assertfail |

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| | | | |
|---|---|---|---|
| | d | | |
| __rhadd | __double2uint_rn | __ll2float_rz | __work_group_barrier |
| __sad | __double2uint_ru | __longlong_as_double | __barrier |
| __uhadd | __double2uint_rz | __uint2double_rn | __syncthreads |
| __umul24 | __double2ull_rd | __uint2float_rd | __syncthreads_count |
| umul24 | __double2ull_rn | __uint2float_rn | __syncthreads_and |
| __umul64hi | __double2ull_ru | __uint2float_ru | __syncthreads_or |
| mul64hi | __double2ull_rz | __uint2float_rz | __smid |
| umul64hi | __double_as_longlong | __uint_as_float | __hip_memcpy |
| umul64hi | __float2half_rn | __ull2double_rd | __hip_memset |
| mul64hi | __half2float | __ull2double_rn | memcpy |
| | __half | __ull2double_ru | memset |

原子函数支持列表

| | |
|---|---|
| atomicCAS | atomicMin |
| atomicCAS_system | atomicMin_system |
| atomicAdd | atomicOr |
| atomicAdd_system | atomicOr_system |
| atomicAddNoRet | atomicXor |
| atomicSub | atomicXor_system |
| atomicSub_system | atomicInc |
| atomicExch | atomicDec |
| atomicExch_system | atomicAnd |
| | atomicAnd_system |

# 4 CUDA API 支持列表

HIP 系统提供了对 CUDA 函数的直接支持，以下 CUDA 函数可以直接使用。

| | |
|---|---|
| cudaDeviceReset | cudaGetMipmappedArrayLevel |
| cudaDeviceSetCacheConfig | cudaMemcpy3D |
| cudaDeviceSetSharedMemConfig | cudaArrayGetInfo |
| cudaDeviceGetByPCIBusId | cudaMemcpy |
| cudaDeviceGetPCIBusId | cudaMemcpyPeer |
| cudaIpcGetEventHandle | cudaMemcpyToArray |
| cudaIpcOpenEventHandle | cudaMemcpyFromArray |
| cudaIpcGetMemHandle | cudaMemcpy2D |
| cudaIpcOpenMemHandle | cudaMemcpy2DToArray |
| cudaIpcCloseMemHandle | cudaMemcpy2DFromArray |
| cudaChooseDevice | cudaMemcpyToSymbol |
| cudaSetDevice | cudaMemcpyFromSymbol |
| cudaSetDeviceFlags | cudaMemcpyPeerAsync |
| cudaGetDeviceFlags | cudaMemcpyToSymbolAsync |
| cudaStreamCreate | cudaMemcpyFromSymbolAsync |
| cudaStreamAddCallback | cudaMemset |
| cudaStreamSynchronize | cudaMemset2D |
| cudaStreamQuery | cudaMemset3D |
| cudaEventCreate | cudaGetSymbolAddress |
| cudaEventQuery | cudaGetSymbolSize |
| cudaEventSynchronize | cudaMemPrefetchAsync |
| cudaEventElapsedTime | cudaMemAdvise |
| cudaLaunchKernel | cudaMemRangeGetAttribute |
| cudaLaunchCooperativeKernel | cudaMemRangeGetAttributes |
| cudaLaunchCooperativeKernelMultiDevice | cudaPointerGetAttributes |

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| | |
|---|---|
| cudaFuncSetCacheConfig | cudaDeviceCanAccessPeer |
| cudaConfigureCall | cudaDeviceEnablePeerAccess |
| cudaSetupArgument | cudaDeviceDisablePeerAccess |
| cudaMallocHost | cudaGetChannelDesc |
| cudaMallocPitch | cudaBindTexture |
| cudaMallocArray | cudaBindTexture2D |
| cudaFreeHost | cudaBindTextureToArray |
| cudaFreeArray | cudaBindTextureToMipmappedArray |
| cudaFreeMipmappedArray | cudaUnbindTexture |
| cudaHostAlloc | cudaGetTextureAlignmentOffset |
| cudaHostRegister | cudaGetTextureReference |
| cudaHostUnregister | cudaCreateTextureObject |
| cudaHostGetDevicePointer | cudaDestroyTextureObject |
| cudaHostGetFlags | cudaGetTextureObjectResourceDesc |
| cudaMalloc3D | cudaGetTextureObjectTextureDesc |
| cudaMalloc3DArray | cudaGetTextureObjectResourceViewDesc |
| cudaMallocMipmappedArray | cudaDriverGetVersion |
| cudaDeviceReset | cudaGetMipmappedArrayLevel |

# 5 数学库兼容性列表

## 5.1 BLAS

## 5.2 BLAS Data types

| type | CUDA | HIP | CUDA version* |
|---|---|---|---|
| define | CUBLAS_VER_MAJOR | | 10.1 Update 2 |
| define | CUBLAS_VER_MIN | | 10.1 Update 2 |

| | | | |
|---|---|---|---|
| | OR | | |
| define | CUBLAS_VER_PATCH | | 10.1 Update 2 |
| define | CUBLAS_VER_BUILD | | 10.1 Update 2 |
| define | CUBLAS_VERSION | | 10.1 Update 2 |
| enum | cublasStatus_t | hipblasStatus_t | |
| 0 | CUBLAS_STATUS_SUCCESS | HIPBLAS_STATUS_SUCCESS | |
| 1 | CUBLAS_STATUS_NOT_INITIALIZED | HIPBLAS_STATUS_NOT_INITIALIZED | |
| 3 | CUBLAS_STATUS_ALLOC_FAILED | HIPBLAS_STATUS_ALLOC_FAILED | |
| 7 | CUBLAS_STATUS_INVALID_VALUE | HIPBLAS_STATUS_INVALID_VALUE | |
| 8 | CUBLAS_STATUS_ARCH_MISMATCH | HIPBLAS_STATUS_ARCH_MISMATCH | |
| 11 | CUBLAS_STATUS_MAPPING_ERROR | HIPBLAS_STATUS_MAPPING_ERROR | |
| 13 | CUBLAS_STATUS_EXECUTION_FAILED | HIPBLAS_STATUS_EXECUTION_FAILED | |
| 14 | CUBLAS_STATUS_INTERNAL_ERROR | HIPBLAS_STATUS_INTERNAL_ERROR | |
| 15 | CUBLAS_STATUS_NOT_SUPPORTED | HIPBLAS_STATUS_NOT_SUPPORTED | |
| 16 | CUBLAS_STATUS_LICENSE_ERROR | | |
| enum | cublasOperation_t | hipblasOperation_t | |

| | | | |
|---|---|---|---|
| 0 | CUBLAS_OP_N | HIPBLAS_OP_N | |
| 1 | CUBLAS_OP_T | HIPBLAS_OP_T | |
| 2 | CUBLAS_OP_C | HIPBLAS_OP_C | |
| 2 | CUBLAS_OP_HERMITAN | HIPBLAS_OP_C | 10.1 |
| 3 | CUBLAS_OP_CONJG | | 10.1 |
| enum | cublasFillMode_t | hipblasFillMode_t | |
| 0 | CUBLAS_FILL_MODE_LOWER | HIPBLAS_FILL_MODE_LOWER | |
| 1 | CUBLAS_FILL_MODE_UPPER | HIPBLAS_FILL_MODE_UPPER | |
| 2 | CUBLAS_FILL_MODE_FULL | HIPBLAS_FILL_MODE_FULL | 10.1 |
| enum | cublasDiagType_t | hipblasDiagType_t | |
| 0 | CUBLAS_DIAG_NON_UNIT | HIPBLAS_DIAG_NON_UNIT | |
| 1 | CUBLAS_DIAG_UNIT | HIPBLAS_DIAG_UNIT | |
| enum | cublasSideMode_t | hipblasSideMode_t | |
| 0 | CUBLAS_SIDE_LEFT | HIPBLAS_SIDE_LEFT | |
| 1 | CUBLAS_SIDE_RIGHT | HIPBLAS_SIDE_RIGHT | |
| enum | cublasPointerMode_t | hipblasPointerMode_t | |
| 0 | CUBLAS_POINTER_MODE_HOST | HIPBLAS_POINTER_MODE_HOST | |
| 1 | CUBLAS_POINTER_ | HIPBLAS_POINTER | |

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|---|---|---|---|
| | MODE_DEVICE | _MODE_DEVICE | |
| enum | cublasAtomicsMode_t | hipblasAtomicsMode_t | |
| 0 | CUBLAS_ATOMICS_NOT_ALLOWED | HIPBLAS_ATOMICS_NOT_ALLOWED | |
| 1 | CUBLAS_ATOMICS_ALLOWED | HIPBLAS_ATOMICS_ALLOWED | |
| enum | cublasGemmAlgo_t | hipblasGemmAlgo_t | 8 |
| -1 | CUBLAS_GEMM_DFALT | HIPBLAS_GEMM_DEFAULT | 8 |
| -1 | CUBLAS_GEMM_DEFAULT | HIPBLAS_GEMM_DEFAULT | 8 |
| 0 | CUBLAS_GEMM_ALGO0 | | 8 |
| 1 | CUBLAS_GEMM_ALGO1 | | 8 |
| 2 | CUBLAS_GEMM_ALGO2 | | 8 |
| 3 | CUBLAS_GEMM_ALGO3 | | 8 |
| 4 | CUBLAS_GEMM_ALGO4 | | 8 |
| 5 | CUBLAS_GEMM_ALGO5 | | 8 |
| 6 | CUBLAS_GEMM_ALGO6 | | 8 |
| 7 | CUBLAS_GEMM_ALGO7 | | 8 |
| 8 | CUBLAS_GEMM_AL | | 9 |

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|---|---|---|---|
| | GO8 | | |
| 9 | CUBLAS_GEMM_ALGO9 | | 9 |
| 10 | CUBLAS_GEMM_ALGO10 | | 9 |
| 11 | CUBLAS_GEMM_ALGO11 | | 9 |
| 12 | CUBLAS_GEMM_ALGO12 | | 9 |
| 13 | CUBLAS_GEMM_ALGO13 | | 9 |
| 14 | CUBLAS_GEMM_ALGO14 | | 9 |
| 15 | CUBLAS_GEMM_ALGO15 | | 9 |
| 16 | CUBLAS_GEMM_ALGO16 | | 9 |
| 17 | CUBLAS_GEMM_ALGO17 | | 9 |
| 18 | CUBLAS_GEMM_ALGO18 | | 9.2 |
| 19 | CUBLAS_GEMM_ALGO19 | | 9.2 |
| 20 | CUBLAS_GEMM_ALGO20 | | 9.2 |
| 21 | CUBLAS_GEMM_ALGO21 | | 9.2 |
| 22 | CUBLAS_GEMM_ALGO22 | | 9.2 |

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|---|---|---|---|
| 23 | CUBLAS_GEMM_ALGO23 | | 9.2 |
| 99 | CUBLAS_GEMM_DEFAULT_TENSOR_OP | | 9 |
| 99 | CUBLAS_GEMM_DFALT_TENSOR_OP | | 9 |
| 100 | CUBLAS_GEMM_ALGO0_TENSOR_OP | | 9 |
| 101 | CUBLAS_GEMM_ALGO1_TENSOR_OP | | 9 |
| 102 | CUBLAS_GEMM_ALGO2_TENSOR_OP | | 9 |
| 103 | CUBLAS_GEMM_ALGO3_TENSOR_OP | | 9 |
| 104 | CUBLAS_GEMM_ALGO4_TENSOR_OP | | 9 |
| 105 | CUBLAS_GEMM_ALGO5_TENSOR_OP | | 9.2 |
| 106 | CUBLAS_GEMM_ALGO6_TENSOR_OP | | 9.2 |
| 107 | CUBLAS_GEMM_ALGO7_TENSOR_OP | | 9.2 |
| 108 | CUBLAS_GEMM_ALGO8_TENSOR_OP | | 9.2 |
| 109 | CUBLAS_GEMM_ALGO9_TENSOR_OP | | 9.2 |
| 110 | CUBLAS_GEMM_ALGO10_TENSOR_OP | | 9.2 |
| 111 | CUBLAS_GEMM_AL | | 9.2 |

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|---|---|---|---|
| | GO11_TENSOR_OP | | |
| 112 | CUBLAS_GEMM_ALGO12_TENSOR_OP | | 9.2 |
| 113 | CUBLAS_GEMM_ALGO13_TENSOR_OP | | 9.2 |
| 114 | CUBLAS_GEMM_ALGO14_TENSOR_OP | | 9.2 |
| 115 | CUBLAS_GEMM_ALGO15_TENSOR_OP | | 9.2 |
| enum | cublasMath_t | | 9 |
| 0 | CUBLAS_DEFAULT_MATH | | 9 |
| 1 | CUBLAS_TENSOR_OP_MATH | | 9 |
| enum* | cublasDataType_t | | 7.5 |
| struct | cublasContext | | |
| struct* | cublasHandle_t | hipblasHandle_t | |

### 5.2.1 BLAS Level-1 Function Reference

| CUDA | HIP | CUDA version* |
|---|---|---|
| cublasCaxpy | hipblasCaxpy | |
| cublasCaxpy_v2 | hipblasCaxpy | |
| cublasCcopy | hipblasCcopy | |
| cublasCcopy_v2 | hipblasCcopy | |
| cublasCdotc | hipblasCdotc | |
| cublasCdotc_v2 | hipblasCdotc | |
| cublasCdotu | hipblasCdotu | |
| cublasCdotu_v2 | hipblasCdotu | |
| cublasCrot | hipblasCrot | |

| | | |
|---|---|---|
| cublasCrot_v2 | hipblasCrot | |
| cublasCrotg | hipblasCrotg | |
| cublasCrotg_v2 | hipblasCrotg | |
| cublasCscal | hipblasCscal | |
| cublasCscal_v2 | hipblasCscal | |
| cublasCsrot | hipblasCsrot | |
| cublasCsrot_v2 | hipblasCsrot | |
| cublasCsscal | hipblasCsscal | |
| cublasCsscal_v2 | hipblasCsscal | |
| cublasCswap | hipblasCswap | |
| cublasCswap_v2 | hipblasCswap | |
| cublasDasum | hipblasDasum | |
| cublasDasum_v2 | hipblasDasum | |
| cublasDaxpy | hipblasDaxpy | |
| cublasDaxpy_v2 | hipblasDaxpy | |
| cublasDcopy | hipblasDcopy | |
| cublasDcopy_v2 | hipblasDcopy | |
| cublasDdot | hipblasDdot | |
| cublasDdot_v2 | hipblasDdot | |
| cublasDnrm2 | hipblasDnrm2 | |
| cublasDnrm2_v2 | hipblasDnrm2 | |
| cublasDrot | hipblasDrot | |
| cublasDrot_v2 | hipblasDrot | |
| cublasDrotg | hipblasDrotg | |
| cublasDrotg_v2 | hipblasDrotg | |
| cublasDrotm | hipblasDrotm | |
| cublasDrotm_v2 | hipblasDrotm | |
| cublasDrotmg | hipblasDrotmg | |

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|---|---|---|
| cublasDrotmg_v2 | hipblasDrotmg | |
| cublasDscal | hipblasDscal | |
| cublasDscal_v2 | hipblasDscal | |
| cublasDswap | hipblasDswap | |
| cublasDswap_v2 | hipblasDswap | |
| cublasDzasum | hipblasDzasum | |
| cublasDzasum_v2 | hipblasDzasum | |
| cublasDznrm2 | hipblasDznrm2 | |
| cublasDznrm2_v2 | hipblasDznrm2 | |
| cublasIcamax | hipblasIcamax | |
| cublasIcamax_v2 | hipblasIcamax | |
| cublasIcamin | hipblasIcamin | |
| cublasIcamin_v2 | hipblasIcamin | |
| cublasIdamax | hipblasIdamax | |
| cublasIdamax_v2 | hipblasIdamax | |
| cublasIdamin | hipblasIdamin | |
| cublasIdamin_v2 | hipblasIdamin | |
| cublasIsamax | hipblasIsamax | |
| cublasIsamax_v2 | hipblasIsamax | |
| cublasIsamin | hipblasIsamin | |
| cublasIsamin_v2 | hipblasIsamin | |
| cublasIzamax | hipblasIzamax | |
| cublasIzamax_v2 | hipblasIzamax | |
| cublasIzamin | hipblasIzamin | |
| cublasIzamin_v2 | hipblasIzamin | |
| cublasNrm2Ex | hipblasNrm2Ex | 8 |
| cublasSasum | hipblasSasum | |
| cublasSasum_v2 | hipblasSasum | |

| | | |
|---|---|---|
| cublasSaxpy | hipblasSaxpy | |
| cublasSaxpy_v2 | hipblasSaxpy | |
| cublasScasum | hipblasScasum | |
| cublasScasum_v2 | hipblasScasum | |
| cublasScnrm2 | hipblasScnrm2 | |
| cublasScnrm2_v2 | hipblasScnrm2 | |
| cublasScopy | hipblasScopy | |
| cublasScopy_v2 | hipblasScopy | |
| cublasSdot | hipblasSdot | |
| cublasSdot_v2 | hipblasSdot | |
| cublasSnrm2 | hipblasSnrm2 | |
| cublasSnrm2_v2 | hipblasSnrm2 | |
| cublasSrot | hipblasSrot | |
| cublasSrot_v2 | hipblasSrot | |
| cublasSrotg | hipblasSrotg | |
| cublasSrotg_v2 | hipblasSrotg | |
| cublasSrotm | hipblasSrotm | |
| cublasSrotm_v2 | hipblasSrotm | |
| cublasSrotmg | hipblasSrotmg | |
| cublasSrotmg_v2 | hipblasSrotmg | |
| cublasSscal | hipblasSscal | |
| cublasSscal_v2 | hipblasSscal | |
| cublasSswap | hipblasSswap | |
| cublasSswap_v2 | hipblasSswap | |
| cublasZaxpy | hipblasZaxpy | |
| cublasZaxpy_v2 | hipblasZaxpy | |
| cublasZcopy | hipblasZcopy | |
| cublasZcopy_v2 | hipblasZcopy | |

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|---|---|---|
| cublasZdotc | hipblasZdotc | |
| cublasZdotc_v2 | hipblasZdotc | |
| cublasZdotu | hipblasZdotu | |
| cublasZdotu_v2 | hipblasZdotu | |
| cublasZdrot | hipblasZdrot | |
| cublasZdrot_v2 | hipblasZdrot | |
| cublasZdscal | hipblasZdscal | |
| cublasZdscal_v2 | hipblasZdscal | |
| cublasZrot | hipblasZrot | |
| cublasZrot_v2 | hipblasZrot | |
| cublasZrotg | hipblasZrotg | |
| cublasZrotg_v2 | hipblasZrotg | |
| cublasZscal | hipblasZscal | |
| cublasZscal_v2 | hipblasZscal | |
| cublasZswap | hipblasZswap | |
| cublasZswap_v2 | hipblasZswap | |

### 5.2.2 BLAS Level-2 Function Reference

| CUDA | HIP | CUDA version* |
|---|---|---|
| cublasCgbmv | hipblasCgbmv | |
| cublasCgbmv_v2 | hipblasCgbmv | |
| cublasCgemv | hipblasCgemv | |
| cublasCgemv_v2 | hipblasCgemv | |
| cublasCgerc | hipblasCgerc | |
| cublasCgerc_v2 | hipblasCgerc | |
| cublasCgeru | hipblasCgeru | |
| cublasCgeru_v2 | hipblasCgeru | |
| cublasChbmv | hipblasChbmv | |
| cublasChbmv_v2 | hipblasChbmv | |

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|---|---|---|
| cublasChemv | hipblasChemv | |
| cublasChemv_v2 | hipblasChemv | |
| cublasCher | hipblasCher | |
| cublasCher2 | hipblasCher2 | |
| cublasCher2_v2 | hipblasCher2 | |
| cublasCher_v2 | hipblasCher | |
| cublasChpmv | hipblasChpmv | |
| cublasChpmv_v2 | hipblasChpmv | |
| cublasChpr | hipblasChpr | |
| cublasChpr2 | hipblasChpr2 | |
| cublasChpr2_v2 | hipblasChpr2 | |
| cublasChpr_v2 | hipblasChpr | |
| cublasCsymv | hipblasCsymv | |
| cublasCsymv_v2 | hipblasCsymv | |
| cublasCsyr | hipblasCsyr | |
| cublasCsyr2 | hipblasCsyr2 | |
| cublasCsyr2_v2 | hipblasCsyr2 | |
| cublasCsyr_v2 | hipblasCsyr | |
| cublasCtbmv | hipblasCtbmv | |
| cublasCtbmv_v2 | hipblasCtbmv | |
| cublasCtbsv | hipblasCtbsv | |
| cublasCtbsv_v2 | hipblasCtbsv | |
| cublasCtpmv | hipblasCtpmv | |
| cublasCtpmv_v2 | hipblasCtpmv | |
| cublasCtpsv | hipblasCtpsv | |
| cublasCtpsv_v2 | hipblasCtpsv | |
| cublasCtrmv | hipblasCtrmv | |
| cublasCtrmv_v2 | hipblasCtrmv | |

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|---|---|---|
| cublasCtrsv | hipblasCtrsv | |
| cublasCtrsv_v2 | hipblasCtrsv | |
| cublasDgbmv | hipblasDgbmv | |
| cublasDgbmv_v2 | hipblasDgbmv | |
| cublasDgemv | hipblasDgemv | |
| cublasDgemv_v2 | hipblasDgemv | |
| cublasDger | hipblasDger | |
| cublasDger_v2 | hipblasDger | |
| cublasDsbmv | hipblasDsbmv | |
| cublasDsbmv_v2 | hipblasDsbmv | |
| cublasDspmv | hipblasDspmv | |
| cublasDspmv_v2 | hipblasDspmv | |
| cublasDspr | hipblasDspr | |
| cublasDspr2 | hipblasDspr2 | |
| cublasDspr2_v2 | hipblasDspr2 | |
| cublasDspr_v2 | hipblasDspr | |
| cublasDsymv | hipblasDsymv | |
| cublasDsymv_v2 | hipblasDsymv | |
| cublasDsyr | hipblasDsyr | |
| cublasDsyr2 | hipblasDsyr2 | |
| cublasDsyr2_v2 | hipblasDsyr2 | |
| cublasDsyr_v2 | hipblasDsyr | |
| cublasDtbmv | hipblasDtbmv | |
| cublasDtbmv_v2 | hipblasDtbmv | |
| cublasDtbsv | hipblasDtbsv | |
| cublasDtbsv_v2 | hipblasDtbsv | |
| cublasDtpmv | hipblasDtpmv | |
| cublasDtpmv_v2 | hipblasDtpmv | |

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|---|---|---|
| cublasDtpsv | hipblasDtpsv | |
| cublasDtpsv_v2 | hipblasDtpsv | |
| cublasDtrmv | hipblasDtrmv | |
| cublasDtrmv_v2 | hipblasDtrmv | |
| cublasDtrsv | hipblasDtrsv | |
| cublasDtrsv_v2 | hipblasDtrsv | |
| cublasSgbmv | hipblasSgbmv | |
| cublasSgbmv_v2 | hipblasSgbmv | |
| cublasSgemv | hipblasSgemv | |
| cublasSgemv_v2 | hipblasSgemv | |
| cublasSger | hipblasSger | |
| cublasSger_v2 | hipblasSger | |
| cublasSsbmv | hipblasSsbmv | |
| cublasSsbmv_v2 | hipblasSsbmv | |
| cublasSspmv | hipblasSspmv | |
| cublasSspmv_v2 | hipblasSspmv | |
| cublasSspr | hipblasSspr | |
| cublasSspr2 | hipblasSspr2 | |
| cublasSspr2_v2 | hipblasSspr2 | |
| cublasSspr_v2 | hipblasSspr | |
| cublasSsymv | hipblasSsymv | |
| cublasSsymv_v2 | hipblasSsymv | |
| cublasSsyr | hipblasSsyr | |
| cublasSsyr2 | hipblasSsyr2 | |
| cublasSsyr2_v2 | hipblasSsyr2 | |
| cublasSsyr_v2 | hipblasSsyr | |
| cublasStbmv | hipblasStbmv | |
| cublasStbmv_v2 | hipblasStbmv | |

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|---|---|---|
| cublasStbsv | hipblasStbsv | |
| cublasStbsv_v2 | hipblasStbsv | |
| cublasStpmv | hipblasStpmv | |
| cublasStpmv_v2 | hipblasStpmv | |
| cublasStpsv | hipblasStpsv | |
| cublasStpsv_v2 | hipblasStpsv | |
| cublasStrmv | hipblasStrmv | |
| cublasStrmv_v2 | hipblasStrmv | |
| cublasStrsv | hipblasStrsv | |
| cublasStrsv_v2 | hipblasStrsv | |
| cublasZgbmv | hipblasZgbmv | |
| cublasZgbmv_v2 | hipblasZgbmv | |
| cublasZgemv | hipblasZgemv | |
| cublasZgemv_v2 | hipblasZgemv | |
| cublasZgerc | hipblasZgerc | |
| cublasZgerc_v2 | hipblasZgerc | |
| cublasZgeru | hipblasZgeru | |
| cublasZgeru_v2 | hipblasZgeru | |
| cublasZhbmv | hipblasZhbmv | |
| cublasZhbmv_v2 | hipblasZhbmv | |
| cublasZhemv | hipblasZhemv | |
| cublasZhemv_v2 | hipblasZhemv | |
| cublasZher | hipblasZher | |
| cublasZher2 | hipblasZher2 | |
| cublasZher2_v2 | hipblasZher2 | |
| cublasZher_v2 | hipblasZher | |
| cublasZhpmv | hipblasZhpmv | |
| cublasZhpmv_v2 | hipblasZhpmv | |

| | | |
|---|---|---|
| cublasZhpr | hipblasZhpr | |
| cublasZhpr2 | hipblasZhpr2 | |
| cublasZhpr2_v2 | hipblasZhpr2 | |
| cublasZhpr_v2 | hipblasZhpr | |
| cublasZsymv | hipblasZsymv | |
| cublasZsymv_v2 | hipblasZsymv | |
| cublasZsyr | hipblasZsyr | |
| cublasZsyr2 | hipblasZsyr2 | |
| cublasZsyr2_v2 | hipblasZsyr2 | |
| cublasZsyr_v2 | hipblasZsyr | |
| cublasZtbmv | hipblasZtbmv | |
| cublasZtbmv_v2 | hipblasZtbmv | |
| cublasZtbsv | hipblasZtbsv | |
| cublasZtbsv_v2 | hipblasZtbsv | |
| cublasZtpmv | hipblasZtpmv | |
| cublasZtpmv_v2 | hipblasZtpmv | |
| cublasZtpsv | hipblasZtpsv | |
| cublasZtpsv_v2 | hipblasZtpsv | |
| cublasZtrmv | hipblasZtrmv | |
| cublasZtrmv_v2 | hipblasZtrmv | |
| cublasZtrsv | hipblasZtrsv | |
| cublasZtrsv_v2 | hipblasZtrsv | |

### 5.2.3 BLAS Level-3 Function Reference

| CUDA | HIP | CUDA version* |
|---|---|---|
| cublasCgemm | hipblasCgemm | |
| cublasCgemm3m | | 8 |
| cublasCgemm3mBatched | | 8 |
| cublasCgemm3mEx | | 8 |

| | | |
|---|---|---|
| cublasCgemm3mStridedBatched | | 8 |
| cublasCgemmBatched | hipblasCgemmBatched | |
| cublasCgemmStridedBatched | hipblasCgemmStridedBatched | 8 |
| cublasCgemm_v2 | hipblasCgemm | |
| cublasChemm | hipblasChemm | |
| cublasChemm_v2 | hipblasChemm | |
| cublasCher2k | hipblasCher2k | |
| cublasCher2k_v2 | hipblasCher2k | |
| cublasCherk | hipblasCherk | |
| cublasCherk_v2 | hipblasCherk | |
| cublasCherkx | hipblasCherkx | |
| cublasCsymm | hipblasCsymm | |
| cublasCsymm_v2 | hipblasCsymm | |
| cublasCsyr2k | hipblasCsyr2k | |
| cublasCsyr2k_v2 | hipblasCsyr2k | |
| cublasCsyrk | hipblasCsyrk | |
| cublasCsyrk_v2 | hipblasCsyrk | |
| cublasCsyrkx | hipblasCsyrkx | |
| cublasCtrmm | hipblasCtrmm | |
| cublasCtrmm_v2 | hipblasCtrmm | |
| cublasCtrsm | hipblasCtrsm | |
| cublasCtrsm_v2 | hipblasCtrsm | |
| cublasDgemm | hipblasDgemm | |
| cublasDgemmBatched | hipblasDgemmBatched | |
| cublasDgemmStridedBatched | hipblasDgemmStridedBatched | 8 |

| | | |
|---|---|---|
| cublasDgemm_v2 | hipblasDgemm | |
| cublasDsymm | hipblasDsymm | |
| cublasDsymm_v2 | hipblasDsymm | |
| cublasDsyr2k | hipblasDsyr2k | |
| cublasDsyr2k_v2 | hipblasDsyr2k | |
| cublasDsyrk | hipblasDsyrk | |
| cublasDsyrk_v2 | hipblasDsyrk | |
| cublasDsyrkx | hipblasDsyrkx | |
| cublasDtrmm | hipblasDtrmm | |
| cublasDtrmm_v2 | hipblasDtrmm | |
| cublasDtrsm | hipblasDtrsm | |
| cublasDtrsm_v2 | hipblasDtrsm | |
| cublasHgemm | hipblasHgemm | 7.5 |
| cublasHgemmBatched | hipblasHgemmBatched | 9 |
| cublasHgemmStridedBatched | hipblasHgemmStridedBatched | 8 |
| cublasSgemm | hipblasSgemm | |
| cublasSgemmBatched | hipblasSgemmBatched | |
| cublasSgemmStridedBatched | hipblasSgemmStridedBatched | 8 |
| cublasSgemm_v2 | hipblasSgemm | |
| cublasSsymm | hipblasSsymm | |
| cublasSsymm_v2 | hipblasSsymm | |
| cublasSsyr2k | hipblasSsyr2k | |
| cublasSsyr2k_v2 | hipblasSsyr2k | |
| cublasSsyrk | hipblasSsyrk | |
| cublasSsyrk_v2 | hipblasSsyrk | |
| cublasSsyrkx | hipblasSsyrkx | |

| | | |
|---|---|---|
| cublasStrmm | hipblasStrmm | |
| cublasStrmm_v2 | hipblasStrmm | |
| cublasStrsm | hipblasStrsm | |
| cublasStrsm_v2 | hipblasStrsm | |
| cublasZgemm | hipblasZgemm | |
| cublasZgemm3m | | 8 |
| cublasZgemmBatched | hipblasZgemmBatched | |
| cublasZgemmStridedBatched | hipblasZgemmStridedBatched | 8 |
| cublasZgemm_v2 | hipblasZgemm | |
| cublasZhemm | hipblasZhemm | |
| cublasZhemm_v2 | hipblasZhemm | |
| cublasZher2k | hipblasZher2k | |
| cublasZher2k_v2 | hipblasZher2k | |
| cublasZherk | hipblasZherk | |
| cublasZherk_v2 | hipblasZherk | |
| cublasZherkx | hipblasZherkx | |
| cublasZsymm | hipblasZsymm | |
| cublasZsymm_v2 | hipblasZsymm | |
| cublasZsyr2k | hipblasZsyr2k | |
| cublasZsyr2k_v2 | hipblasZsyr2k | |
| cublasZsyrk | hipblasZsyrk | |
| cublasZsyrk_v2 | hipblasZsyrk | |
| cublasZsyrkx | hipblasZsyrkx | |
| cublasZtrmm | hipblasZtrmm | |
| cublasZtrmm_v2 | hipblasZtrmm | |
| cublasZtrsm | hipblasZtrsm | |
| cublasZtrsm_v2 | hipblasZtrsm | |

### 5.2.4 BLAS Helper Function Referenc

| CUDA | HIP | CUDA version* |
|---|---|---|
| cublasAlloc | | |
| cublasCreate | hipblasCreate | |
| cublasCreate_v2 | hipblasCreate | |
| cublasDestroy | hipblasDestroy | |
| cublasDestroy_v2 | hipblasDestroy | |
| cublasFree | | |
| cublasGetAtomicsMode | hipblasGetAtomicsMode | |
| cublasGetCudartVersion | | 10.1 |
| cublasGetError | | |
| cublasGetLoggerCallback | | 9.2 |
| cublasGetMathMode | | 9 |
| cublasGetMatrix | hipblasGetMatrix | |
| cublasGetMatrixAsync | hipblasGetMatrixAsync | |
| cublasGetPointerMode | hipblasGetPointerMode | |
| cublasGetPointerMode_v2 | hipblasGetPointerMode | |
| cublasGetProperty | | |
| cublasGetSmCountTarget | | 11.3 |
| cublasGetStatusName | | 11.4 |
| cublasGetStatusString | | 11.4 |
| cublasGetStream | hipblasGetStream | |
| cublasGetStream_v2 | hipblasGetStream | |
| cublasGetVector | hipblasGetVector | |
| cublasGetVectorAsync | hipblasGetVectorAsync | |
| cublasGetVersion | | |

| | | |
|---|---|---|
| cublasGetVersion_v2 | | |
| cublasInit | | |
| cublasLogCallback | | 9.2 |
| cublasLoggerConfigure | | 9.2 |
| cublasMigrateComputeType | | 11 |
| cublasSetAtomicsMode | hipblasSetAtomicsMode | |
| cublasSetKernelStream | | |
| cublasSetLoggerCallback | | 9.2 |
| cublasSetMathMode | | |
| cublasSetMatrix | hipblasSetMatrix | |
| cublasSetMatrixAsync | hipblasSetMatrixAsync | |
| cublasSetPointerMode | hipblasSetPointerMode | |
| cublasSetPointerMode_v2 | hipblasSetPointerMode | |
| cublasSetSmCountTarget | | 11.3 |
| cublasSetStream | hipblasSetStream | |
| cublasSetStream_v2 | hipblasSetStream | |
| cublasSetVector | hipblasSetVector | |
| cublasSetVectorAsync | hipblasSetVectorAsync | |
| cublasShutdown | | |
| cublasXerbla | | |

### 5.2.5 BLAS-like Extension

| CUDA | HIP | CUDA version* |
|---|---|---|
| cublasAsumEx | | 10.1 |
| cublasAxpyEx | hipblasAxpyEx | 8 |
| cublasCdgmm | hipblasCdgmm | |
| cublasCgeam | hipblasCgeam | |

| | | |
|---|---|---|
| cublasCgelsBatched | hipblasCgelsBatched | |
| cublasCgemmEx | | 8 |
| cublasCgeqrfBatched | hipblasCgeqrfBatched | |
| cublasCgetrfBatched | hipblasCgetrfBatched | |
| cublasCgetriBatched | hipblasCgetriBatched | |
| cublasCgetrsBatched | hipblasCgetrsBatched | |
| cublasCherk3mEx | | 8 |
| cublasCherkEx | | 8 |
| cublasCmatinvBatched | | |
| cublasCopyEx | | 10.1 |
| cublasCsyrk3mEx | | 8 |
| cublasCsyrkEx | | 8 |
| cublasCtpttr | | |
| cublasCtrsmBatched | hipblasCtrsmBatched | |
| cublasCtrttp | | |
| cublasDdgmm | hipblasDdgmm | |
| cublasDgeam | hipblasDgeam | |
| cublasDgelsBatched | hipblasDgelsBatched | |
| cublasDgeqrfBatched | hipblasDgeqrfBatched | |
| cublasDgetrfBatched | hipblasDgetrfBatched | |
| cublasDgetriBatched | hipblasDgetriBatched | |
| cublasDgetrsBatched | hipblasDgetrsBatched | |
| cublasDmatinvBatched | | |
| cublasDotEx | hipblasDotEx | 8 |
| cublasDotcEx | hipblasDotcEx | 8 |
| cublasDtpttr | | |
| cublasDtrsmBatched | hipblasDtrsmBatched | |
| cublasDtrttp | | |

| | | |
|---|---|---|
| cublasGemmBatchedEx | hipblasGemmBatchedEx | 9.1 |
| cublasGemmEx | hipblasGemmEx | 8 |
| cublasGemmStridedBatchedEx | hipblasGemmStridedBatchedEx | 9.1 |
| cublasIamaxEx | | 10.1 |
| cublasIaminEx | | 10.1 |
| cublasRotEx | hipblasRotEx | 10.1 |
| cublasRotgEx | | 10.1 |
| cublasRotmEx | | 10.1 |
| cublasRotmgEx | | 10.1 |
| cublasScalEx | hipblasScalEx | 8 |
| cublasSdgmm | hipblasSdgmm | |
| cublasSgeam | hipblasSgeam | |
| cublasSgelsBatched | hipblasSgelsBatched | |
| cublasSgemmEx | | 7.5 |
| cublasSgeqrfBatched | hipblasSgeqrfBatched | |
| cublasSgetrfBatched | hipblasSgetrfBatched | |
| cublasSgetriBatched | hipblasSgetriBatched | |
| cublasSgetrsBatched | hipblasSgetrsBatched | |
| cublasSmatinvBatched | | |
| cublasStpttr | | |
| cublasStrsmBatched | hipblasStrsmBatched | |
| cublasStrttp | | |
| cublasSwapEx | | 10.1 |
| cublasUint8gemmBias | | 8 |
| cublasZdgmm | hipblasZdgmm | |
| cublasZgeam | hipblasZgeam | |
| cublasZgelsBatched | hipblasZgelsBatched | |

| | | |
|---|---|---|
| cublasZgeqrfBatched | hipblasZgeqrfBatched | |
| cublasZgetrfBatched | hipblasZgetrfBatched | |
| cublasZgetriBatched | hipblasZgetriBatched | |
| cublasZgetrsBatched | hipblasZgetrsBatched | |
| cublasZmatinvBatched | | |
| cublasZtpttr | | |
| cublasZtrsmBatched | hipblasZtrsmBatched | |
| cublasZtrttp | | |

## 5.3 MIOpen

| CUDA | HIP | CUDA version * |
|---|---|---|
| cudnnActivationBackward | hipdnnActivationBackward | |
| cudnnActivationForward | hipdnnActivationForward | |
| cudnnAddTensor | hipdnnAddTensor | |
| cudnnAdvInferVersionCheck | | 8.0.1 |
| cudnnAdvTrainVersionCheck | | 8.0.1 |
| cudnnBackendCreateDescriptor | | 8.0.1 |
| cudnnBackendDestroyDescriptor | | 8.0.1 |
| cudnnBackendExecute | | 8.0.1 |
| cudnnBackendFinalize | | 8.0.1 |
| cudnnBackendGetAttribute | | 8.0.1 |
| cudnnBackendInitialize | | 8.0.1 |
| cudnnBackendSetAttribute | | 8.0.1 |
| cudnnBatchNormalizationBackward | hipdnnBatchNormalizationBackward | |
| cudnnBatchNormalizationBackwardEx | | 7.4.1 |

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| | | |
|---|---|---|
| cudnnBatchNormalizationForwardInference | hipdnnBatchNormalizationForwardInference | |
| cudnnBatchNormalizationForwardTraining | hipdnnBatchNormalizationForwardTraining | |
| cudnnBatchNormalizationForwardTrainingEx | | 7.4.1 |
| cudnnBuildRNNDynamic | | 8.0.1 |
| cudnnCTCLoss | | 7.0.5 |
| cudnnCTCLoss_v8 | | 8.0.1 |
| cudnnCnnInferVersionCheck | | 8.0.2 |
| cudnnCnnTrainVersionCheck | | 8.0.2 |
| cudnnConvolutionBackwardBias | hipdnnConvolutionBackwardBias | |
| cudnnConvolutionBackwardData | hipdnnConvolutionBackwardData | |
| cudnnConvolutionBackwardFilter | hipdnnConvolutionBackwardFilter | |
| cudnnConvolutionBiasActivationForward | | 6.0.0 |
| cudnnConvolutionForward | hipdnnConvolutionForward | |
| cudnnCreate | hipdnnCreate | |
| cudnnCreateActivationDescriptor | hipdnnCreateActivationDescriptor | |
| cudnnCreateAttnDescriptor | | 7.5.0 |
| cudnnCreateCTCLossDescriptor | | 7.0.5 |
| cudnnCreateConvolutionDescriptor | hipdnnCreateConvolutionDescriptor | |
| cudnnCreateDropoutDescriptor | hipdnnCreateDropoutDescrip | |

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| | | |
|---|---|---|
| | tor | |
| cudnnCreateFilterDescriptor | hipdnnCreateFilterDescriptor | |
| cudnnCreateFusedOpsConstParamPack | | 7.6.0 |
| cudnnCreateFusedOpsPlan | | 7.6.0 |
| cudnnCreateFusedOpsVariantParamPack | | 7.6.0 |
| cudnnCreateLRNDescriptor | hipdnnCreateLRNDescriptor | |
| cudnnCreateOpTensorDescriptor | hipdnnCreateOpTensorDescriptor | |
| cudnnCreatePoolingDescriptor | hipdnnCreatePoolingDescriptor | |
| cudnnCreateRNNDataDescriptor | | 7.2.1 |
| cudnnCreateRNNDescriptor | hipdnnCreateRNNDescriptor | |
| cudnnCreateReduceTensorDescriptor | hipdnnCreateReduceTensorDescriptor | |
| cudnnCreateSeqDataDescriptor | | 7.5.0 |
| cudnnCreateSpatialTransformerDescriptor | | 5.0.0 |
| cudnnCreateTensorDescriptor | hipdnnCreateTensorDescriptor | |
| cudnnCreateTensorTransformDescriptor | | 7.5.0 |
| cudnnDeriveBNTensorDescriptor | hipdnnDeriveBNTensorDescriptor | |
| cudnnDeriveNormTensorDescriptor | | 8.0.1 |
| cudnnDestroy | hipdnnDestroy | |
| cudnnDestroyActivationDescriptor | hipdnnDestroyActivationDes | |

| | | |
|---|---|---|
| | criptor | |
| cudnnDestroyAttnDescriptor | | 7.5.0 |
| cudnnDestroyCTCLossDescriptor | | 7.0.5 |
| cudnnDestroyConvolutionDescriptor | hipdnnDestroyConvolutionDescriptor | |
| cudnnDestroyDropoutDescriptor | hipdnnDestroyDropoutDescriptor | |
| cudnnDestroyFilterDescriptor | hipdnnDestroyFilterDescriptor | |
| cudnnDestroyFusedOpsConstParamPack | | 7.6.0 |
| cudnnDestroyFusedOpsPlan | | 7.6.0 |
| cudnnDestroyFusedOpsVariantParamPack | | 7.6.0 |
| cudnnDestroyLRNDescriptor | hipdnnDestroyLRNDescriptor | |
| cudnnDestroyOpTensorDescriptor | hipdnnDestroyOpTensorDescriptor | |
| cudnnDestroyPoolingDescriptor | hipdnnDestroyPoolingDescriptor | |
| cudnnDestroyRNNDataDescriptor | | 7.2.1 |
| cudnnDestroyRNNDescriptor | hipdnnDestroyRNNDescriptor | |
| cudnnDestroyReduceTensorDescriptor | hipdnnDestroyReduceTensorDescriptor | |
| cudnnDestroySeqDataDescriptor | | 7.5.0 |
| cudnnDestroySpatialTransformerDescriptor | | 5.0.0 |

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| | | |
|---|---|---|
| cudnnDestroyTensorDescriptor | hipdnnDestroyTensorDescriptor | |
| cudnnDestroyTensorTransformDescriptor | | 7.5.0 |
| cudnnDivisiveNormalizationBackward | | 3.0.0 |
| cudnnDivisiveNormalizationForward | | 3.0.0 |
| cudnnDropoutBackward | | 5.0.0 |
| cudnnDropoutForward | | 5.0.0 |
| cudnnDropoutGetReserveSpaceSize | | 5.0.0 |
| cudnnDropoutGetStatesSize | hipdnnDropoutGetStatesSize | |
| cudnnFindConvolutionBackwardDataAlgorithm | hipdnnFindConvolutionBackwardDataAlgorithm | |
| cudnnFindConvolutionBackwardDataAlgorithmEx | hipdnnFindConvolutionBackwardDataAlgorithmEx | |
| cudnnFindConvolutionBackwardFilterAlgorithm | hipdnnFindConvolutionBackwardFilterAlgorithm | |
| cudnnFindConvolutionBackwardFilterAlgorithmEx | hipdnnFindConvolutionBackwardFilterAlgorithmEx | |
| cudnnFindConvolutionForwardAlgorithm | hipdnnFindConvolutionForwardAlgorithm | |
| cudnnFindConvolutionForwardAlgorithmEx | hipdnnFindConvolutionForwardAlgorithmEx | |
| cudnnFusedOpsExecute | | 7.6.0 |
| cudnnGetActivationDescriptor | hipdnnGetActivationDescriptor | |
| cudnnGetActivationDescriptorSwishBeta | | 8.2.0 |

| | | |
|---|---|---|
| cudnnGetAttnDescriptor | | 7.5.0 |
| cudnnGetBatchNormalizationBackwardExWorkspaceSize | | 7.4.1 |
| cudnnGetBatchNormalizationForwardTrainingExWorkspaceSize | | 7.4.1 |
| cudnnGetBatchNormalizationTrainingExReserveSpaceSize | | 7.4.1 |
| cudnnGetCTCLossDescriptor | | 7.0.5 |
| cudnnGetCTCLossDescriptorEx | | 7.5.0 |
| cudnnGetCTCLossDescriptor_v8 | | 8.0.1 |
| cudnnGetCTCLossWorkspaceSize | | 7.0.5 |
| cudnnGetCTCLossWorkspaceSize_v8 | | 8.0.1 |
| cudnnGetCallback | | 7.1.3 |
| cudnnGetConvolution2dDescriptor | hipdnnGetConvolution2dDescriptor | |
| cudnnGetConvolution2dForwardOutputDim | hipdnnGetConvolution2dForwardOutputDim | |
| cudnnGetConvolutionBackwardDataAlgorithmMaxCount | | 7.0.5 |
| cudnnGetConvolutionBackwardDataAlgorithm_v7 | | 7.0.5 |
| cudnnGetConvolutionBackwardDataWorkspaceSize | hipdnnGetConvolutionBackwardDataWorkspaceSize | |
| cudnnGetConvolutionBackwardFilterAlgorithmMaxCount | | 7.0.5 |
| cudnnGetConvolutionBackwardFilterAlgorithm_v7 | | 7.0.5 |
| cudnnGetConvolutionBackwardFilter | hipdnnGetConvolutionBackw | |

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| | | |
|---|---|---|
| WorkspaceSize | ardFilterWorkspaceSize | |
| cudnnGetConvolutionForwardAlgorithmMaxCount | | 7.0.5 |
| cudnnGetConvolutionForwardAlgorithm_v7 | | 7.0.5 |
| cudnnGetConvolutionForwardWorkspaceSize | hipdnnGetConvolutionForwardWorkspaceSize | |
| cudnnGetConvolutionGroupCount | | 7.0.5 |
| cudnnGetConvolutionMathType | | 7.0.5 |
| cudnnGetConvolutionNdDescriptor | | 2.0.0 |
| cudnnGetConvolutionNdForwardOutputDim | | 2.0.0 |
| cudnnGetConvolutionReorderType | | 7.6.0 |
| cudnnGetCudartVersion | | 6.0.0 |
| cudnnGetDropoutDescriptor | | 7.0.5 |
| cudnnGetErrorString | hipdnnGetErrorString | |
| cudnnGetFilter4dDescriptor | hipdnnGetFilter4dDescriptor | |
| cudnnGetFilterNdDescriptor | hipdnnGetFilterNdDescriptor | |
| cudnnGetFilterSizeInBytes | | 7.6.0 |
| cudnnGetFoldedConvBackwardDataDescriptors | | 7.6.0 |
| cudnnGetFusedOpsConstParamPackAttribute | | 7.6.0 |
| cudnnGetFusedOpsVariantParamPackAttribute | | 7.6.0 |
| cudnnGetLRNDescriptor | hipdnnGetLRNDescriptor | |
| cudnnGetMultiHeadAttnBuffers | | 7.5.0 |
| cudnnGetMultiHeadAttnWeights | | 7.5.0 |

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| | | |
|---|---|---|
| cudnnGetNormalizationBackwardWorkspaceSize | | 8.0.1 |
| cudnnGetNormalizationForwardTrainingWorkspaceSize | | 8.0.1 |
| cudnnGetNormalizationTrainingReserveSpaceSize | | 8.0.1 |
| cudnnGetOpTensorDescriptor | hipdnnGetOpTensorDescriptor | |
| cudnnGetPooling2dDescriptor | hipdnnGetPooling2dDescriptor | |
| cudnnGetPooling2dForwardOutputDim | hipdnnGetPooling2dForwardOutputDim | |
| cudnnGetPoolingNdDescriptor | | 2.0.0 |
| cudnnGetPoolingNdForwardOutputDim | | 2.0.0 |
| cudnnGetProperty | | 6.0.0 |
| cudnnGetRNNDataDescriptor | | 7.2.1 |
| cudnnGetRNNDescriptor_v8 | | 8.0.1 |
| cudnnGetRNNTempSpaceSizes | | 8.0.1 |
| cudnnGetRNNTrainingReserveSize | hipdnnGetRNNTrainingReserveSize | |
| cudnnGetRNNWeightParams | | 8.0.1 |
| cudnnGetRNNWeightSpaceSize | | 8.0.1 |
| cudnnGetRNNWorkspaceSize | hipdnnGetRNNWorkspaceSize | |
| cudnnGetReduceTensorDescriptor | hipdnnGetReduceTensorDescriptor | |
| cudnnGetReductionIndicesSize | | 6.0.0 |

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| | | |
|---|---|---|
| cudnnGetReductionWorkspaceSize | hipdnnGetReductionWorkspaceSize | |
| cudnnGetSeqDataDescriptor | | 7.5.0 |
| cudnnGetStream | hipdnnGetStream | |
| cudnnGetTensor4dDescriptor | hipdnnGetTensor4dDescriptor | |
| cudnnGetTensorNdDescriptor | hipdnnGetTensorNdDescriptor | |
| cudnnGetTensorSizeInBytes | | 6.0.0 |
| cudnnGetTensorTransformDescriptor | | 7.5.0 |
| cudnnGetVersion | hipdnnGetVersion | |
| cudnnIm2Col | | 2.0.0 |
| cudnnInitTransformDest | | 7.5.0 |
| cudnnLRNCrossChannelBackward | hipdnnLRNCrossChannelBackward | |
| cudnnLRNCrossChannelForward | hipdnnLRNCrossChannelForward | |
| cudnnMakeFusedOpsPlan | | 7.6.0 |
| cudnnMultiHeadAttnBackwardData | | 7.5.0 |
| cudnnMultiHeadAttnBackwardWeights | | 7.5.0 |
| cudnnMultiHeadAttnForward | | 7.5.0 |
| cudnnNormalizationBackward | | 8.0.1 |
| cudnnNormalizationForwardInference | | 8.0.1 |
| cudnnNormalizationForwardTraining | | 8.0.1 |
| cudnnOpTensor | hipdnnOpTensor | |
| cudnnOpsInferVersionCheck | | 8.0.1 |
| cudnnOpsTrainVersionCheck | | 8.0.1 |

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| | | |
|---|---|---|
| cudnnPoolingBackward | hipdnnPoolingBackward | |
| cudnnPoolingForward | hipdnnPoolingForward | |
| cudnnQueryRuntimeError | | 7.0.5 |
| cudnnRNNBackwardData | hipdnnRNNBackwardData | |
| cudnnRNNBackwardData_v8 | | 8.0.2 |
| cudnnRNNBackwardWeights | hipdnnRNNBackwardWeights | |
| cudnnRNNBackwardWeights_v8 | | 8.0.2 |
| cudnnRNNForward | | 8.0.1 |
| cudnnRNNForwardInference | hipdnnRNNForwardInference | |
| cudnnRNNForwardTraining | hipdnnRNNForwardTraining | |
| cudnnRNNGetClip_v8 | | 8.0.1 |
| cudnnRNNSetClip_v8 | | 8.0.1 |
| cudnnReduceTensor | hipdnnReduceTensor | |
| cudnnReorderFilterAndBias | | 7.6.0 |
| cudnnRestoreDropoutDescriptor | | 7.0.5 |
| cudnnScaleTensor | hipdnnScaleTensor | |
| cudnnSetActivationDescriptor | hipdnnSetActivationDescriptor | |
| cudnnSetActivationDescriptorSwishBeta | | 8.2.0 |
| cudnnSetAttnDescriptor | | 7.5.0 |
| cudnnSetCTCLossDescriptor | | 7.0.5 |
| cudnnSetCTCLossDescriptorEx | | 7.6.0 |
| cudnnSetCTCLossDescriptor_v8 | | 8.0.1 |
| cudnnSetCallback | | 7.1.3 |
| cudnnSetConvolution2dDescriptor | hipdnnSetConvolution2dDescriptor | |

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| | | |
|---|---|---|
| cudnnSetConvolutionGroupCount | hipdnnSetConvolutionGroupCount | |
| cudnnSetConvolutionMathType | hipdnnSetConvolutionMathType | |
| cudnnSetConvolutionNdDescriptor | hipdnnSetConvolutionNdDescriptor | |
| cudnnSetConvolutionReorderType | | 7.6.0 |
| cudnnSetDropoutDescriptor | hipdnnSetDropoutDescriptor | |
| cudnnSetFilter4dDescriptor | hipdnnSetFilter4dDescriptor | |
| cudnnSetFilterNdDescriptor | hipdnnSetFilterNdDescriptor | |
| cudnnSetFusedOpsConstParamPackAttribute | | 7.6.0 |
| cudnnSetFusedOpsVariantParamPackAttribute | | 7.6.0 |
| cudnnSetLRNDescriptor | hipdnnSetLRNDescriptor | |
| cudnnSetOpTensorDescriptor | hipdnnSetOpTensorDescriptor | |
| cudnnSetPooling2dDescriptor | hipdnnSetPooling2dDescriptor | |
| cudnnSetPoolingNdDescriptor | hipdnnSetPoolingNdDescriptor | |
| cudnnSetRNNDataDescriptor | | 7.2.1 |
| cudnnSetRNNDescriptor_v8 | | 8.0.1 |
| cudnnSetReduceTensorDescriptor | hipdnnSetReduceTensorDescriptor | |
| cudnnSetSeqDataDescriptor | | 7.5.0 |
| cudnnSetSpatialTransformerNdDescriptor | | 5.0.0 |

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| | | |
|---|---|---|
| cudnnSetStream | hipdnnSetStream | |
| cudnnSetTensor | hipdnnSetTensor | |
| cudnnSetTensor4dDescriptor | hipdnnSetTensor4dDescriptor | |
| cudnnSetTensor4dDescriptorEx | hipdnnSetTensor4dDescriptorEx | |
| cudnnSetTensorNdDescriptor | hipdnnSetTensorNdDescriptor | |
| cudnnSetTensorNdDescriptorEx | | 6.0.0 |
| cudnnSetTensorTransformDescriptor | | 7.5.0 |
| cudnnSoftmaxBackward | hipdnnSoftmaxBackward | |
| cudnnSoftmaxForward | hipdnnSoftmaxForward | |
| cudnnSpatialTfGridGeneratorBackward | | 5.0.0 |
| cudnnSpatialTfGridGeneratorForward | | 5.0.0 |
| cudnnSpatialTfSamplerBackward | | 5.0.0 |
| cudnnSpatialTfSamplerForward | | 5.0.0 |
| cudnnTransformFilter | | 7.6.0 |
| cudnnTransformTensor | | 2.0.0 |
| cudnnTransformTensorEx | | 7.5.0 |

# 5.4 FFT

## 5.4.1 Data types

| CUDA | HIP | CUDAversion* |
|---|---|---|
| CUFFT_ALLOC_FAILED | HIPFFT_ALLOC_FAILED | |
| CUFFT_C2C | HIPFFT_C2C | |
| CUFFT_C2R | HIPFFT_C2R | |
| CUFFT_CB_LD_COMPLEX | HIPFFT_CB_LD_COMPLEX | |
| CUFFT_CB_LD_COMPLEX_DOUBLE | HIPFFT_CB_LD_COMPLEX_DOUBLE | |

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| | | |
|---|---|---|
| CUFFT_CB_LD_REAL | HIPFFT_CB_LD_REAL | |
| CUFFT_CB_LD_REAL_DOUBLE | HIPFFT_CB_LD_REAL_DOUBLE | |
| CUFFT_CB_ST_COMPLEX | HIPFFT_CB_ST_COMPLEX | |
| CUFFT_CB_ST_COMPLEX_DOUBLE | HIPFFT_CB_ST_COMPLEX_DOUBLE | |
| CUFFT_CB_ST_REAL | HIPFFT_CB_ST_REAL | |
| CUFFT_CB_ST_REAL_DOUBLE | HIPFFT_CB_ST_REAL_DOUBLE | |
| CUFFT_CB_UNDEFINED | HIPFFT_CB_UNDEFINED | |
| CUFFT_COMPATIBILITY_DEFAULT | | |
| CUFFT_COMPATIBILITY_FFTW_PADDING | | |
| CUFFT_COPY_DEVICE_TO_DEVICE | | |
| CUFFT_COPY_DEVICE_TO_HOST | | |
| CUFFT_COPY_HOST_TO_DEVICE | | |
| CUFFT_COPY_UNDEFINED | | |
| CUFFT_D2Z | HIPFFT_D2Z | |
| CUFFT_EXEC_FAILED | HIPFFT_EXEC_FAILED | |
| CUFFT_FORMAT_UNDEFINED | | |
| CUFFT_FORWARD | HIPFFT_FORWARD | |
| CUFFT_INCOMPLETE_PARAMETER_LIST | HIPFFT_INCOMPLETE_PARAMETER_LIST | |

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| | | |
|---|---|---|
| CUFFT_INTERNAL_ERROR | HIPFFT_INTERNAL_ERRO R | |
| CUFFT_INVALID_DEVICE | HIPFFT_INVALID_DEVICE | |
| CUFFT_INVALID_PLAN | HIPFFT_INVALID_PLAN | |
| CUFFT_INVALID_SIZE | HIPFFT_INVALID_SIZE | |
| CUFFT_INVALID_TYPE | HIPFFT_INVALID_TYPE | |
| CUFFT_INVALID_VALUE | HIPFFT_INVALID_VALUE | |
| CUFFT_INVERSE | HIPFFT_BACKWARD | |
| CUFFT_LICENSE_ERROR | | |
| CUFFT_NOT_IMPLEMENTE D | HIPFFT_NOT_IMPLEMENT ED | |
| CUFFT_NOT_SUPPORTED | HIPFFT_NOT_SUPPORTED | 8 |
| CUFFT_NO_WORKSPACE | HIPFFT_NO_WORKSPACE | |
| CUFFT_PARSE_ERROR | HIPFFT_PARSE_ERROR | |
| CUFFT_QUERY_1D_FACTOR S | | |
| CUFFT_QUERY_UNDEFINE D | | |
| CUFFT_R2C | HIPFFT_R2C | |
| CUFFT_SETUP_FAILED | HIPFFT_SETUP_FAILED | |
| CUFFT_SUCCESS | HIPFFT_SUCCESS | |
| CUFFT_UNALIGNED_DATA | HIPFFT_UNALIGNED_DAT A | |
| CUFFT_VERSION | | 10.2 |
| CUFFT_VER_BUILD | | 10.2 |
| CUFFT_VER_MAJOR | HIPFFT_MAJOR_VERSION | 10.2 |
| CUFFT_VER_MINOR | HIPFFT_MINOR_VERSION | 10.2 |
| CUFFT_VER_PATCH | HIPFFT_PATCH_LEVEL | 10.2 |

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| | | |
|---|---|---|
| CUFFT_WORKAREA_MINIMAL | | 9.2 |
| CUFFT_WORKAREA_PERFORMANCE | | |
| CUFFT_WORKAREA_USER | | 9.2 |
| CUFFT_XT_FORMAT_1D_INPUT_SHUFFLED | | |
| CUFFT_XT_FORMAT_INPLACE | | |
| CUFFT_XT_FORMAT_INPLACE_SHUFFLED | | |
| CUFFT_XT_FORMAT_INPUT | | |
| CUFFT_XT_FORMAT_OUTPUT | | |
| CUFFT_Z2D | HIPFFT_Z2D | |
| CUFFT_Z2Z | HIPFFT_Z2Z | |
| MAX_CUFFT_ERROR | | |
| cuFFTFORWARD | HIPFFT_FORWARD | 10.2 |
| cuFFTINVERSE | HIPFFT_BACKWARD | 10.2 |
| cufftCompatibility | | |
| cufftCompatibility_t | | |
| cufftComplex | hipfftComplex | |
| cufftDoubleComplex | hipfftDoubleComplex | |
| cufftDoubleReal | hipfftDoubleReal | |
| cufftHandle | hipfftHandle | |
| cufftReal | hipfftReal | |
| cufftResult | hipfftResult | |
| cufftResult_t | hipfftResult_t | |

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| | | |
|---|---|---|
| cufftType | hipfftType | |
| cufftType_t | hipfftType_t | |
| cufftXt1dFactors | | |
| cufftXt1dFactors_t | | |
| cufftXtCallbackType | hipfftXtCallbackType | |
| cufftXtCallbackType_t | hipfftXtCallbackType_t | |
| cufftXtCopyType | | |
| cufftXtCopyType_t | | |
| cufftXtQueryType | | |
| cufftXtQueryType_t | | |
| cufftXtSubFormat | | |
| cufftXtSubFormat_t | | |
| cufftXtWorkAreaPolicy | | 9.2 |
| cufftXtWorkAreaPolicy_t | | 9.2 |

### 5.4.2 APIfunctions

| CUDA | HIP | CUDAversion* |
|---|---|---|
| cufftCallbackLoadC | hipfftCallbackLoadC | |
| cufftCallbackLoadD | hipfftCallbackLoadD | |
| cufftCallbackLoadR | hipfftCallbackLoadR | |
| cufftCallbackLoadZ | hipfftCallbackLoadZ | |
| cufftCallbackStoreC | hipfftCallbackStoreC | |
| cufftCallbackStoreD | hipfftCallbackStoreD | |
| cufftCallbackStoreR | hipfftCallbackStoreR | |
| cufftCallbackStoreZ | hipfftCallbackStoreZ | |
| cufftCreate | hipfftCreate | |
| cufftDestroy | hipfftDestroy | |
| cufftEstimate1d | hipfftEstimate1d | |

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| | | |
|---|---|---|
| cufftEstimate2d | hipfftEstimate2d | |
| cufftEstimate3d | hipfftEstimate3d | |
| cufftEstimateMany | hipfftEstimateMany | |
| cufftExecC2C | hipfftExecC2C | |
| cufftExecC2R | hipfftExecC2R | |
| cufftExecD2Z | hipfftExecD2Z | |
| cufftExecR2C | hipfftExecR2C | |
| cufftExecZ2D | hipfftExecZ2D | |
| cufftExecZ2Z | hipfftExecZ2Z | |
| cufftGetProperty | hipfftGetProperty | 8 |
| cufftGetSize | hipfftGetSize | |
| cufftGetSize1d | hipfftGetSize1d | |
| cufftGetSize2d | hipfftGetSize2d | |
| cufftGetSize3d | hipfftGetSize3d | |
| cufftGetSizeMany | hipfftGetSizeMany | |
| cufftGetSizeMany64 | hipfftGetSizeMany64 | 7.5 |
| cufftGetVersion | hipfftGetVersion | |
| cufftMakePlan1d | hipfftMakePlan1d | |
| cufftMakePlan2d | hipfftMakePlan2d | |
| cufftMakePlan3d | hipfftMakePlan3d | |
| cufftMakePlanMany | hipfftMakePlanMany | |
| cufftMakePlanMany64 | hipfftMakePlanMany64 | 7.5 |
| cufftPlan1d | hipfftPlan1d | |
| cufftPlan2d | hipfftPlan2d | |
| cufftPlan3d | hipfftPlan3d | |
| cufftPlanMany | hipfftPlanMany | |
| cufftSetAutoAllocation | hipfftSetAutoAllocation | |
| cufftSetStream | hipfftSetStream | |

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| | | |
|---|---|---|
| cufftSetWorkArea | hipfftSetWorkArea | |
| cufftXtClearCallback | hipfftXtClearCallback | |
| cufftXtExec | | 8 |
| cufftXtExecDescriptor | | |
| cufftXtExecDescriptorC2C | | |
| cufftXtExecDescriptorC2R | | |
| cufftXtExecDescriptorD2Z | | |
| cufftXtExecDescriptorR2C | | |
| cufftXtExecDescriptorZ2D | | |
| cufftXtExecDescriptorZ2Z | | |
| cufftXtFree | | |
| cufftXtGetSizeMany | | 8 |
| cufftXtMakePlanMany | | 8 |
| cufftXtMalloc | | |
| cufftXtMemcpy | | |
| cufftXtQueryPlan | | |
| cufftXtSetCallback | hipfftXtSetCallback | |
| cufftXtSetCallbackSharedSize | hipfftXtSetCallbackSharedSize | |
| cufftXtSetGPUs | | |
| cufftXtSetWorkArea | | |
| cufftXtSetWorkAreaPolicy | | 9.2 |

## 5.5 SOLVER

| CUDA | HIP | CUDA version * |
|---|---|---|
| cusolverDnCreate | hipsolverDnCreate | 10.2 |
| cusolverDnDestroy | hipsolverDnDestroy | |

开发者社区

| | | |
|---|---|---|
| cusolverDnSetStream | hipsolverDnSetStream | |
| cusolverDnGetStream | hipsolverDnGetStream | |
| cusolverDnCreateSyevjInfo | hipsolverDnCreateSyevjInfo | |
| cusolverDnDestroySyevjInfo | hipsolverDnDestroySyevjInfo | |
| cusolverDnXsyevjSetTolerance | hipsolverDnXsyevjSetTolerance | |
| cusolverDnXsyevjSetMaxSweeps | hipsolverDnXsyevjSetMaxSweeps | |
| cusolverDnXsyevjSetSortEig | hipsolverDnXsyevjSetSortEig | |
| cusolverDnXsyevjGetResidual | hipsolverDnXsyevjGetResidual | |
| cusolverDnXsyevjGetSweeps | hipsolverDnXsyevjGetSweeps | |
| cusolverDnCreateGesvdjInfo | hipsolverDnCreateGesvdjInfo | |
| cusolverDnDestroyGesvdjInfo | hipsolverDnDestroyGesvdjInfo | |
| cusolverDnXgesvdjSetTolerance | hipsolverDnXgesvdjSetTolerance | |
| cusolverDnXgesvdjSetMaxSweeps | hipsolverDnXgesvdjSetMaxSweeps | |
| cusolverDnXgesvdjSetSortEig | hipsolverDnXgesvdjSetSortEig | |
| cusolverDnXgesvdjGetResidual | hipsolverDnXgesvdjGetResidual | |
| cusolverDnXgesvdjGetSweeps | hipsolverDnXgesvdjGetSweeps | |
| cusolverDnIRSParamsCreate | | |
| cusolverDnIRSParamsDestroy | | |
| cusolverDnIRSParamsSetSolverPrecisions | | |
| cusolverDnIRSParamsSetSolverMainPrecision | | |
| cusolverDnIRSParamsSetSolverLowestPrecision | | |
| cusolverDnIRSParamsSetRefinementSolver | | |
| cusolverDnIRSParamsSetTol | | |
| cusolverDnIRSParamsSetTolInner | | |

| | | |
|---|---|---|
| cusolverDnIRSParamsSetMaxIters | | |
| cusolverDnIRSParamsSetMaxItersInner | | |
| cusolverDnIRSParamsEnableFallback | | |
| cusolverDnIRSParamsDisableFallback | | |
| cusolverDnIRSParamsGetMaxIters | | |
| cusolverDnIRSInfosCreate | | |
| cusolverDnIRSInfosDestroy | | |
| cusolverDnIRSInfosGetMaxIters | | |
| cusolverDnIRSInfosGetNiters | | |
| cusolverDnIRSInfosGetOuterNiters | | |
| cusolverDnIRSInfosRequestResidual | | |
| cusolverDnIRSInfosGetResidualHistory | | |
| cusolverDnCreateParams | | |
| cusolverDnDestroyParams | | |
| cusolverDnSetAdvOptions | | |
| cusolverDnSpotrf_bufferSize | hipsolverDnSpotrf_bufferSize | |
| cusolverDnDpotrf_bufferSize | hipsolverDnDpotrf_bufferSize | |
| cusolverDnCpotrf_bufferSize | hipsolverDnCpotrf_bufferSize | |
| cusolverDnZpotrf_bufferSize | hipsolverDnZpotrf_bufferSize | |
| cusolverDnSpotrf | hipsolverDnSpotrf | |
| cusolverDnDpotrf | hipsolverDnDpotrf | |
| cusolverDnCpotrf | hipsolverDnCpotrf | |
| cusolverDnZpotrf | hipsolverDnZpotrf | |

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|---|---|---|
| cusolverDnSpotrs | hipsolverDnSpotrs | |
| cusolverDnDpotrs | hipsolverDnDpotrs | |
| cusolverDnCpotrs | hipsolverDnCpotrs | |
| cusolverDnZpotrs | hipsolverDnZpotrs | |
| cusolverDnSpotri_bufferSize | hipsolverDnSpotri_bufferSize | |
| cusolverDnDpotri_bufferSize | hipsolverDnDpotri_bufferSize | |
| cusolverDnCpotri_bufferSize | hipsolverDnCpotri_bufferSize | |
| cusolverDnZpotri_bufferSize | hipsolverDnZpotri_bufferSize | |
| cusolverDnSpotri | hipsolverDnSpotri | |
| cusolverDnDpotri | hipsolverDnDpotri | |
| cusolverDnCpotri | hipsolverDnCpotri | |
| cusolverDnZpotri | hipsolverDnZpotri | |
| cusolverDnSgetrf_bufferSize | hipsolverDnSgetrf_bufferSize | |
| cusolverDnDgetrf_bufferSize | hipsolverDnDgetrf_bufferSize | |
| cusolverDnCgetrf_bufferSize | hipsolverDnCgetrf_bufferSize | |
| cusolverDnZgetrf_bufferSize | hipsolverDnZgetrf_bufferSize | |
| cusolverDnSgetrf | hipsolverDnSgetrf | |
| cusolverDnDgetrf | hipsolverDnDgetrf | |
| cusolverDnCgetrf | hipsolverDnCgetrf | |
| cusolverDnZgetrf | hipsolverDnZgetrf | |
| cusolverDnSgetrs | hipsolverDnSgetrs | |
| cusolverDnDgetrs | hipsolverDnDgetrs | |
| cusolverDnCgetrs | hipsolverDnCgetrs | |
| cusolverDnZgetrs | hipsolverDnZgetrs | |
| cusolverDnZZgesv_bufferSize | hipsolverDnZZgesv_bufferSize | |
| cusolverDnZCgesv_bufferSize | | |
| cusolverDnZKgesv_bufferSize | | |
| cusolverDnZEgesv_bufferSize | | |

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|---|---|---|
| cusolverDnZYgesv_bufferSize | | |
| cusolverDnCCgesv_bufferSize | hipsolverCCgesv_bufferSize | |
| cusolverDnCKgesv_bufferSize | | |
| cusolverDnCEgesv_bufferSize | | |
| cusolverDnCYgesv_bufferSize | | |
| cusolverDnDDgesv_bufferSize | hipsolverDnDDgesv_bufferSize | |
| cusolverDnDSgesv_bufferSize | | |
| cusolverDnDHgesv_bufferSize | | |
| cusolverDnDBgesv_bufferSize | | |
| cusolverDnDXgesv_bufferSize | | |
| cusolverDnSSgesv_bufferSize | hipsolverDnSSgesv_bufferSize | |
| cusolverDnSHgesv_bufferSize | | |
| cusolverDnSBgesv_bufferSize | | |
| cusolverDnSXgesv_bufferSize | | |
| cusolverDnZZgesv | hipsolverDnZZgesv | |
| cusolverDnZCgesv | | |
| cusolverDnZKgesv | | |
| cusolverDnZEgesv | | |
| cusolverDnZYgesv | | |
| cusolverDnCCgesv | hipsolverDnCCgesv | |
| cusolverDnCKgesv | | |
| cusolverDnCEgesv | | |
| cusolverDnCYgesv | | |
| cusolverDnDDgesv | hipsolverDnDDgesv | |
| cusolverDnDSgesv | | |
| cusolverDnDHgesv | | |
| cusolverDnDBgesv | | |
| cusolverDnDXgesv | | |

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|---|---|---|
| cusolverDnSSgesv | hipsolverDnSSgesv | |
| cusolverDnSHgesv | | |
| cusolverDnSBgesv | | 9 |
| cusolverDnSXgesv | | |
| cusolverDnIRSXgesv_bufferSize | | |
| cusolverDnIRSXgesv | | |
| cusolverDnSgeqrf_bufferSize | hipsolverDnSgeqrf_bufferSize | |
| cusolverDnDgeqrf_bufferSize | hipsolverDnDgeqrf_bufferSize | |
| cusolverDnCgeqrf_bufferSize | hipsolverDnCgeqrf_bufferSize | |
| cusolverDnZgeqrf_bufferSize | hipsolverDnZgeqrf_bufferSize | |
| cusolverDnSgeqrf | hipsolverDnSgeqrf | |
| cusolverDnDgeqrf | hipsolverDnDgeqrf | |
| cusolverDnCgeqrf | hipsolverDnCgeqrf | |
| cusolverDnZgeqrf | hipsolverDnZgeqrf | |
| cusolverDnZZgels_bufferSize | hipsolverDnZZgels_bufferSize | |
| cusolverDnZCgels_bufferSize | | |
| cusolverDnZKgels_bufferSize | | |
| cusolverDnZEgels_bufferSize | | |
| cusolverDnZYgels_bufferSize | | |
| cusolverDnCCgels_bufferSize | hipsolverDnCCgels_bufferSize | |
| cusolverDnCKgels_bufferSize | | |
| cusolverDnCEgels_bufferSize | | |
| cusolverDnCYgels_bufferSize | | |
| cusolverDnDDgels_bufferSize | hipsolverDnDDgels_bufferSize | |
| cusolverDnDSgels_bufferSize | | |
| cusolverDnDHgels_bufferSize | | |
| cusolverDnDBgels_bufferSize | | |
| cusolverDnDXgels_bufferSize | | |

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|---|---|---|
| cusolverDnSSgels_bufferSize | hipsolverDnSSgels_bufferSize | |
| cusolverDnSHgels_bufferSize | | |
| cusolverDnSBgels_bufferSize | | |
| cusolverDnSXgels_bufferSize | | |
| cusolverDnZZgels | hipsolverDnZZgels | |
| cusolverDnZCgels | | |
| cusolverDnZKgels | | |
| cusolverDnZEgels | | |
| cusolverDnZYgels | | |
| cusolverDnCCgels | hipsolverDnCCgels | |
| cusolverDnCKgels | | |
| cusolverDnCEgels | | |
| cusolverDnCYgels | | |
| cusolverDnDDgels | hipsolverDnDDgels | |
| cusolverDnDSgels | | |
| cusolverDnDHgels | | |
| cusolverDnDBgels | | |
| cusolverDnDXgels | | |
| cusolverDnSSgels | hipsolverDnSSgels | |
| cusolverDnSHgels | | |
| cusolverDnSBgels | | |
| cusolverDnSXgels | | |
| cusolverDnIRSXgels_bufferSize | | |
| cusolverDnIRSXgels | | |
| cusolverDnSormqr_bufferSize | hipsolverDnSormqr_bufferSize | |
| cusolverDnDormqr_bufferSize | hipsolverDnDormqr_bufferSize | |
| cusolverDnCunmqr_bufferSize | hipsolverDnCunmqr_bufferSize | |
| cusolverDnZunmqr_bufferSize | hipsolverDnZunmqr_bufferSize | |

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|---|---|---|
| cusolverDnSormqr | hipsolverDnSormqr | |
| cusolverDnDormqr | hipsolverDnDormqr | |
| cusolverDnCunmqr | hipsolverDnCunmqr | |
| cusolverDnZunmqr | hipsolverDnZunmqr | |
| cusolverDnSorgqr_bufferSize | hipsolverDnSorgqr_bufferSiz | |
| cusolverDnDorgqr_bufferSize | hipsolverDnDorgqr_bufferSiz | |
| cusolverDnCungqr_bufferSize | hipsolverDnCungqr_bufferSize | |
| cusolverDnZungqr_bufferSize | hipsolverDnZungqr_bufferSize | |
| cusolverDnSorgqr | hipsolverDnSorgqr | |
| cusolverDnDorgqr | hipsolverDnDorgqr | |
| cusolverDnCungqr | hipsolverDnCungqr | |
| cusolverDnZungqr | hipsolverDnZungqr | |
| cusolverDnSsytrf_bufferSize | hipsolverDnSsytrf_bufferSiz | |
| cusolverDnDsytrf_bufferSize | hipsolverDnDsytrf_bufferSiz | |
| cusolverDnCsytrf_bufferSize | hipsolverDnCsytrf_bufferSiz | |
| cusolverDnZsytrf_bufferSize | hipsolverDnZsytrf_bufferSiz | |
| cusolverDnSsytrf | hipsolverDnSsytrf | |
| cusolverDnDsytrf | hipsolverDnDsytrf | |
| cusolverDnCsytrf | hipsolverDnCsytrf | |
| cusolverDnZsytrf | hipsolverDnZsytrf | |
| cusolverDnSpotrfBatched | hipsolverDnSpotrfBatched | |
| cusolverDnDpotrfBatched | hipsolverDnDpotrfBatched | |
| cusolverDnCpotrfBatched | hipsolverDnCpotrfBatched | |
| cusolverDnZpotrfBatched | hipsolverDnZpotrfBatched | |
| cusolverDnSpotrsBatched | hipsolverDnSpotrsBatched | |
| cusolverDnDpotrsBatched | hipsolverDnDpotrsBatched | |
| cusolverDnCpotrsBatched | hipsolverDnCpotrsBatched | |
| cusolverDnZpotrsBatched | hipsolverDnZpotrsBatched | |

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|---|---|---|
| cusolverDnSgebrd_bufferSize | hipsolverDnSgebrd_bufferSize | |
| cusolverDnDgebrd_bufferSize | hipsolverDnDgebrd_bufferSize | |
| cusolverDnCgebrd_bufferSize | hipsolverDnCgebrd_bufferSize | |
| cusolverDnZgebrd_bufferSize | hipsolverDnZgebrd_bufferSize | |
| cusolverDnSgebrd | hipsolverDnSgebrd | |
| cusolverDnDgebr | hipsolverDnDgebrd | |
| cusolverDnCgebrd | hipsolverDnCgebrd | |
| cusolverDnZgebrd | hipsolverDnZgebrd | |
| cusolverDnSorgbr_bufferSize | hipsolverDnSorgbr_bufferSize | |
| cusolverDnDorgbr_bufferSize | hipsolverDnDorgbr_bufferSize | |
| cusolverDnCungbr_bufferSize | hipsolverDnCungbr_bufferSize | |
| cusolverDnZungbr_bufferSize | hipsolverDnZungbr_bufferSize | |
| cusolverDnSorgbr | hipsolverDnSorgbr | |
| cusolverDnDorgbr | hipsolverDnDorgbr | |
| cusolverDnCungbr | hipsolverDnCungbr | |
| cusolverDnZungbr | hipsolverDnZungbr | |
| cusolverDnSsytrd_bufferSize | hipsolverDnSsytrd_bufferSize | |
| cusolverDnDsytrd_bufferSize | hipsolverDnDsytrd_bufferSize | |
| cusolverDnChetrd_bufferSize | hipsolverDnChetrd_bufferSize | |
| cusolverDnZhetrd_bufferSize | hipsolverDnZhetrd_bufferSize | |
| cusolverDnSsytrd | hipsolverDnSsytrd | |
| cusolverDnDsytrd | hipsolverDnDsytrd | |
| cusolverDnChetrd | hipsolverDnChetrd | |
| cusolverDnZhetrd | hipsolverDnZhetrd | |
| cusolverDnSormtr_bufferSize | hipsolverDnSormtr_bufferSize | |
| cusolverDnDormtr_bufferSize | hipsolverDnDormtr_bufferSize | |
| cusolverDnCunmtr_bufferSize | hipsolverDnCunmtr_bufferSize | |
| cusolverDnZunmtr_bufferSize | hipsolverDnZunmtr_bufferSize | |

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|---|---|---|
| cusolverDnSormtr | hipsolverDnSormtr | |
| cusolverDnDormtr | hipsolverDnDormtr | |
| cusolverDnCunmtr | hipsolverDnCunmtr | |
| cusolverDnZunmtr | hipsolverDnZunmtr | |
| cusolverDnSorgtr_bufferSiz | hipsolverDnSorgtr_bufferSize | |
| cusolverDnDorgtr_bufferSize | hipsolverDnDorgtr_bufferSize | |
| cusolverDnCungtr_bufferSize | hipsolverDnCungtr_bufferSize | |
| cusolverDnZungtr_bufferSize | hipsolverDnZungtr_bufferSize | |
| cusolverDnSorgtr | hipsolverDnSorgtr | |
| cusolverDnDorgtr | hipsolverDnDorgtr | |
| cusolverDnCungtr | hipsolverDnCungtr | |
| cusolverDnZungtr | hipsolverDnZungtr | |
| cusolverDnSgesvd_bufferSize | hipsolverDnSgesvd_bufferSize | |
| cusolverDnDgesvd_bufferSize | hipsolverDnDgesvd_bufferSize | |
| cusolverDnCgesvd_bufferSize | hipsolverDnCgesvd_bufferSize | |
| cusolverDnZgesvd_bufferSize | hipsolverDnZgesvd_bufferSize | |
| cusolverDnSgesvd | hipsolverDnSgesvd | |
| cusolverDnDgesvd | hipsolverDnDgesvd | |
| cusolverDnCgesvd | hipsolverDnCgesvd | |
| cusolverDnZgesvd | hipsolverDnZgesvd | |
| cusolverDnSgesvdj_bufferSize | hipsolverDnSgesvdj_bufferSize | |
| cusolverDnDgesvdj_bufferSize | hipsolverDnDgesvdj_bufferSize | |
| cusolverDnCgesvdj_bufferSize | hipsolverDnCgesvdj_bufferSize | |
| cusolverDnZgesvdj_bufferSize | hipsolverDnZgesvdj_bufferSize | |
| cusolverDnSgesvdj | hipsolverDnSgesvdj | |
| cusolverDnDgesvdj | hipsolverDnDgesvdj | |
| cusolverDnCgesvdj | hipsolverDnCgesvdj | |
| cusolverDnZgesvdj | hipsolverDnZgesvdj | |

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|---|---|---|
| cusolverDnSgesvdjBatched_bufferSize | hipsolverDnSgesvdjBatched_bufferSize | |
| cusolverDnDgesvdjBatched_bufferSize | hipsolverDnDgesvdjBatched_bufferSize | |
| cusolverDnCgesvdjBatched_bufferSize | hipsolverDnCgesvdjBatched_bufferSize | |
| cusolverDnZgesvdjBatched_bufferSize | hipsolverDnZgesvdjBatched_bufferSize | |
| cusolverDnSgesvdjBatched | hipsolverDnSgesvdjBatched | |
| cusolverDnDgesvdjBatched | hipsolverDnDgesvdjBatched | |
| cusolverDnCgesvdjBatched | hipsolverDnCgesvdjBatched | |
| cusolverDnZgesvdjBatched | hipsolverDnZgesvdjBatched | |
| cusolverDnSgesvdaStridedBatched_bufferSize | hipsolverDnSgesvdaStridedBatched_bufferSize | |
| cusolverDnDgesvdaStridedBatched_bufferSize | hipsolverDnDgesvdaStridedBatched_bufferSize | |
| cusolverDnCgesvdaStridedBatched_bufferSize | hipsolverDnCgesvdaStridedBatched_bufferSize | |
| cusolverDnZgesvdaStridedBatched_bufferSize | hipsolverDnZgesvdaStridedBatched_bufferSize | |
| cusolverDnSgesvdaStridedBatched | hipsolverDnSgesvdaStridedBatched | |
| cusolverDnDgesvdaStridedBatched | hipsolverDnDgesvdaStridedBatched | |
| cusolverDnCgesvdaStridedBatched | hipsolverDnCgesvdaStridedBatched | |
| cusolverDnZgesvdaStridedBatched | hipsolverDnZgesvdaStridedBatched | |
| cusolverDnSsyevd_bufferSize | hipsolverDnSsyevd_bufferSize | |

| | | |
|---|---|---|
| cusolverDnDsyevd_bufferSize | hipsolverDnDsyevd_bufferSize | |
| cusolverDnCheevd_bufferSize | hipsolverDnCheevd_bufferSize | |
| cusolverDnZheevd_bufferSize | hipsolverDnZheevd_bufferSize | |
| cusolverDnSsyevd | hipsolverDnSsyevd | |
| cusolverDnDsyevd | hipsolverDnDsyevd | |
| cusolverDnCheevd | hipsolverDnCheevd | |
| cusolverDnZheevd | hipsolverDnZheevd | |
| cusolverDnSsyevdx_bufferSize | hipsolverDnSsyevdx_bufferSize | |
| cusolverDnDsyevdx_bufferSize | hipsolverDnDsyevdx_bufferSize | |
| cusolverDnCheevdx_bufferSize | hipsolverDnCheevdx_bufferSize | |
| cusolverDnZheevdx_bufferSize | hipsolverDnZheevdx_bufferSize | |
| cusolverDnSsyevdx | hipsolverDnSsyevdx | |
| cusolverDnDsyevdx | hipsolverDnDsyevdx | |
| cusolverDnCheevdx | hipsolverDnCheevdx | |
| cusolverDnZheevdx | hipsolverDnZheevdx | |
| cusolverDnSsygvd_bufferSize | hipsolverDnSsygvd_bufferSize | |
| cusolverDnDsygvd_bufferSize | hipsolverDnDsygvd_bufferSize | |
| cusolverDnChegvd_bufferSize | hipsolverDnChegvd_bufferSize | |
| cusolverDnZhegvd_bufferSize | hipsolverDnZhegvd_bufferSize | |
| cusolverDnSsygvd | hipsolverDnSsygvd | |
| cusolverDnDsygvd | hipsolverDnDsygvd | |
| cusolverDnChegvd | hipsolverDnChegvd | |
| cusolverDnZhegvd | hipsolverDnZhegvd | |
| cusolverDnSsygvdx_bufferSize | hipsolverDnSsygvdx_bufferSize | |
| cusolverDnDsygvdx_bufferSize | hipsolverDnDsygvdx_bufferSize | |
| cusolverDnChegvdx_bufferSize | hipsolverDnChegvdx_bufferSize | |
| cusolverDnZhegvdx_bufferSize | hipsolverDnZhegvdx_bufferSize | |
| cusolverDnSsygvdx | hipsolverDnSsygvdx | |

| | | |
|---|---|---|
| cusolverDnDsygvdx | hipsolverDnDsygvdx | |
| cusolverDnChegvdx | hipsolverDnChegvdx | |
| cusolverDnZhegvdx | hipsolverDnZhegvdx | |
| cusolverDnSsyevj_bufferSize | hipsolverDnSsyevj_bufferSize | |
| cusolverDnDsyevj_bufferSize | hipsolverDnDsyevj_bufferSize | |
| cusolverDnCheevj_bufferSize | hipsolverDnCheevj_bufferSize | |
| cusolverDnZheevj_bufferSize | hipsolverDnZheevj_bufferSize | |
| cusolverDnSsyevj | hipsolverDnSsyevj | |
| cusolverDnDsyevj | hipsolverDnDsyevj | |
| cusolverDnCheevj | hipsolverDnCheevj | |
| cusolverDnZheevj | hipsolverDnZheevj | |
| cusolverDnSsygvj_bufferSize | hipsolverDnSsygvj_bufferSize | |
| cusolverDnDsygvj_bufferSize | hipsolverDnDsygvj_bufferSize | |
| cusolverDnChegvj_bufferSize | hipsolverDnChegvj_bufferSize | |
| cusolverDnZhegvj_bufferSize | hipsolverDnZhegvj_bufferSize | |
| cusolverDnSsygvj | hipsolverDnSsygvj | |
| cusolverDnDsygvj | hipsolverDnDsygvj | |
| cusolverDnChegvj | hipsolverDnChegvj_ | |
| cusolverDnZhegvj | hipsolverDnZhegvj_ | |
| cusolverDnSsyevjBatched_bufferSize | hipsolverDnSsyevjBatched_bufferSize | |
| cusolverDnDsyevjBatched_bufferSize | hipsolverDnDsyevjBatched_bufferSize | |
| cusolverDnCheevjBatched_bufferSize | hipsolverDnCheevjBatched_bufferSize | |
| cusolverDnZheevjBatched_bufferSize | hipsolverDnZheevjBatched_bufferSize | |
| cusolverDnSsyevjBatched | hipsolverDnSsyevjBatched | |

| | | |
|---|---|---|
| cusolverDnDsyevjBatched | hipsolverDnDsyevjBatched | |
| cusolverDnCheevjBatched | hipsolverDnCheevjBatched | |
| cusolverDnZheevjBatched | hipsolverDnZheevjBatched | |
| cusolverDnXpotrf_bufferSize | | |
| cusolverDnXpotrf | | |
| cusolverDnXpotrs | | |
| cusolverDnXgetrf_bufferSize | | |
| cusolverDnXgetrf | | |
| cusolverDnXgetrs | | |
| cusolverDnXgeqrf_bufferSize | | |
| cusolverDnXgeqrf | | |
| cusolverDnXsytrs_bufferSize | | |
| cusolverDnXsytrs | | |
| cusolverDnXtrtri_bufferSize | | |
| cusolverDnXtrtri | | |
| cusolverDnXgesvd_bufferSize | | |
| cusolverDnXgesvd | | |
| cusolverDnXgesvdp_bufferSize | | |
| cusolverDnXgesvdp | | |
| cusolverDnXgesvdr_bufferSize | | |
| cusolverDnXgesvdr | | |
| cusolverDnXsyevd_bufferSize | | |
| cusolverDnXsyevd | | |
| cusolverDnXsyevdx_bufferSize | | |
| cusolverDnXsyevdx | | |
| cusolverSpCreate | | |
| cusolverSpDestroy | | |
| cusolverSpSetStream | | |

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| | | |
|---|---|---|
| cusolverSpXcsrissymHost | | |
| cusolverSpScsrlsvlu | | |
| cusolverSpDcsrlsvlu | | |
| cusolverSpCcsrlsvlu | | |
| cusolverSpZcsrlsvlu | | |
| cusolverSpScsrlsvqr | | |
| cusolverSpDcsrlsvqr | | |
| cusolverSpCcsrlsvqr | | |
| cusolverSpZcsrlsvqr | | |
| cusolverSpScsrlsvchol | | |
| cusolverSpDcsrlsvchol | | |
| cusolverSpCcsrlsvchol | | |
| cusolverSpZcsrlsvchol | | |
| cusolverSpScsrlsqvqr | | |
| cusolverSpDcsrlsqvqr | | |
| cusolverSpCcsrlsqvqr | | |
| cusolverSpZcsrlsqvqr | | |
| cusolverSpScsreigvsi | | |
| cusolverSpDcsreigvsi | | |
| cusolverSpCcsreigvsi | | |
| cusolverSpZcsreigvsi | | |
| solverspScsreigs | | |
| cusolverSpDcsreigs | | |
| cusolverSpCcsreigs | | |
| cusolverSpZcsreigs | | |
| cusolverSpXcsrsymrcmHost | | |
| cusolverSpXcsrsymmdqHost | | |
| cusolverSpXcsrsymamdHost | | |

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| | | |
|---|---|---|
| cusolverSpXcsrmetisndHost | | |
| cusolverSpScsrzfdHost | | |
| cusolverSpDcsrzfdHost | | |
| cusolverSpCcsrzfdHost | | |
| cusolverSpZcsrzfdHost | | |
| cusolverSpXcsrperm_bufferSizeHost | | |
| cusolverSpXcsrpermHost | | |
| cusolverSpCreateCsrqrInfo | | |
| cusolverSpDestroyCsrqrInfo | | |
| cusolverSpXcsrqrAnalysisBatched | | |
| cusolverSpScsrqrBufferInfoBatched | | |
| cusolverSpDcsrqrBufferInfoBatched | | |
| cusolverSpCcsrqrBufferInfoBatched | | |
| cusolverSpZcsrqrBufferInfoBatched | | |
| cusolverSpScsrqrsvBatched | | |
| cusolverSpDcsrqrsvBatched | | |
| cusolverSpCcsrqrsvBatched | | |
| cusolverSpZcsrqrsvBatched | | |
| cusolverRfAccessBundledFactors | | |
| cusolverRfAnalyze | | |
| cusolverRfSetupDevice | | |
| cusolverRfSetupHost | | |
| cusolverRfCreate | | |

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| | | |
|---|---|---|
| cusolverRfExtractBundledFactorsHost | | |
| cusolverRfExtractSplitFactorsHost | | |
| cusolverRfDestroy | | |
| cusolverRfGetMatrixFormat | | |
| cusolverRfGetNumericProperties | | |
| cusolverRfGetNumericBoostReport | | |
| cusolverRfGetResetValuesFastMode | | |
| cusolverRfGet_Algs | | |
| cusolverRfRefactor | | |
| cusolverRfResetValues | | |
| cusolverRfSetMatrixFormat | | |
| cusolverRfSetNumericProperties | | |
| cusolverRfSetResetValuesFastMode | | |
| cusolverRfSetAlgs | | |
| cusolverRfSolve | | |
| cusolverRfBatchSetupHost | | |
| cusolverRfBatchAnalyze | | |
| cusolverRfBatchResetValues | | |
| cusolverRfBatchRefactor | | |
| cusolverRfBatchSolve | | |
| cusolverRfBatchZeroPivot | | |

# 5.6 SPARSE

## 5.6.1 hipSPARSE 数据类型支持

| type | CUDA | HIP | CUDAversion* |
|---|---|---|---|
| define | CUSPARSE_VER_MAJ | | 10.1Update2 |

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| | | | |
|---|---|---|---|
| | OR | | |
| define | CUSPARSE_VER_MINOR | | 10.1Update2 |
| define | CUSPARSE_VER_PATCH | | 10.1Update2 |
| define | CUSPARSE_VER_BUILD | | 10.1Update2 |
| define | CUSPARSE_VERSION | | 10.1Update2 |
| enum | cusparseAction_t | hipsparseAction_t | |
| 0 | CUSPARSE_ACTION_SYMBOLIC | HIPSPARSE_ACTION_SYMBOLIC | |
| 1 | CUSPARSE_ACTION_NUMERIC | HIPSPARSE_ACTION_NUMERIC | |
| enum | cusparseDirection_t | hipsparseDirection_t | |
| 0 | CUSPARSE_DIRECTION_ROW | HIPSPARSE_DIRECTION_ROW | |
| 1 | CUSPARSE_DIRECTION_COLUMN | HIPSPARSE_DIRECTION_COLUMN | |
| enum | cusparseHybPartition_t | hipsparseHybPartition_t | |
| 0 | CUSPARSE_HYB_PARTITION_AUTO | HIPSPARSE_HYB_PARTITION_AUTO | |
| 1 | CUSPARSE_HYB_PARTITION_USER | HIPSPARSE_HYB_PARTITION_USER | |
| 2 | CUSPARSE_HYB_PARTITION_MAX | HIPSPARSE_HYB_PARTITION_MAX | |
| enum | cusparseDiagType_t | hipsparseDiagType_t | |
| 0 | CUSPARSE_DIAG_TYPE_NON_UNIT | HIPSPARSE_DIAG_TYPE_NON_UNIT | |

| | | | |
|---|---|---|---|
| 1 | CUSPARSE_DIAG_TYPE_UNIT | HIPSPARSE_DIAG_TYPE_UNIT | |
| enum | cusparseFillMode_t | hipsparseFillMode_t | |
| 0 | CUSPARSE_FILL_MODE_LOWER | HIPSPARSE_FILL_MODE_LOWER | |
| 1 | CUSPARSE_FILL_MODE_UPPER | HIPSPARSE_FILL_MODE_UPPER | |
| enum | cusparseIndexBase_t | hipsparseIndexBase_t | |
| 0 | CUSPARSE_INDEX_BASE_ZERO | HIPSPARSE_INDEX_BASE_ZERO | |
| 1 | CUSPARSE_INDEX_BASE_ONE | HIPSPARSE_INDEX_BASE_ONE | |
| enum | cusparseMatrixType_t | hipsparseMatrixType_t | |
| 0 | CUSPARSE_MATRIX_TYPE_GENERAL | HIPSPARSE_MATRIX_TYPE_GENERAL | |
| 1 | CUSPARSE_MATRIX_TYPE_SYMMETRIC | HIPSPARSE_MATRIX_TYPE_SYMMETRIC | |
| 2 | CUSPARSE_MATRIX_TYPE_HERMITIAN | HIPSPARSE_MATRIX_TYPE_HERMITIAN | |
| 3 | CUSPARSE_MATRIX_TYPE_TRIANGULAR | HIPSPARSE_MATRIX_TYPE_TRIANGULAR | |
| enum | cusparseOperation_t | hipsparseOperation_t | |
| 0 | CUSPARSE_OPERATION_NON_TRANSPOSE | HIPSPARSE_OPERATION_NON_TRANSPOSE | |
| 1 | CUSPARSE_OPERATION_TRANSPOSE | HIPSPARSE_OPERATION_TRANSPOSE | |
| 2 | CUSPARSE_OPERATION_CONJUGATE_TRANSPOSE | HIPSPARSE_OPERATION_CONJUGATE_TRANSPOSE | |

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|---|---|---|---|
| enum | cusparsePointerMode_t | hipsparsePointerMode_t | |
| 0 | CUSPARSE_POINTER_MODE_HOST | HIPSPARSE_POINTER_MODE_HOST | |
| 1 | CUSPARSE_POINTER_MODE_DEVICE | HIPSPARSE_POINTER_MODE_DEVICE | |
| enum | cusparseAlgMode_t | | 8.0 |
| 0 | CUSPARSE_ALG0 | | 8.0 |
| 1 | CUSPARSE_ALG1 | | 8.0 |
| 0 | CUSPARSE_ALG_NAIVE | | 9.2 |
| 1 | CUSPARSE_ALG_MERGE_PATH | | 9.2 |
| enum | cusparseSolvePolicy_t | hipsparseSolvePolicy_t | |
| 0 | CUSPARSE_SOLVE_POLICY_NO_LEVEL | HIPSPARSE_SOLVE_POLICY_NO_LEVEL | |
| 1 | CUSPARSE_SOLVE_POLICY_USE_LEVEL | HIPSPARSE_SOLVE_POLICY_USE_LEVEL | |
| enum | cusparseStatus_t | hipsparseMatrixType_t | |
| 0 | CUSPARSE_STATUS_SUCCESS | HIPSPARSE_STATUS_SUCCESS | |
| 1 | CUSPARSE_STATUS_NOT_INITIALIZED | HIPSPARSE_STATUS_NOT_INITIALIZED | |
| 2 | CUSPARSE_STATUS_ALLOC_FAILED | HIPSPARSE_STATUS_ALLOC_FAILED | |
| 3 | CUSPARSE_STATUS_INVALID_VALUE | HIPSPARSE_STATUS_INVALID_VALUE | |
| 4 | CUSPARSE_STATUS_ARCH_MISMATCH | HIPSPARSE_STATUS_ARCH_MISMATCH | |

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|---|---|---|---|
| 5 | CUSPARSE_STATUS_MAPPING_ERROR | HIPSPARSE_STATUS_MAPPING_ERROR | |
| 6 | CUSPARSE_STATUS_EXECUTION_FAILED | HIPSPARSE_STATUS_EXECUTION_FAILED | |
| 7 | CUSPARSE_STATUS_INTERNAL_ERROR | HIPSPARSE_STATUS_INTERNAL_ERROR | |
| 8 | CUSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED | HIPSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED | |
| 9 | CUSPARSE_STATUS_ZERO_PIVOT | HIPSPARSE_STATUS_ZERO_PIVOT | |
| struct | cusparseContext | | |
| typedef | cusparseHandle_t | hipsparseHandle_t | |
| struct | cusparseHybMat | | |
| typedef | cusparseHybMat_t | hipsparseHybMat_t | |
| struct | cusparseMatDescr | | |
| typedef | cusparseMatDescr_t | hipsparseMatDescr_t | |
| struct | cusparseSolveAnalysisInfo | | |
| typedef | cusparseSolveAnalysisInfo_t | | |
| struct | csrsv2Info | | |
| typedef | csrsv2Info_t | csrsv2Info_t | |
| struct | csrsm2Info | csrsm2Info | 9.2 |

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| | | | |
|---|---|---|---|
| typedef | csrsm2Info_t | csrsm2Info_t | |
| struct | bsrsv2Info | | |
| typedef | bsrsv2Info_t | | |
| struct | bsrsm2Info | | |
| typedef | bsrsm2Info_t | | |
| struct | bsric02Info | | |
| typedef | bsric02Info_t | | |
| struct | csrilu02Info | | |
| typedef | csrilu02Info_t | csrilu02Info_t | |
| struct | bsrilu02Info | | |
| typedef | bsrilu02Info_t | | |
| struct | csru2csrInfo | | |
| typedef | csru2csrInfo_t | | |
| struct | csrgemm2Info | csrgemm2Info | |
| typedef | csrgemm2Info_t | csrgemm2Info_t | |
| struct | cusparseColorInfo | | |
| typedef | cusparseColorInfo_t | | |
| struct | pruneInfo | | 9.0 |
| typed | pruneInfo_t | | 9.0 |

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| | | | |
|---|---|---|---|
| ef | | | |
| enum | cusparseCsr2CscAlg_t | | 10.1 |
| 1 | CUSPARSE_CSR2CSC_ALG1 | | 10.1 |
| 2 | CUSPARSE_CSR2CSC_ALG2 | | 10.1 |
| enum | cusparseFormat_t | | 10.1 |
| 1 | CUSPARSE_FORMAT_CSR | | 10.1 |
| 2 | CUSPARSE_FORMAT_CSC | | 10.1 |
| 3 | CUSPARSE_FORMAT_COO | | 10.1 |
| 4 | CUSPARSE_FORMAT_COO_AOS | | 10.1 |
| enum | cusparseOrder_t | | 10.1 |
| 1 | CUSPARSE_ORDER_COL | | 10.1 |
| 2 | CUSPARSE_ORDER_ROW | | 10.1 |
| enum | cusparseSpMVAlg_t | | 10.1 |
| 0 | CUSPARSE_MV_ALG_DEFAULT | | 10.1 |
| 1 | CUSPARSE_COOMV_ALG | | 10.1 |
| 2 | CUSPARSE_CSRMV_ALG1 | | 10.1 |
| 3 | CUSPARSE_CSRMV_A | | 10.1 |

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| | | | |
|---|---|---|---|
| | LG2 | | |
| enum | cusparseSpMMAlg_t | | 10.1 |
| 0 | CUSPARSE_MM_ALG_DEFAULT | | 10.1 |
| 1 | CUSPARSE_COOMM_ALG1 | | 10.1 |
| 2 | CUSPARSE_COOMM_ALG2 | | 10.1 |
| 3 | CUSPARSE_COOMM_ALG3 | | 10.1 |
| 4 | CUSPARSE_CSRMM_ALG1 | | 10.1 |
| enum | cusparseIndexType_t | | 10.1 |
| 1 | CUSPARSE_INDEX_16U | | 10.1 |
| 2 | CUSPARSE_INDEX_32I | | 10.1 |
| 3 | CUSPARSE_INDEX_64I | | 10.1 |
| struct | cusparseSpMatDescr | | 10.1 |
| typedef | cusparseSpMatDescr_t | | 10.1 |
| struct | cusparseDnMatDescr | | 10.1 |
| typedef | cusparseDnMatDescr_t | | 10.1 |
| struct | cusparseSpVecDescr | | 10.1 |
| typedef | cusparseSpVecDescr_t | | 10.1 |

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| | | | |
|---|---|---|---|
| struct | cusparseDnVecDescr | | 10.1 |
| typedef | cusparseDnVecDescr_t | | 10.1 |

### 5.6.2 hipSPARSE 辅助函数

| CUDA | HIP | CUDAversion * |
|---|---|---|
| cusparseCreate | hipsparseCreate | |
| cusparseCreateSolveAnalysisInfo | | |
| cusparseCreateHybMat | hipsparseCreateHybMat | |
| cusparseCreateMatDescr | hipsparseCreateMatDescr | |
| cusparseDestroy | hipsparseDestroy | |
| cusparseDestroySolveAnalysisInfo | | |
| cusparseDestroyHybMat | hipsparseDestroyHybMat | |
| cusparseDestroyMatDescr | hipsparseDestroyMatDescr | |
| cusparseGetLevelInfo | | |
| cusparseGetMatDiagType | hipsparseGetMatDiagType | |
| cusparseGetMatFillMode | hipsparseGetMatFillMode | |
| cusparseGetMatIndexBase | hipsparseGetMatIndexBase | |
| cusparseGetMatType | hipsparseGetMatType | |
| cusparseGetPointerMode | hipsparseGetPointerMode | |
| cusparseGetVersion | hipsparseGetVersion | |
| cusparseSetMatDiagType | hipsparseSetMatDiagType | |
| cusparseSetMatIndexBase | hipsparseSetMatIndexBase | |
| cusparseSetMatFillMode | hipsparseSetMatFillMode | |
| cusparseSetMatType | hipsparseSetMatType | |
| cusparseSetPointerMode | hipsparseSetPointerMode | |

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| | | |
|---|---|---|
| cusparseSetStream | hipsparseSetStream | |
| cusparseGetStream | hipsparseGetStream | 8 |
| cusparseCreateCsrsv2Info | hipsparseCreateCsrsv2Info | |
| cusparseDestroyCsrsv2Info | hipsparseDestroyCsrsv2Info | |
| cusparseCreateCsrsm2Info | hipsparseCreateCsrsm2Info | 9.2 |
| cusparseDestroyCsrsm2Info | hipsparseDestroyCsrsm2Info | 9.2 |
| cusparseCreateCsric02Info | hipsparseCreateCsric02Info | |
| cusparseDestroyCsric02Info | hipsparseDestroyCsric02Info | |
| cusparseCreateCsrilu02Info | hipsparseCreateCsrilu02Info | |
| cusparseDestroyCsrilu02Info | hipsparseDestroyCsrilu02Info | |
| cusparseCreateBsrsv2Info | hipsparseCreateBsrsv2Info | |
| cusparseDestroyBsrsv2Info | hipsparseDestroyBsrsv2Info | |
| cusparseCreateBsrsm2Info | hipsparseCreateBsrsm2Info | |
| cusparseDestroyBsrsm2Info | hipsparseDestroyBsrsm2Info | |
| cusparseCreateBsric02Info | hipsparseCreateBsric02Info | |
| cusparseDestroyBsric02Info | hipsparseDestroyBsric02Info | |
| cusparseCreateBsrilu02Info | hipsparseCreateBsrilu02Info | |
| cusparseDestroyBsrilu02Info | hipsparseDestroyBsrilu02Info | |
| cusparseCreateCsrgemm2Info | hipsparseCreateCsrgemm2Info | |
| cusparseDestroyCsrgemm2Info | hipsparseDestroyCsrgemm2Info | |
| cusparseCreatePruneInfo | hipsparseCreatePruneInfo | 9 |

### 5.6.3 Level1 支持函数列表

| CUDA | HIP | CUDAversion |
|---|---|---|
| cusparseSaxpyi | hipsparseSaxpyi | |
| cusparseDaxpyi | hipsparseDaxpyi | |
| cusparseCaxpyi | hipsparseCaxpyi | |
| cusparseZaxpyi | hipsparseZaxpyi | |

| | | |
|---|---|---|
| cusparseSdoti | hipsparseSdoti | |
| cusparseDdoti | hipsparseDdoti | |
| cusparseCdoti | hipsparseCdoti | |
| cusparseZdoti | hipsparseZdoti | |
| cusparseCdotci | hipsparseCdotci | |
| cusparseZdotci | hipsparseZdotci | |
| cusparseSgthr | hipsparseSgthr | |
| cusparseDgthr | hipsparseDgthr | |
| cusparseCgthr | hipsparseCgthr | |
| cusparseZgthr | hipsparseZgthr | |
| cusparseSgthrz | hipsparseSgthrz | |
| cusparseDgthrz | hipsparseDgthrz | |
| cusparseCgthrz | hipsparseCgthrz | |
| cusparseZgthrz | hipsparseZgthrz | |
| cusparseSroti | hipsparseSroti | |
| cusparseDroti | hipsparseDroti | |
| cusparseSsctr | hipsparseSsctr | |
| cusparseDsctr | hipsparseDsctr | |
| cusparseCsctr | hipsparseCsctr | |
| cusparseZsctr | hipsparseZsctr | |

### 5.6.4 Level2 支持函数列表

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSbsrmv | hipsparseSbsrmv | |
| cusparseDbsrmv | hipsparseDbsrmv | |
| cusparseCbsrmv | hipsparseCbsrmv | |
| cusparseZbsrmv | hipsparseZbsrmv | |
| cusparseSbsrxmv | hipsparseSbsrxmv | |

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| | | |
|---|---|---|
| cusparseDbsrxmv | hipsparseDbsrxmv | |
| cusparseCbsrxmv | hipsparseCbsrxmv | |
| cusparseZbsrxmv | hipsparseZbsrxmv | |
| cusparseScsrmv | hipsparseScsrmv | |
| cusparseDcsrmv | hipsparseDcsrmv | |
| cusparseCcsrmv | hipsparseCcsrmv | |
| cusparseZcsrmv | hipsparseZcsrmv | |
| cusparseCsrmvEx | | 8 |
| cusparseCsrmvEx_bufferSize | | 8 |
| cusparseScsrmv_mp | | 8 |
| cusparseDcsrmv_mp | | 8 |
| cusparseCcsrmv_mp | | 8 |
| cusparseZcsrmv_mp | | 8 |
| cusparseSgemvi | hipsparseSgemvi | 7.5 |
| cusparseDgemvi | hipsparseDgemvi | 7.5 |
| cusparseCgemvi | hipsparseCgemvi | 7.5 |
| cusparseZgemvi | hipsparseZgemvi | 7.5 |
| cusparseSgemvi_bufferSize | hipsparseSgemvi_bufferSize | 7.5 |
| cusparseDgemvi_bufferSize | hipsparseDgemvi_bufferSize | 7.5 |
| cusparseCgemvi_bufferSize | hipsparseCgemvi_bufferSize | 7.5 |
| cusparseZgemvi_bufferSize | hipsparseZgemvi_bufferSize | 7.5 |
| cusparseSbsrsv2_bufferSize | hipsparseSbsrsv2_bufferSize | |
| cusparseSbsrsv2_bufferSizeExt | hipsparseSbsrsv2_bufferSizeExt | |
| cusparseDbsrsv2_bufferSize | hipsparseDbsrsv2_bufferSize | |
| cusparseDbsrsv2_bufferSizeExt | hipsparseDbsrsv2_bufferSizeExt | |
| cusparseCbsrsv2_bufferSize | hipsparseCbsrsv2_bufferSize | |
| cusparseCbsrsv2_bufferSizeExt | hipsparseCbsrsv2_bufferSizeExt | |
| cusparseZbsrsv2_bufferSize | hipsparseZbsrsv2_bufferSize | |

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| | | |
|---|---|---|
| cusparseZbsrsv2_bufferSizeExt | hipsparseZbsrsv2_bufferSizeExt | |
| cusparseSbsrsv2_analysis | hipsparseSbsrsv2_analysis | |
| cusparseDbsrsv2_analysis | hipsparseDbsrsv2_analysis | |
| cusparseCbsrsv2_analysis | hipsparseCbsrsv2_analysis | |
| cusparseZbsrsv2_analysis | hipsparseZbsrsv2_analysis | |
| cusparseXbsrsv2_zeroPivot | hipsparseXbsrsv2_zeroPivot | |
| cusparseSbsrsv2_solve | hipsparseSbsrsv2_solve | |
| cusparseDbsrsv2_solve | hipsparseDbsrsv2_solve | |
| cusparseCbsrsv2_solve | hipsparseCbsrsv2_solve | |
| cusparseZbsrsv2_solve | hipsparseZbsrsv2_solve | |
| cusparseScsrsv_analysis | | |
| cusparseDcsrsv_analysis | | |
| cusparseCcsrsv_analysis | | |
| cusparseZcsrsv_analysis | | |
| cusparseCsrsv_analysisEx | | 8 |
| cusparseScsrsv_solve | | |
| cusparseDcsrsv_solve | | |
| cusparseCcsrsv_solve | | |
| cusparseZcsrsv_solve | | |
| cusparseCsrsv_solveEx | | 8 |
| cusparseScsrsv2_bufferSize | hipsparseScsrsv2_bufferSize | |
| cusparseScsrsv2_bufferSizeExt | hipsparseScsrsv2_bufferSizeExt | |
| cusparseDcsrsv2_bufferSize | hipsparseDcsrsv2_bufferSize | |
| cusparseDcsrsv2_bufferSizeExt | hipsparseDcsrsv2_bufferSizeExt | |
| cusparseCcsrsv2_bufferSize | hipsparseCcsrsv2_bufferSize | |
| cusparseCcsrsv2_bufferSizeExt | hipsparseCcsrsv2_bufferSizeExt | |
| cusparseZcsrsv2_bufferSize | hipsparseZcsrsv2_bufferSize | |
| cusparseZcsrsv2_bufferSizeExt | hipsparseZcsrsv2_bufferSizeExt | |

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| | | |
|---|---|---|
| cusparseScsrsv2_analysis | hipsparseScsrsv2_analysis | |
| cusparseDcsrsv2_analysis | hipsparseDcsrsv2_analysis | |
| cusparseCcsrsv2_analysis | hipsparseCcsrsv2_analysis | |
| cusparseZcsrsv2_analysis | hipsparseZcsrsv2_analysis | |
| cusparseScsrsv2_solve | hipsparseScsrsv2_solve | |
| cusparseDcsrsv2_solve | hipsparseDcsrsv2_solve | |
| cusparseCcsrsv2_solve | hipsparseCcsrsv2_solve | |
| cusparseZcsrsv2_solve | hipsparseZcsrsv2_solve | |
| cusparseXcsrsv2_zeroPivot | hipsparseXcsrsv2_zeroPivot | |
| cusparseShybmv | hipsparseShybmv | |
| cusparseDhybmv | hipsparseDhybmv | |
| cusparseChybmv | hipsparseChybmv | |
| cusparseZhybmv | hipsparseZhybmv | |
| cusparseShybsv_analysis | | |
| cusparseDhybsv_analysis | | |
| cusparseChybsv_analysis | | |
| cusparseZhybsv_analysis | | |
| cusparseShybsv_solve | | |
| cusparseDhybsv_solve | | |

### 5.6.5 Level3 支持函数列表

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseDcsrmm | hipsparseDcsrmm | |
| cusparseCcsrmm | hipsparseCcsrmm | |
| cusparseZcsrmm | hipsparseZcsrmm | |
| cusparseScsrmm2 | hipsparseScsrmm2 | |
| cusparseDcsrmm2 | hipsparseDcsrmm2 | |
| cusparseCcsrmm2 | hipsparseCcsrmm2 | |

| | | |
|---|---|---|
| cusparseZcsrmm2 | hipsparseZcsrmm2 | |
| cusparseScsrsm_analysis | | |
| cusparseDcsrsm_analysis | | |
| cusparseCcsrsm_analysis | | |
| cusparseZcsrsm_analysis | | |
| cusparseScsrsm_solve | | |
| cusparseDcsrsm_solve | | |
| cusparseCcsrsm_solve | | |
| cusparseZcsrsm_solve | | |
| cusparseScsrsm2_bufferSizeExt | hipsparseScsrsm2_bufferSizeExt | 9.2 |
| cusparseDcsrsm2_bufferSizeExt | hipsparseDcsrsm2_bufferSizeExt | 9.2 |
| cusparseCcsrsm2_bufferSizeExt | hipsparseCcsrsm2_bufferSizeExt | 9.2 |
| cusparseZcsrsm2_bufferSizeExt | hipsparseZcsrsm2_bufferSizeExt | 9.2 |
| cusparseScsrsm2_analysis | hipsparseScsrsm2_analysis | 9.2 |
| cusparseDcsrsm2_analysis | hipsparseDcsrsm2_analysis | 9.2 |
| cusparseCcsrsm2_analysis | hipsparseCcsrsm2_analysis | 9.2 |
| cusparseZcsrsm2_analysis | hipsparseZcsrsm2_analysis | 9.2 |
| cusparseScsrsm2_solve | hipsparseScsrsm2_solve | 9.2 |
| cusparseDcsrsm2_solve | hipsparseDcsrsm2_solve | 9.2 |
| cusparseCcsrsm2_solve | hipsparseCcsrsm2_solve | 9.2 |
| cusparseZcsrsm2_solve | hipsparseZcsrsm2_solve | 9.2 |
| cusparseXcsrsm2_zeroPivot | hipsparseXcsrsm2_zeroPivot | 9.2 |
| cusparseSbsrmm | hipsparseSbsrmm | |
| cusparseDbsrmm | hipsparseDbsrmm | |
| cusparseCbsrmm | hipsparseCbsrmm | |
| cusparseZbsrmm | hipsparseZbsrmm | |
| cusparseSbsrsm2_bufferSize | hipsparseSbsrsm2_bufferSize | |
| cusparseSbsrsm2_bufferSizeExt | | |

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|---|---|---|
| cusparseDbsrsm2_bufferSize | hipsparseDbsrsm2_bufferSize | |
| cusparseDbsrsm2_bufferSizeExt | | |
| cusparseCbsrsm2_bufferSize | hipsparseCbsrsm2_bufferSize | |
| cusparseCbsrsm2_bufferSizeExt | | |
| cusparseZbsrsm2_bufferSize | hipsparseZbsrsm2_bufferSize | |
| cusparseZbsrsm2_bufferSizeExt | | |
| cusparseSbsrsm2_analysis | hipsparseSbsrsm2_analysis | |
| cusparseDbsrsm2_analysis | hipsparseDbsrsm2_analysis | |
| cusparseCbsrsm2_analysis | hipsparseCbsrsm2_analysis | |
| cusparseZbsrsm2_analysis | hipsparseZbsrsm2_analysis | |
| cusparseSbsrsm2_solve | hipsparseSbsrsm2_solve | |
| cusparseDbsrsm2_solve | hipsparseDbsrsm2_solve | |
| cusparseCbsrsm2_solve | hipsparseCbsrsm2_solve | |
| cusparseZbsrsm2_solve | hipsparseZbsrsm2_solve | |
| cusparseXbsrsm2_zeroPivot | hipsparseXcsrsm2_zeroPivot | |
| cusparseSgemmi | hipsparseSgemmi | 8 |
| cusparseDgemmi | hipsparseDgemmi | 8 |
| cusparseCgemmi | hipsparseCgemmi | 8 |
| cusparseZgemmi | hipsparseZgemmi | 8 |

### 5.6.6 其他函数列表

| CUDA | HIP | CUDA version * |
|---|---|---|
| cusparseXcsrgeamNnz | hipsparseXcsrgeamNnz | |
| cusparseScsrgeam | hipsparseScsrgeam | |
| cusparseDcsrgeam | hipsparseDcsrgeam | |
| cusparseCcsrgeam | hipsparseCcsrgeam | |
| cusparseZcsrgeam | hipsparseZcsrgeam | |

| | | |
|---|---|---|
| cusparseXcsrgeam2Nnz | hipsparseXcsrgeam2Nnz | 9.2 |
| cusparseScsrgeam2 | hipsparseScsrgeam2 | 9.2 |
| cusparseDcsrgeam2 | hipsparseDcsrgeam2 | 9.2 |
| cusparseCcsrgeam2 | hipsparseCcsrgeam2 | 9.2 |
| cusparseZcsrgeam2 | hipsparseZcsrgeam2 | 9.2 |
| cusparseScsrgeam2_bufferSizeExt | hipsparseScsrgeam2_bufferSizeExt | 9.2 |
| cusparseDcsrgeam2_bufferSizeExt | hipsparseDcsrgeam2_bufferSizeExt | 9.2 |
| cusparseCcsrgeam2_bufferSizeExt | hipsparseCcsrgeam2_bufferSizeExt | 9.2 |
| cusparseZcsrgeam2_bufferSizeExt | hipsparseZcsrgeam2_bufferSizeExt | 9.2 |
| cusparseXcsrgemmNnz | hipsparseXcsrgemmNnz | |
| cusparseScsrgemm | hipsparseScsrgemm | |
| cusparseDcsrgemm | hipsparseDcsrgemm | |
| cusparseCcsrgemm | hipsparseCcsrgemm | |
| cusparseZcsrgemm | hipsparseZcsrgemm | |
| cusparseXcsrgemm2Nnz | hipsparseXcsrgemm2Nnz | |
| cusparseScsrgemm2 | hipsparseScsrgemm2 | |
| cusparseDcsrgemm2 | hipsparseDcsrgemm2 | |
| cusparseCcsrgemm2 | hipsparseCcsrgemm2 | |
| cusparseZcsrgemm2 | hipsparseZcsrgemm2 | |
| cusparseScsrgemm2_bufferSizeExt | hipsparseScsrgemm2_bufferSizeExt | |
| cusparseDcsrgemm2_bufferSizeExt | hipsparseDcsrgemm2_bufferSizeExt | |
| cusparseCcsrgemm2_bufferSizeExt | hipsparseCcsrgemm2_bufferSizeExt | |
| cusparseZcsrgemm2_bufferSizeExt | hipsparseZcsrgemm2_bufferSizeExt | |

### 5.6.7 预设条件函数接口

乔里斯基分解

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseScsric0 | | |
| cusparseDcsric0 | | |
| cusparseCcsric0 | | |
| cusparseZcsric0 | | |
| cusparseScsric02_bufferSize | hipsparseScsric02_bufferSize | |
| cusparseScsric02_bufferSizeExt | hipsparseScsric02_bufferSizeExt | |
| cusparseDcsric02_bufferSize | hipsparseDcsric02_bufferSize | |
| cusparseDcsric02_bufferSizeExt | hipsparseDcsric02_bufferSizeExt | |
| cusparseCcsric02_bufferSize | hipsparseCcsric02_bufferSize | |
| cusparseCcsric02_bufferSizeExt | hipsparseCcsric02_bufferSizeExt | |
| cusparseZcsric02_bufferSize | hipsparseZcsric02_bufferSize | |
| cusparseZcsric02_bufferSizeExt | hipsparseZcsric02_bufferSizeExt | |
| cusparseScsric02_analysis | hipsparseScsric02_analysis | |
| cusparseDcsric02_analysis | hipsparseDcsric02_analysis | |
| cusparseCcsric02_analysis | hipsparseCcsric02_analysis | |
| cusparseZcsric02_analysis | hipsparseZcsric02_analysis | |
| cusparseScsric02 | hipsparseScsric02 | |
| cusparseDcsric02 | hipsparseDcsric02 | |
| cusparseCcsric02 | hipsparseCcsric02 | |
| cusparseZcsric02 | hipsparseZcsric02 | |
| cusparseXcsric02_zeroPivot | hipsparseXcsric02_zeroPivot | |
| cusparseSbsric02_bufferSize | hipsparseSbsric02_bufferSize | |
| cusparseSbsric02_bufferSizeExt | | |
| cusparseDbsric02_bufferSize | hipsparseDbsric02_bufferSize | |

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|---|---|---|
| cusparseDbsric02_bufferSizeExt | | |
| cusparseCbsric02_bufferSize | hipsparseCbsric02_bufferSize | |
| cusparseCbsric02_bufferSizeExt | | |
| cusparseZbsric02_bufferSize | hipsparseZbsric02_bufferSize | |
| cusparseZbsric02_bufferSizeExt | | |
| cusparseSbsric02_analysis | hipsparseSbsric02_analysis | |
| cusparseDbsric02_analysis | hipsparseDbsric02_analysis | |
| cusparseCbsric02_analysis | hipsparseCbsric02_analysis | |
| cusparseZbsric02_analysis | hipsparseZbsric02_analysis | |
| cusparseSbsric02 | hipsparseSbsric02 | |
| cusparseDbsric02 | hipsparseDbsric02 | |
| cusparseCbsric02 | hipsparseCbsric02 | |
| cusparseZbsric02 | hipsparseZbsric02 | |
| cusparseXbsric02_zeroPivot | hipsparseXbsric02_zeroPivot | |

LU 分解

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseScsrilu0 | | |
| cusparseDcsrilu0 | | |
| cusparseCcsrilu0 | | |
| cusparseZcsrilu0 | | |
| cusparseCsrilu0Ex | | 8 |
| cusparseScsrilu02_numericBoost | hipsparseScsrilu02_numericBoost | |
| cusparseDcsrilu02_numericBoost | hipsparseDcsrilu02_numericBoost | |
| cusparseCcsrilu02_numericBoost | hipsparseCcsrilu02_numericBoost | |
| cusparseZcsrilu02_numericBoost | hipsparseZcsrilu02_numericBoost | |
| cusparseXcsrilu02_zeroPivot | hipsparseXcsrilu02_zeroPivot | |
| cusparseScsrilu02_bufferSize | hipsparseScsrilu02_bufferSize | |

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| | | |
|---|---|---|
| cusparseScsrilu02_bufferSizeExt | hipsparseScsrilu02_bufferSizeExt | |
| cusparseDcsrilu02_bufferSize | hipsparseDcsrilu02_bufferSize | |
| cusparseDcsrilu02_bufferSizeExt | hipsparseDcsrilu02_bufferSizeExt | |
| cusparseCcsrilu02_bufferSize | hipsparseCcsrilu02_bufferSize | |
| cusparseCcsrilu02_bufferSizeExt | hipsparseCcsrilu02_bufferSizeExt | |
| cusparseZcsrilu02_bufferSize | hipsparseZcsrilu02_bufferSize | |
| cusparseZcsrilu02_bufferSizeExt | hipsparseZcsrilu02_bufferSizeExt | |
| cusparseScsrilu02_analysis | hipsparseScsrilu02_analysis | |
| cusparseDcsrilu02_analysis | hipsparseDcsrilu02_analysis | |
| cusparseCcsrilu02_analysis | hipsparseCcsrilu02_analysis | |
| cusparseZcsrilu02_analysis | hipsparseZcsrilu02_analysis | |
| cusparseScsrilu02 | hipsparseScsrilu02 | |
| cusparseDcsrilu02 | hipsparseDcsrilu02 | |
| cusparseCcsrilu02 | hipsparseCcsrilu02 | |
| cusparseZcsrilu02 | hipsparseZcsrilu02 | |
| cusparseXbsric02_zeroPivot | hipsparseXcsrilu02_zeroPivot | |
| cusparseSbsrilu02_numericBoost | hipsparseSbsrilu02_numericBoost | |
| cusparseDbsrilu02_numericBoost | hipsparseDbsrilu02_numericBoost | |
| cusparseCbsrilu02_numericBoost | hipsparseCbsrilu02_numericBoost | |
| cusparseZbsrilu02_numericBoost | hipsparseZbsrilu02_numericBoost | |
| cusparseSbsrilu02_bufferSize | hipsparseSbsrilu02_bufferSize | |
| cusparseSbsrilu02_bufferSizeExt | | |
| cusparseDbsrilu02_bufferSize | hipsparseDbsrilu02_bufferSize | |
| cusparseDbsrilu02_bufferSizeExt | | |
| cusparseCbsrilu02_bufferSize | hipsparseCbsrilu02_bufferSize | |
| cusparseCbsrilu02_bufferSizeExt | | |
| cusparseZbsrilu02_bufferSize | hipsparseZbsrilu02_bufferSize | |
| cusparseZbsrilu02_bufferSizeExt | | |

| | | |
|---|---|---|
| cusparseSbsrilu02_analysis | hipsparseSbsrilu02_analysis | |
| cusparseDbsrilu02_analysis | hipsparseDbsrilu02_analysis | |
| cusparseCbsrilu02_analysis | hipsparseCbsrilu02_analysis | |
| cusparseZbsrilu02_analysis | hipsparseZbsrilu02_analysis | |
| cusparseSbsrilu02 | hipsparseSbsrilu02 | |
| cusparseDbsrilu02 | hipsparseDbsrilu02 | |
| cusparseCbsrilu02 | hipsparseCbsrilu02 | |
| cusparseZbsrilu02 | hipsparseZbsrilu02 | |
| cusparseXbsrilu02_zeroPivot | hipsparseXbsrilu02_zeroPivot | |

三对角矩阵求解

| CUDA | HIP | CUDA version * |
|---|---|---|
| cusparseSgtsv | | |
| cusparseDgtsv | | |
| cusparseCgtsv | | |
| cusparseZgtsv | | |
| cusparseSgtsv_nopivot | | |
| cusparseDgtsv_nopivot | | |
| cusparseCgtsv_nopivot | | |
| cusparseZgtsv_nopivot | | |
| cusparseSgtsv2_bufferSizeExt | hipsparseSgtsv2_bufferSizeExt | 9 |
| cusparseDgtsv2_bufferSizeExt | hipsparseDgtsv2_bufferSizeExt | 9 |
| cusparseCgtsv2_bufferSizeExt | hipsparseCgtsv2_bufferSizeExt | 9 |
| cusparseZgtsv2_bufferSizeExt | hipsparseZgtsv2_bufferSizeExt | 9 |
| cusparseSgtsv2 | hipsparseSgtsv2 | 9 |
| cusparseDgtsv2 | hipsparseDgtsv2 | 9 |
| cusparseCgtsv2 | hipsparseCgtsv2 | 9 |

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|---|---|---|
| cusparseZgtsv2 | hipsparseZgtsv2 | 9 |
| cusparseSgtsv2_nopivot_bufferSizeExt | hipsparseSgtsv2_nopivot_bufferSizeExt | 9 |
| cusparseDgtsv2_nopivot_bufferSizeExt | hipsparseDgtsv2_nopivot_bufferSizeExt | 9 |
| cusparseCgtsv2_nopivot_bufferSizeExt | hipsparseCgtsv2_nopivot_bufferSizeExt | 9 |
| cusparseZgtsv2_nopivot_bufferSizeExt | hipsparseZgtsv2_nopivot_bufferSizeExt | 9 |
| cusparseSgtsv2_nopivot | hipsparseSgtsv2_nopivot | 9 |
| cusparseDgtsv2_nopivot | hipsparseDgtsv2_nopivot | 9 |
| cusparseCgtsv2_nopivot | hipsparseCgtsv2_nopivot | 9 |
| cusparseZgtsv2_nopivot | hipsparseZgtsv2_nopivot | 9 |

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSgtsvStridedBatch | | |
| cusparseDgtsvStridedBatch | | |
| cusparseCgtsvStridedBatch | | |
| cusparseZgtsvStridedBatch | | |
| cusparseSgtsv2StridedBatch_bufferSizeExt | hipsparseSgtsv2StridedBatch_bufferSizeExt | 9 |
| cusparseDgtsv2StridedBatch_bufferSizeExt | hipsparseDgtsv2StridedBatch_bufferSizeExt | 9 |
| cusparseCgtsv2StridedBatch_bufferSizeExt | hipsparseCgtsv2StridedBatch_bufferSizeExt | 9 |

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|---|---|---|
| cusparseZgtsv2StridedBatch_bufferSizeExt | hipsparseZgtsv2StridedBatch_bufferSizeExt | 9 |
| cusparseSgtsv2StridedBatch | hipsparseSgtsv2StridedBatch | 9 |
| cusparseDgtsv2StridedBatch | hipsparseDgtsv2StridedBatch | 9 |
| cusparseCgtsv2StridedBatch | hipsparseCgtsv2StridedBatch | 9 |
| cusparseZgtsv2StridedBatch | hipsparseZgtsv2StridedBatch | 9 |
| cusparseSgtsvInterleavedBatch_bufferSizeExt | hipsparseSgtsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseDgtsvInterleavedBatch_bufferSizeExt | hipsparseDgtsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseCgtsvInterleavedBatch_bufferSizeExt | hipsparseCgtsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseZgtsvInterleavedBatch_bufferSizeExt | hipsparseZgtsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseSgtsvInterleavedBatch | hipsparseSgtsvInterleavedBatch | 9.2 |
| cusparseDgtsvInterleavedBatch | hipsparseDgtsvInterleavedBatch | 9.2 |
| cusparseCgtsvInterleavedBatch | hipsparseCgtsvInterleavedBatch | 9.2 |
| cusparseZgtsvInterleavedBatch | hipsparseZgtsvInterleavedBatch | 9.2 |

PentadiagonalSolve

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSgpsvInterleavedBatch_bufferSizeExt | hipsparseSgpsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseDgpsvInterleavedBatch_bufferSizeExt | hipsparseDgpsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseCgpsvInterleavedBatch_bufferSizeExt | hipsparseCgpsvInterleavedBatch_bufferSizeExt | 9.2 |
| cusparseZgpsvInterleavedBatch_bufferSizeExt | hipsparseZgpsvInterleavedBatch_bufferSizeExt | 9.2 |

| | | |
|---|---|---|
| cusparseSgpsvInterleavedBatch | hipsparseSgpsvInterleavedBatch | 9.2 |
| cusparseDgpsvInterleavedBatch | hipsparseDgpsvInterleavedBatch | 9.2 |
| cusparseCgpsvInterleavedBatch | hipsparseCgpsvInterleavedBatch | 9.2 |
| cusparseZgpsvInterleavedBatch | hipsparseZgpsvInterleavedBatch | 9.2 |

矩阵重排

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseScsrcolor | hipsparseScsrcolor | 9.2 |
| cusparseDcsrcolor | hipsparseDcsrcolor | 9.2 |
| cusparseCcsrcolor | hipsparseCcsrcolor | 9.2 |
| cusparseZcsrcolor | hipsparseZcsrcolor | |

格式转换

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSbsr2csr | hipsparseSbsr2csr | |
| cusparseDbsr2csr | hipsparseDbsr2csr | |
| cusparseCbsr2csr | hipsparseCbsr2csr | |
| cusparseZbsr2csr | hipsparseZbsr2csr | |
| cusparseSgebsr2gebsc_bufferSize | hipsparseSgebsr2gebsc_bufferSize | |
| cusparseSgebsr2gebsc_bufferSizeExt | | |
| cusparseDgebsr2gebsc_bufferSize | hipsparseDgebsr2gebsc_bufferSize | |
| cusparseDgebsr2gebsc_bufferSizeExt | | |
| cusparseCgebsr2gebsc_bufferSize | hipsparseCgebsr2gebs | |

| | | |
|---|---|---|
| | c_bufferSize | |
| cusparseCgebsr2gebsc_bufferSizeExt | | |
| cusparseZgebsr2gebsc_bufferSize | hipsparseZgebsr2gebsc_bufferSize | |
| cusparseZgebsr2gebsc_bufferSizeExt | | |
| cusparseSgebsr2gebsc | hipsparseSgebsr2gebsc | |
| cusparseDgebsr2gebsc | hipsparseDgebsr2gebsc | |
| cusparseCgebsr2gebsc | hipsparseCgebsr2gebsc | |
| cusparseZgebsr2gebsc | hipsparseZgebsr2gebsc | |
| cusparseSgebsr2gebsr_bufferSize | hipsparseSgebsr2gebsr_bufferSize | |
| cusparseSgebsr2gebsr_bufferSizeExt | | |
| cusparseDgebsr2gebsr_bufferSize | hipsparseDgebsr2gebsr_bufferSize | |
| cusparseDgebsr2gebsr_bufferSizeExt | | |
| cusparseCgebsr2gebsr_bufferSize | hipsparseCgebsr2gebsr_bufferSize | |
| cusparseCgebsr2gebsr_bufferSizeExt | | |
| cusparseZgebsr2gebsr_bufferSize | hipsparseZgebsr2gebsr_bufferSize | |
| cusparseZgebsr2gebsr_bufferSizeExt | | |
| cusparseXgebsr2gebsrNnz | hipsparseXgebsr2gebsrNnz | |
| cusparseSgebsr2gebsr | hipsparseSgebsr2gebs | |

| | | |
|---|---|---|
| | r | |
| cusparseDgebsr2gebsr | hipsparseDgebsr2gebsr | |
| cusparseCgebsr2gebsr | hipsparseCgebsr2gebsr | |
| cusparseZgebsr2gebsr | hipsparseZgebsr2gebsr | |
| cusparseXgebsr2csr | hipsparseXgebsr2csr | |
| cusparseSgebsr2csr | hipsparseSgebsr2csr | |
| cusparseDgebsr2csr | hipsparseDgebsr2csr | |
| cusparseCgebsr2csr | hipsparseCgebsr2csr | |
| cusparseZgebsr2csr | hipsparseZgebsr2csr | |
| cusparseScsr2gebsr_bufferSize | hipsparseScsr2gebsr_bufferSize | |
| cusparseScsr2gebsr_bufferSizeExt | | |
| cusparseDcsr2gebsr_bufferSize | hipsparseDcsr2gebsr_bufferSize | |
| cusparseDcsr2gebsr_bufferSizeExt | | |
| cusparseCcsr2gebsr_bufferSize | hipsparseCcsr2gebsr_bufferSize | |
| cusparseCcsr2gebsr_bufferSizeExt | | |
| cusparseZcsr2gebsr_bufferSize | hipsparseZcsr2gebsr_bufferSize | |
| cusparseZcsr2gebsr_bufferSizeExt | | |
| cusparseXcsr2gebsrNnz | hipsparseXcsr2gebsrNnz | |
| cusparseScsr2gebsr | hipsparseScsr2gebsr | |
| cusparseDcsr2gebsr | hipsparseDcsr2gebsr | |

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|---|---|---|
| cusparseCcsr2gebsr | hipsparseCcsr2gebsr | |
| cusparseZcsr2gebsr | hipsparseZcsr2gebsr | |
| cusparseXcoo2csr | hipsparseXcoo2csr | |
| cusparseScsc2dense | hipsparseScsc2dense | |
| cusparseDcsc2dense | hipsparseDcsc2dense | |
| cusparseCcsc2dense | hipsparseCcsc2dense | |
| cusparseZcsc2dense | hipsparseZcsc2dense | |
| cusparseScsc2hyb | | |
| cusparseDcsc2hyb | | |
| cusparseCcsc2hyb | | |
| cusparseZcsc2hyb | | |
| cusparseXcsr2bsrNnz | hipsparseXcsr2bsrNnz | |
| cusparseScsr2bsr | hipsparseScsr2bsr | |
| cusparseDcsr2bsr | hipsparseDcsr2bsr | |
| cusparseCcsr2bsr | hipsparseCcsr2bsr | |
| cusparseZcsr2bsr | hipsparseZcsr2bsr | |
| cusparseXcsr2coo | hipsparseXcsr2coo | |
| cusparseScsr2csc | hipsparseScsr2csc | |
| cusparseDcsr2csc | hipsparseDcsr2csc | |
| cusparseCcsr2csc | hipsparseCcsr2csc | |
| cusparseZcsr2csc | hipsparseZcsr2csc | |
| cusparseCsr2cscEx | | 8 |
| cusparseCsr2cscEx2 | | 10.1 |
| cusparseCsr2cscEx2_bufferSize | | 10.1 |
| cusparseScsr2dense | hipsparseScsr2dense | |
| cusparseDcsr2dense | hipsparseScsr2dense | |
| cusparseCcsr2dense | hipsparseCcsr2dense | |
| cusparseZcsr2dense | hipsparseZcsr2dense | |

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|---|---|---|
| cusparseScsr2csr_compress | hipsparseScsr2csr_compress | 8 |
| cusparseDcsr2csr_compress | hipsparseDcsr2csr_compress | 8 |
| cusparseCcsr2csr_compress | hipsparseCcsr2csr_compress | 8 |
| cusparseZcsr2csr_compress | hipsparseZcsr2csr_compress | 8 |
| cusparseScsr2hyb | hipsparseScsr2hyb | |
| cusparseDcsr2hyb | hipsparseDcsr2hyb | |
| cusparseCcsr2hyb | hipsparseCcsr2hyb | |
| cusparseZcsr2hyb | hipsparseZcsr2hyb | |
| cusparseSdense2csc | hipsparseSdense2csc | |
| cusparseDdense2csc | hipsparseDdense2csc | |
| cusparseCdense2csc | hipsparseCdense2csc | |
| cusparseZdense2csc | hipsparseZdense2csc | |
| cusparseSdense2csr | hipsparseSdense2csr | |
| cusparseDdense2csr | hipsparseDdense2csr | |
| cusparseCdense2csr | hipsparseCdense2csr | |
| cusparseZdense2csr | hipsparseZdense2csr | |
| cusparseSdense2hyb | | |
| cusparseDdense2hyb | | |
| cusparseCdense2hyb | | |
| cusparseZdense2hyb | | |
| cusparseShyb2csc | | |
| cusparseDhyb2csc | | |
| cusparseChyb2csc | | |
| cusparseZhyb2csc | | |

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|---|---|---|
| cusparseShyb2csr | hipsparseShyb2csr | |
| cusparseDhyb2csr | hipsparseDhyb2csr | |
| cusparseChyb2csr | hipsparseChyb2csr | |
| cusparseZhyb2csr | hipsparseZhyb2csr | |
| cusparseShyb2dense | | |
| cusparseDhyb2dense | | |
| cusparseChyb2dense | | |
| cusparseZhyb2dense | | |
| cusparseSnnz | cusparseSnnz | |
| cusparseDnnz | cusparseDnnz | |
| cusparseCnnz | cusparseCnnz | |
| cusparseZnnz | cusparseZnnz | |
| cusparseCreateIdentityPermutation | hipsparseCreateIdentityPermutation | |
| cusparseXcoosort_bufferSizeExt | hipsparseXcoosort_bufferSizeExt | |
| cusparseXcoosortByRow | hipsparseXcoosortByRow | |
| cusparseXcoosortByColumn | hipsparseXcoosortByColumn | |
| cusparseXcsrsort_bufferSizeExt | hipsparseXcsrsort_bufferSizeExt | |
| cusparseXcsrsort | hipsparseXcsrsort | |
| cusparseXcscsort_bufferSizeExt | hipsparseXcscsort_bufferSizeExt | |
| cusparseXcscsort | hipsparseXcscsort | |
| cusparseCreateCsru2csrInfo | hipsparseCreateCsru2csrInfo | |

| | | |
|---|---|---|
| cusparseDestroyCsru2csrInfo | hipsparseDestroyCsru2csrInfo | |
| cusparseScsru2csr_bufferSizeExt | hipsparseScsru2csr_bufferSizeExt | |
| cusparseDcsru2csr_bufferSizeExt | hipsparseDcsru2csr_bufferSizeExt | |
| cusparseCcsru2csr_bufferSizeExt | hipsparseCcsru2csr_bufferSizeExt | |
| cusparseZcsru2csr_bufferSizeExt | hipsparseZcsru2csr_bufferSizeExt | |
| cusparseScsru2csr | hipsparseScsru2csr | |
| cusparseDcsru2csr | hipsparseDcsru2csr | |
| cusparseCcsru2csr | hipsparseCcsru2csr | |
| cusparseZcsru2csr | hipsparseZcsru2csr | |
| cusparseScsr2csru | hipsparseScsr2csru | |
| cusparseDcsr2csru | hipsparseDcsr2csru | |
| cusparseCcsr2csru | hipsparseCcsr2csru | |
| cusparseZcsr2csru | hipsparseZcsr2csru | |
| cusparseHpruneDense2csr | | 9 |
| cusparseSpruneDense2csr | hipsparseSpruneDense2csr | 9 |
| cusparseDpruneDense2csr | hipsparseDpruneDense2csr | 9 |
| cusparseHpruneDense2csr_bufferSizeExt | hipsparseSpruneDense2csr_bufferSizeExt | 9 |
| cusparseSpruneDense2csr_bufferSizeExt | hipsparseSpruneDense2csr_bufferSizeExt | 9 |
| cusparseDpruneDense2csr_bufferSizeExt | hipsparseDpruneDens | 9 |

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|---|---|---|
| | e2csr_bufferSizeExt | |
| cusparseHpruneDense2csrNnz | | 9 |
| cusparseSpruneDense2csrNnz | hipsparseSpruneDense2csrNnz | 9 |
| cusparseDpruneDense2csrNnz | hipsparseDpruneDense2csrNnz | 9 |
| cusparseHpruneCsr2csr | | 9 |
| cusparseSpruneCsr2csr | hipsparseSpruneCsr2csr | 9 |
| cusparseDpruneCsr2csr | hipsparseDpruneCsr2csr | 9 |
| cusparseHpruneCsr2csr_bufferSizeExt | | 9 |
| cusparseSpruneCsr2csr_bufferSizeExt | hipsparseSpruneCsr2csr_bufferSizeExt | 9 |
| cusparseDpruneCsr2csr_bufferSizeExt | hipsparseDpruneCsr2csr_bufferSizeExt | 9 |
| cusparseHpruneCsr2csrNnz | | 9 |
| cusparseSpruneCsr2csrNnz | hipsparseSpruneCsr2csrNnz | 9 |
| cusparseDpruneCsr2csrNnz | hipsparseDpruneCsr2csrNnz | 9 |
| cusparseHpruneDense2csrByPercentage | | 9 |
| cusparseSpruneDense2csrByPercentage | hipsparseSpruneDense2csrByPercentage | 9 |
| cusparseDpruneDense2csrByPercentage | hipsparseDpruneDense2csrByPercentage | 9 |
| cusparseHpruneDense2csrByPercentage_bufferSizeExt | | 9 |

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|---|---|---|
| cusparseSpruneDense2csrByPercentage_bufferSizeExt | hipsparseSpruneDense2csrByPercentage_bufferSizeExt | 9 |
| cusparseDpruneDense2csrByPercentage_bufferSizeExt | hipsparseDpruneDense2csrByPercentage_bufferSizeExt | 9 |
| cusparseHpruneDense2csrNnzByPercentage | | 9 |
| cusparseSpruneDense2csrNnzByPercentage | hipsparseSpruneDense2csrNnzByPercentage | 9 |
| cusparseDpruneDense2csrNnzByPercentage | hipsparseDpruneDense2csrNnzByPercentage | 9 |
| cusparseHpruneCsr2csrByPercentage | | 9 |
| cusparseSpruneCsr2csrByPercentage | hipsparseSpruneCsr2csrByPercentage | 9 |
| cusparseDpruneCsr2csrByPercentage | hipsparseDpruneCsr2csrByPercentage | 9 |
| cusparseHpruneCsr2csrByPercentage_bufferSizeExt | | 9 |
| cusparseSpruneCsr2csrByPercentage_bufferSizeExt | hipsparseSpruneCsr2csrByPercentage_bufferSizeExt | 9 |
| cusparseDpruneCsr2csrByPercentage_bufferSizeExt | hipsparseDpruneCsr2csrByPercentage_bufferSizeExt | 9 |
| cusparseHpruneCsr2csrNnzByPercentage | | 9 |
| cusparseSpruneCsr2csrNnzByPercentage | hipsparseSpruneCsr2csrNnzByPercentage | 9 |

| | | |
|---|---|---|
| cusparseDpruneCsr2csrNnzByPercentage | hipsparseDpruneCsr2csrNnzByPercentage | 9 |
| cusparseZnnz | hipsparseZnnz | |
| cusparseCnnz | hipsparseCnnz | |
| cusparseDnnz | hipsparseDnnz | |
| cusparseSnnz | hipsparseSnnz | |
| cusparseSnnz_compress | hipsparseSnnz_compress | 8 |
| cusparseDnnz_compress | hipsparseDnnz_compress | 8 |
| cusparseCnnz_compress | hipsparseCnnz_compress | 8 |
| cusparseZnnz_compress | hipsparseZnnz_compress | 8 |

矩阵生成函数

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseCreateCoo | hipsparseCreateCoo | 10.1 |
| cusparseCreateCooAoS | hipsparseCreateCooAoS | 10.1 |
| cusparseCreateCsr | hipsparseCreateCsr | 10.1 |
| cusparseDestroySpMat | hipsparseDestroySpMat | 10.1 |
| cusparseCooGet | hipsparseCooGet | 10.1 |
| cusparseCooAoSGet | hipsparseCooAoSGet | 10.1 |
| cusparseCsrGet | hipsparseCsrGet | 10.1 |
| cusparseSpMatGetFormat | hipsparseSpMatGetFormat | 10.1 |
| cusparseSpMatGetIndexBase | hipsparseSpMatGetIndexBase | 10.1 |
| cusparseSpMatGetValues | hipsparseSpMatGetValues | 10.1 |
| cusparseSpMatSetValues | hipsparseSpMatSetValues | 10.1 |

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|---|---|---|
| cusparseSpMatGetStridedBatch | hipsparseSpMatGetStridedBatch | 10.1 |
| cusparseSpMatSetStridedBatch | hipsparseSpMatSetStridedBatch | 10.1 |
| cusparseSpMatGetNumBatches | | 10.1 |
| cusparseSpMatSetNumBatches | | 10.1 |
| cusparseCreateSpVec | hipsparseCreateSpVec | 10.1 |
| cusparseDestroySpVec | hipsparseDestroySpVec | 10.1 |
| cusparseSpVecGet | hipsparseSpVecGet | 10.1 |
| cusparseSpVecGetIndexBase | hipsparseSpVecGetIndexBase | 10.1 |
| cusparseSpVecGetValues | hipsparseSpVecGetValues | 10.1 |
| cusparseSpVecSetValues | hipsparseSpVecSetValues | 10.1 |
| cusparseCreateCoo | hipsparseCreateCoo | 10.1 |
| cusparseCreateCooAoS | hipsparseCreateCooAoS | 10.1 |
| cusparseCreateCsr | hipsparseCreateCsr | 10.1 |

稠密矩阵和向量生成函数

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseCreateDnMat | hipsparseCreateDnMat | 10.1 |
| cusparseDestroyDnMat | hipsparseDestroyDnMat | 10.1 |
| cusparseDnMatGet | hipsparseDnMatGet | 10.1 |
| cusparseDnMatGetValues | hipsparseDnMatGetValues | 10.1 |
| cusparseDnMatSetValues | hipsparseDnMatSetValues | 10.1 |
| cusparseDnMatSetStridedBatch | hipsparseDnMatSetStridedBatch | 10.1 |
| cusparseDnMatGetStridedBatch | hipsparseDnMatGetStridedBatch | 10.1 |
| cusparseCreateDnVec | hipsparseCreateDnVec | 10.1 |
| cusparseDestroyDnVec | hipsparseDestroyDnVec | 10.1 |
| cusparseDnVecGet | hipsparseDnVecGet | 10.1 |
| cusparseDnVecGetValues | hipsparseDnVecGetValues | 10.1 |
| cusparseDnVecSetValues | hipsparseDnVecSetValues | 10.1 |

SpMM 创建函数

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSpMM | hipsparseSpMM | 10.1 |
| cusparseSpMM_bufferSize | hipsparseSpMM_bufferSize | |

SpMV 创建函数

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSpMV | hipsparseSpMV | 10.1 |
| cusparseSpMV_bufferSize | hipsparseSpMV_bufferSize | 10.1 |

SpVV 创建函数

| CUDA | HIP | CUDA version* |
|---|---|---|
| cusparseSpVV | hipsparseSpVV | |
| cusparseSpVV_bufferSize | hipsparseSpVV_bufferSize | |

# 5.7 RAND

## 5.7.1 Data Types

| type | CUDA | HIP | CUDA version* |
|---|---|---|---|
| enum | curandStatus_t | hiprandStatus_t | 10.2 |
| 0 | CURAND_STATUS_SUCCESS | HIPRAND_STATUS_SUCCESS | |
| 1 | CURAND_STATUS_VERSION_MISMATCH | HIPRAND_STATUS_VERSION_MISMATCH | |
| 2 | CURAND_STATUS_NOT_INITIALIZED | HIPRAND_STATUS_NOT_INITIALIZED | |
| 3 | CURAND_STATUS_ALLOCATION_FAILED | HIPRAND_STATUS_ALLOCATION_FAILED | |
| 4 | CURAND_STATUS_TYPE | HIPRAND_STATUS_TYPE_ | |

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|---|---|---|---|
| | _ERROR | ERROR | |
| 5 | CURAND_STATUS_OUT_OF_RANGE | HIPRAND_STATUS_OUT_OF_RANGE | |
| 6 | CURAND_STATUS_LENGTH_NOT_MULTIPLE | HIPRAND_STATUS_LENGTH_NOT_MULTIPLE | |
| 7 | CURAND_STATUS_DOUBLE_PRECISION_REQUIRED | HIPRAND_STATUS_DOUBLE_PRECISION_REQUIRED | |
| 8 | CURAND_STATUS_LAUNCH_FAILURE | HIPRAND_STATUS_LAUNCH_FAILURE | |
| 9 | CURAND_STATUS_PREEXISTING_FAILURE | HIPRAND_STATUS_PREEXISTING_FAILURE | |
| 10 | CURAND_STATUS_INITIALIZATION_FAILED | HIPRAND_STATUS_INITIALIZATION_FAILED | |
| 11 | CURAND_STATUS_ARCH_MISMATCH | HIPRAND_STATUS_ARCH_MISMATCH | |
| 12 | CURAND_STATUS_INTERNAL_ERROR | HIPRAND_STATUS_INTERNAL_ERROR | |
| enum | curandRngType_t | hiprandRngType_t | |
| 0 | CURAND_RNG_TEST | HIPRAND_RNG_TEST | |
| 1 | CURAND_RNG_PSEUDO_DEFAULT | HIPRAND_RNG_PSEUDO_DEFAULT | |
| 2 | CURAND_RNG_PSEUDO_XORWOW | HIPRAND_RNG_PSEUDO_XORWOW | |
| 3 | CURAND_RNG_PSEUDO_MRG32K3A | HIPRAND_RNG_PSEUDO_MRG32K3A | |
| 4 | CURAND_RNG_PSEUDO_MTGP32 | HIPRAND_RNG_PSEUDO_MTGP32 | |
| 5 | CURAND_RNG_PSEUDO | HIPRAND_RNG_PSEUDO_ | |

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|---|---|---|---|
| | _MT19937 | MT19937 | |
| 6 | CURAND_RNG_PSEUDO_PHILOX4_32_10 | HIPRAND_RNG_PSEUDO_PHILOX4_32_10 | |
| 7 | CURAND_RNG_QUASI_DEFAULT | HIPRAND_RNG_QUASI_DEFAULT | |
| 8 | CURAND_RNG_QUASI_SOBOL32 | HIPRAND_RNG_QUASI_SOBOL32 | |
| 9 | CURAND_RNG_QUASI_SCRAMBLED_SOBOL32 | HIPRAND_RNG_QUASI_SCRAMBLED_SOBOL32 | |
| 10 | CURAND_RNG_QUASI_SOBOL64 | HIPRAND_RNG_QUASI_SOBOL64 | |
| 11 | CURAND_RNG_QUASI_SCRAMBLED_SOBOL64 | HIPRAND_RNG_QUASI_SCRAMBLED_SOBOL64 | |
| enum | curandOrdering_t | | |
| 0 | CURAND_ORDERING_PSEUDO_BEST | | |
| 1 | CURAND_ORDERING_PSEUDO_DEFAULT | | |
| 2 | CURAND_ORDERING_PSEUDO_SEEDED | | |
| 3 | CURAND_ORDERING_QUASI_DEFAULT | | |
| enum | curandDirectionVectorSet_t | | |
| 0 | CURAND_DIRECTION_VECTORS_32_JOEKUO6 | | |
| 1 | CURAND_SCRAMBLED_DIRECTION_VECTORS_32_JOEKUO6 | | |
| 2 | CURAND_DIRECTION_V | | |

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|---|---|---|---|
| | ECTORS_64_JOEKUO6 | | |
| 3 | CURAND_SCRAMBLED_DIRECTION_VECTORS_64_JOEKUO6 | | |
| enum | curandMethod_t | | |
| 0 | CURAND_CHOOSE_BEST | | |
| 1 | CURAND_ITR | | |
| 2 | CURAND_KNUTH | | |
| 3 | CURAND_HITR | | |
| 4 | CURAND_M1 | | |
| 5 | CURAND_M2 | | |
| 6 | CURAND_BINARY_SEARCH | | |
| 7 | CURAND_DISCRETE_GAUSS | | |
| 8 | CURAND_REJECTION | | |
| 9 | CURAND_DEVICE_API | | |
| 10 | CURAND_FAST_REJECTION | | |
| 11 | CURAND_3RD | | |
| 12 | CURAND_DEFINITION | | |
| 13 | CURAND_POISSON | | |
| struct | curandGenerator_t | hiprandGenerator_t | |
| struct | curandDistribution_t | | |
| struct | curandDistributionShift_t | | |
| struct | curandDistributionM2Shift_t | | |

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|---|---|---|---|
| struct | curandHistogramM2_t | | |
| struct | curandHistogramM2K_t | | |
| struct | curandHistogramM2V_t | | |
| struct | curandDiscreteDistribution_t | hiprandDiscreteDistribution_t | |
| | curandDirectionVectors32_t | hiprandDirectionVectors32_t | |
| struct | curandState_t | hiprandState | |
| struct | curandStateTest_t | | |
| struct | curandStateXORWOW_t | hiprandStateXORWOW | |
| struct | curandStateMRG32k3a_t | hiprandStateMRG32k3a | |
| struct | curandStateSobol32_t | hiprandStateSobol32 | |
| struct | curandStateScrambledSobol32_t | hiprandStateScrambledSobol32 | |
| struct | curandStateSobol64_t | hiprandStateSobol64 | |
| struct | curandStateScrambledSobol64_t | hiprandStateScrambledSobol64 | |
| struct | curandStateMtgp32_t | hiprandStateMtgp32 | |
| | | hiprandStatePhilox4_32_10 | |
| struct | normal_args_t | | |
| struct | normal_args_double_t | | |
| struct | mtgp32_params_fast_t | mtgp32_fast_param_t | |
| struct | mtgp32_kernel_params_t | mtgp32_kernel_params_t | |

### 5.7.2 API Functions

| CUDA | HIP | CUDA version * |
|---|---|---|
| curandCreateGenerator | hiprandCreateGenerator | 10.2 |
| curandCreateGeneratorHost | hiprandCreateGeneratorHost | |

| | | |
|---|---|---|
| curandDestroyGenerator | hiprandDestroyGenerator | |
| curandGetVersion | hiprandGetVersion | |
| curandGetProperty | | |
| curandSetStream | hiprandSetStream | |
| curandSetPseudoRandomGeneratorSeed | hiprandSetPseudoRandomGeneratorSeed | |
| curandSetGeneratorOffset | hiprandSetGeneratorOffset | |
| curandSetGeneratorOrdering | | |
| curandSetQuasiRandomGeneratorDimensions | hiprandSetQuasiRandomGeneratorDimensions | |
| curandGenerate | hiprandGenerate | |
| | hiprandGenerateChar | |
| | hiprandGenerateShort | |
| curandGenerateLongLong | | |
| curandGenerateUniform | hiprandGenerateUniform | |
| curandGenerateUniformDouble | hiprandGenerateUniformDouble | |
| | hiprandGenerateUniformHalf | |
| curandGenerateNormal | hiprandGenerateNormal | |
| curandGenerateNormalDouble | hiprandGenerateNormalDouble | |
| | hiprandGenerateNormalHalf | |
| curandGenerateLogNormal | hiprandGenerateLogNormal | |
| curandGenerateLogNormalDouble | hiprandGenerateLogNormalDouble | |
| | hiprandGenerateLogNormalHalf | |
| curandCreatePoissonDistribution | hiprandCreatePoissonDistribution | |
| curandDestroyDistribution | hiprandDestroyDistribution | |
| curandGeneratePoisson | hiprandGeneratePoisson | |
| curandGeneratePoissonMethod | | |
| curandGenerateBinomial | | |

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|---|---|---|
| curandGenerateBinomialMethod | | |
| curandGenerateSeeds | hiprandGenerateSeeds | |
| curandGetDirectionVectors32 | | |
| curandGetScrambleConstants32 | | |
| curandGetDirectionVectors64 | | |
| curandGetScrambleConstants64 | | |
| | | |
| curand_init | hiprand_init | |
| curand | hiprand | |
| curand4 | hiprand4 | |
| curand_MRG32k3a | | |
| curand_discrete | hiprand_discrete | |
| curand_discrete4 | hiprand_discrete4 | |
| curand_log_normal | hiprand_log_normal | |
| curand_log_normal2 | hiprand_log_normal2 | |
| curand_log_normal4 | hiprand_log_normal4 | |
| curand_log_normal_double | hiprand_log_normal_double | |
| curand_log_normal2_double | hiprand_log_normal2_double | |
| curand_log_normal4_double | hiprand_log_normal4_double | |
| curandMakeMTGP32Constants | hiprandMakeMTGP32Constants | |
| curandMakeMTGP32KernelState | hiprandMakeMTGP32KernelState | |
| curand_mtgp32_specific | | |
| curand_mtgp32_single | | |
| curand_mtgp32_single_specific | | |
| curand_normal | hiprand_normal | |
| curand_normal2 | hiprand_normal2 | |
| curand_normal4 | hiprand_normal4 | |
| curand_normal_double | hiprand_normal_double | |

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|---|---|---|
| curand_normal2_double | hiprand_normal2_double | |
| curand_normal4_double | hiprand_normal4_double | |
| curand_uniform | hiprand_uniform | |
| curand_uniform_double | hiprand_uniform_double | |
| curand_uniform4 | hiprand_uniform4 | |
| | hiprand_uniform2_double | |
| curand_uniform4_double | hiprand_uniform4_double | |
| skipahead | skipahead | |
| skipaheadsequence | skipaheadsequence | |
| skipahead_subsequence | skipahead_subsequence | |
| curand_poisson | hiprand_poisson | |
| curand_poisson4 | hiprand_poisson4 | |
| curand_poisson_knuth | | |
| curand_poisson_knuth4 | | |
| curand_poisson_gammainc | | |
| curand_poisson_gammainc4 | | |
| curand_poisson_from_normal | | |
| curand_M2_double | | |
| curand_M2_double4 | | |
| curand_binary_search_double | | |

## 5.8 CUB

hipCUB 在 rocPRIM 的基础上实现的，主要功能接口与 CUB 一致。hipCUB 与 CUB 的后端实现不一样，CUB 内部的 Agent、block/specialization 及 device/dispatch_*相关内容 hipCUB 均不支持。

### 5.8.1 Warp-wide "collective" primitives

| CUDA | HIP | CUDA version* |
|---|---|---|
| cub::WarpExchange | hipcub::WarpExchange | |

| | | |
|---|---|---|
| cub::WarpExchange::BlockedToStriped | hipcub::WarpExchange::BlockedToStriped | |
| cub::WarpExchange::StripedToBlocked | hipcub::WarpExchange::StripedToBlocked | |
| cub::WarpExchange::ScatterToStriped | hipcub::WarpExchange::ScatterToStriped | |
| cub::WarpLoad | hipcub::WarpLoad | |
| cub::WarpLoad::TempStorage | hipcub::WarpLoad::TempStorage | |
| cub::WarpLoad::Load | hipcub::WarpLoad::Load | |
| cub::WarpMergeSort | hipcub::WarpMergeSort | |
| cub::WarpMergeSort::get_member_mask | hipcub::WarpMergeSort::get_member_mask | |
| cub::WarpReduce | hipcub::WarpReduce | |
| cub::WarpReduce::TempStorage | hipcub::WarpReduce::TempStorage | |
| cub::WarpReduce::Sum | hipcub::WarpReduce::Sum | |
| cub::WarpReduce::HeadSegmentedSum | hipcub::WarpReduce::HeadSegmentedSum | |
| cub::WarpReduce::TailSegmentedSum | hipcub::WarpReduce::TailSegmentedSum | |
| cub::WarpReduce::Reduce | hipcub::WarpReduce::Reduce | |
| cub::WarpReduce::HeadSegmentedReduce | hipcub::WarpReduce::HeadSegmentedReduce | |
| cub::WarpReduce::TailSegmentedReduce | hipcub::WarpReduce::TailSegmentedReduce | |
| cub::WarpReduce | hipcub::WarpReduce | |
| cub::WarpScan | hipcub::WarpScan | |
| cub::WarpScan::TempStorage | hipcub::WarpScan::TempStorage | |

| | | |
|---|---|---|
| cub::WarpScan::InclusiveSum | hipcub::WarpScan::InclusiveSum | |
| cub::WarpScan::ExclusiveSum | hipcub::WarpScan::ExclusiveSum | |
| cub::WarpScan::InclusiveScan | hipcub::WarpScan::InclusiveScan | |
| cub::WarpScan::ExclusiveScan | hipcub::WarpScan::ExclusiveScan | |
| cub::WarpScan::Scan | hipcub::WarpScan::Scan | |
| cub::WarpScan::Broadcast | hipcub::WarpScan::Broadcast | |
| cub::WarpStore | hipcub::WarpStore | |
| cub::WarpStore::TempStorage | hipcub::WarpStore::TempStorage | |
| cub::WarpStore::Store | hipcub::WarpStore::Store | |

### 5.8.2 Block-wide "collective" primitives

| CUDA | HIP | CUDA version* |
|---|---|---|
| cub::BlockAdjacentDifference | cub::BlockAdjacentDifference | |
| cub::BlockAdjacentDifference::TempStorage | cub::BlockAdjacentDifference::TempStorage | |
| cub::BlockAdjacentDifference::BlockAdjacentDifference | cub::BlockAdjacentDifference::BlockAdjacentDifference | |
| cub::BlockAdjacentDifference::SubtractLeft | cub::BlockAdjacentDifference::SubtractLeft | |
| cub::BlockAdjacentDifference::SubtractLeftPartialTile | cub::BlockAdjacentDifference::SubtractLeftPartialTile | |
| cub::BlockAdjacentDifference::SubtractRight | cub::BlockAdjacentDifference::SubtractRight | |
| cub::BlockAdjacentDifference::SubtractRightPartialTile | cub::BlockAdjacentDifference::SubtractRightPartialTile | |
| cub::BlockAdjacentDifference::FlagHeads | cub::BlockAdjacentDifference::FlagHeads | |
| cub::BlockAdjacentDifference::Fl | cub::BlockAdjacentDifference::Fla | |

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| | | |
|---|---|---|
| agTails | gTails | |
| cub::BlockAdjacentDifference::FlagHeadsAndTails | cub::BlockAdjacentDifference::FlagHeadsAndTails | |
| cub::BlockDiscontinuity | cub::BlockDiscontinuity | |
| cub::BlockDiscontinuity::TempStorage | cub::BlockDiscontinuity::TempStorage | |
| cub::BlockDiscontinuity::FlagHeads | hipcub::BlockDiscontinuity::FlagHeads | |
| cub::BlockDiscontinuity::FlagTails | hipcub::BlockDiscontinuity::FlagTails | |
| cub::BlockDiscontinuity::FlagHeadsAndTails | hipcub::BlockDiscontinuity::FlagHeadsAndTails | |
| cub::BlockExchange | hipcub::BlockExchange | |
| cub::BlockExchange::TempStorage | hipcub::BlockExchange::TempStorage | |
| cub::BlockExchange::StripedToBlocked | hipcub::BlockExchange::StripedToBlocked | |
| cub::BlockExchange::BlockedToStriped | hipcub::BlockExchange::BlockedToStriped | |
| cub::BlockExchange::WarpStripedToBlocked | hipcub::BlockExchange::WarpStripedToBlocked | |
| cub::BlockExchange::BlockedToWarpStriped | hipcub::BlockExchange::BlockedToWarpStriped | |
| cub::BlockExchange::ScatterToBlocked | hipcub::BlockExchange::ScatterToBlocked | |
| cub::BlockExchange::ScatterToStriped | hipcub::BlockExchange::ScatterToStriped | |
| cub::BlockExchange::ScatterToStripedGuarded | hipcub::BlockExchange::ScatterToStripedGuarded | |

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| | | |
|---|---|---|
| cub::BlockExchange::ScatterToStripedFlagged | hipcub::BlockExchange::ScatterToStripedFlagged | |
| cub::BlockHistogram | hipcub::BlockHistogram | |
| cub::BlockHistogram::TempStorage | hipcub::BlockHistogram::TempStorage | |
| cub::BlockHistogram::InitHistogram | hipcub::BlockHistogram::InitHistogram | |
| cub::BlockHistogram::Histogram | hipcub::BlockHistogram::Histogram | |
| cub::BlockHistogram::Composite | hipcub::BlockHistogram::Composite | |
| cub::BlockLoad | hipcub::BlockLoad | |
| cub::BlockLoad::TempStorage | hipcub::BlockLoad::TempStorage | |
| cub::BlockLoad::Load | hipcub::BlockLoad::Load | |
| cub::BlockMergeSort | hipcub::BlockMergeSort | |
| cub::BlockRadixSort | hipcub::BlockRadixSort | |
| cub::BlockRadixSort::TempStorage | hipcub::BlockRadixSort::TempStorage | |
| cub::BlockRadixSort::Sort | hipcub::BlockRadixSort::Sort | |
| cub::BlockRadixSort::SortDescending | hipcub::BlockRadixSort::SortDescending | |
| cub::BlockRadixSort::SortBlockedToStriped | hipcub::BlockRadixSort::SortBlockedToStriped | |
| cub::BlockRadixSort::SortDescendingBlockedToStriped | hipcub::BlockRadixSort::SortDescendingBlockedToStriped | |
| cub::BlockReduce | hipcub::BlockReduce | |
| cub::BlockReduce::TempStorage | hipcub::BlockReduce::TempStorage | |

| | | |
|---|---|---|
| cub::BlockReduce::Reduce | hipcub::BlockReduce::Reduce | |
| cub::BlockReduce::Sum | hipcub::BlockReduce::Sum | |
| cub::BlockStore | hipcub::BlockStore | |
| cub::BlockStore::TempStorage | hipcub::BlockStore::TempStorage | |
| cub::BlockStore::Store | hipcub::BlockStore::Store | |
| cub::BlockScan | hipcub::BlockScan | |
| cub::BlockScan::TempStorage | hipcub::BlockScan::TempStorage | |
| cub::BlockScan::ExclusiveSum | hipcub::BlockScan::ExclusiveSum | |
| cub::BlockScan::ExclusiveScan | hipcub::BlockScan::ExclusiveScan | |
| cub::BlockScan::InclusiveSum | hipcub::BlockScan::InclusiveSum | |
| cub::BlockScan::InclusiveScan | hipcub::BlockScan::InclusiveScan | |
| cub::BlockShuffle | hipcub::BlockShuffle | |
| cub::BlockShuffle::TempStorage | hipcub::BlockShuffle::TempStorage | |
| cub::BlockShuffle::Offset | hipcub::BlockShuffle::Offset | |
| cub::BlockShuffle::Rotate | hipcub::BlockShuffle::Rotate | |
| cub::BlockShuffle::Up | hipcub::BlockShuffle::Up | |
| cub::BlockShuffle::Down | hipcub::BlockShuffle::Down | |
| cub::BlockStore | hipcub::BlockStore | |
| cub::BlockStore::TempStorage | hipcub::BlockStore::TempStorage | |
| cub::BlockStore::Store | hipcub::BlockStore::Store | |

### 5.8.3 Device-wide primitives

| CUDA | HIP | CUDA version* |
|---|---|---|
| cub::DeviceAdjacentDifference | hipcub::DeviceAdjacentDifference | |
| cub::DeviceAdjacentDifference::SubtractLeftCopy | hipcub::DeviceAdjacentDifference::SubtractLeftCopy | |
| cub::DeviceAdjacentDifference:: | hipcub::DeviceAdjacentDifference: | |

| | | |
|---|---|---|
| SubtractLeft | :SubtractLeft | |
| cub::DeviceAdjacentDifference::SubtractRightCopy | hipcub::DeviceAdjacentDifference::SubtractRightCopy | |
| cub::DeviceAdjacentDifference::SubtractRight | hipcub::DeviceAdjacentDifference::SubtractRight | |
| cub::DeviceHistogram | hipcub::DeviceHistogram | |
| cub::DeviceHistogram::HistogramEven | hipcub::DeviceHistogram::HistogramEven | |
| cub::DeviceHistogram::MultiHistogramEven | hipcub::DeviceHistogram::MultiHistogramEven | |
| cub::DeviceHistogram::HistogramRange | hipcub::DeviceHistogram::HistogramRange | |
| cub::DeviceHistogram::MultiHistogramRange | hipcub::DeviceHistogram::MultiHistogramRange | |
| cub::DeviceMergeSort | hipcub::DeviceMergeSort | |
| cub::DeviceMergeSort::SortKeys | hipcub::DeviceMergeSort::SortKeys | |
| cub::DeviceMergeSort::SortKeysCopy | hipcub::DeviceMergeSort::SortKeysCopy | |
| cub::DeviceMergeSort::StableSortPairs | hipcub::DeviceMergeSort::StableSortPairs | |
| cub::DeviceMergeSort::StableSortKeys | hipcub::DeviceMergeSort::StableSortKeys | |
| cub::DeviceMergeSort::SortPairsCopy | hipcub::DeviceMergeSort::SortPairsCopy | |
| cub::DevicePartition | hipcub::DevicePartition | |
| cub::DevicePartition::Flagged | hipcub::DevicePartition::Flagged | |
| cub::DevicePartition::If | hipcub::DevicePartition::If | |

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| | | |
|---|---|---|
| cub::DeviceRadixSort | hipcub::DeviceRadixSort | |
| cub::DeviceRadixSort::SortPairs | hipcub::DeviceRadixSort::SortPairs | |
| cub::DeviceRadixSort::SortPairs Descending | hipcub::DeviceRadixSort::SortPairs Descending | |
| cub::DeviceRadixSort::SortKeys | hipcub::DeviceRadixSort::SortKeys | |
| cub::DeviceRadixSort::SortKeys Descending | hipcub::DeviceRadixSort::SortKeysDescending | |
| cub::DeviceReduce | hipcub::DeviceReduce | |
| cub::DeviceReduce::Reduce | hipcub::DeviceReduce::Reduce | |
| cub::DeviceReduce::Sum | hipcub::DeviceReduce::Sum | |
| cub::DeviceReduce::Min | hipcub::DeviceReduce::Min | |
| cub::DeviceReduce::ArgMin | hipcub::DeviceReduce::ArgMin | |
| cub::DeviceReduce::Max | hipcub::DeviceReduce::Max | |
| cub::DeviceReduce::ArgMax | hipcub::DeviceReduce::ArgMax | |
| cub::DeviceReduce::ReduceByKey | hipcub::DeviceReduce::ReduceByKey | |
| cub::DeviceRunLengthEncode | hipcub::DeviceRunLengthEncode | |
| cub::DeviceRunLengthEncode::Encode | hipcub::DeviceRunLengthEncode::Encode | |
| cub::DeviceRunLengthEncode::NonTrivialRuns | hipcub::DeviceRunLengthEncode::NonTrivialRuns | |
| cub::DeviceScan | hipcub::DeviceScan | |
| cub::DeviceScan::ExclusiveSum | hipcub::DeviceScan::ExclusiveSum | |
| cub::DeviceScan::ExclusiveScan | hipcub::DeviceScan::ExclusiveScan | |
| cub::DeviceScan::InclusiveSum | hipcub::DeviceScan::InclusiveSum | |
| cub::DeviceScan::InclusiveScan | hipcub::DeviceScan::InclusiveScan | |

| | | |
|---|---|---|
| cub::DeviceScan::ExclusiveSum ByKey | hipcub::DeviceScan::ExclusiveSum ByKey | |
| cub::DeviceScan::ExclusiveScan ByKey | hipcub::DeviceScan::ExclusiveSca nByKey | |
| cub::DeviceScan::InclusiveSumB yKey | hipcub::DeviceScan::InclusiveSum ByKey | |
| cub::DeviceScan::InclusiveScanB yKey | hipcub::DeviceScan::InclusiveScan ByKey | |
| cub::DeviceSelect | hipcub::DeviceSelect | |
| cub::DeviceSelect::Flagged | hipcub::DeviceSelect::Flagged | |
| cub::DeviceSelect::If | hipcub::DeviceSelect::If | |
| cub::DeviceSelect::Unique | hipcub::DeviceSelect::Unique | |
| cub::DeviceSelect::UniqueByKey | hipcub::DeviceSelect::UniqueByKe y | |
| cub::DeviceSpmv | hipcub::DeviceSpmv | |
| cub::DeviceSpmv::CsrMV | hipcub::DeviceSpmv::CsrMV | |
| cub::DeviceSegmentedRadixSort | hipcub::DeviceSegmentedRadixSor t | |
| cub::DeviceSegmentedRadixSort: :SortPairs | hipcub::DeviceSegmentedRadixSor t::SortPairs | |
| cub::DeviceSegmentedRadixSort: :SortPairsDescending | hipcub::DeviceSegmentedRadixSor t::SortPairsDescending | |
| cub::DeviceSegmentedRadixSort: :SortKeys | hipcub::DeviceSegmentedRadixSor t::SortKeys | |
| cub::DeviceSegmentedRadixSort: :SortKeysDescending | hipcub::DeviceSegmentedRadixSor t::SortKeysDescending | |
| cub::DeviceSegmentedReduce | hipcub::DeviceSegmentedReduce | |
| cub::DeviceSegmentedReduce::R | hipcub::DeviceSegmentedReduce:: | |

| | | |
|---|---|---|
| educe | Reduce | |
| cub::DeviceSegmentedReduce::Sum | hipcub::DeviceSegmentedReduce::Sum | |
| cub::DeviceSegmentedReduce::Min | hipcub::DeviceSegmentedReduce::Min | |
| cub::DeviceSegmentedReduce::ArgMin | hipcub::DeviceSegmentedReduce::ArgMin | |
| cub::DeviceSegmentedReduce::Max | hipcub::DeviceSegmentedReduce::Max | |
| cub::DeviceSegmentedReduce::ArgMax | hipcub::DeviceSegmentedReduce::ArgMax | |
| cub::DeviceSegmentedSort | hipcub::DeviceSegmentedSort | |
| cub::DeviceSegmentedSort::SortKeys | hipcub::DeviceSegmentedSort::SortKeys | |
| cub::DeviceSegmentedSort::SortKeysDescending | hipcub::DeviceSegmentedSort::SortKeysDescending | |
| cub::DeviceSegmentedSort::StableSortKeys | hipcub::DeviceSegmentedSort::StableSortKeys | |
| cub::DeviceSegmentedSort::StableSortKeysDescending | hipcub::DeviceSegmentedSort::StableSortKeysDescending | |
| cub::DeviceSegmentedSort::StableSortKeys | hipcub::DeviceSegmentedSort::StableSortKeys | |
| cub::DeviceSegmentedSort::StableSortKeysDescending | hipcub::DeviceSegmentedSort::StableSortKeysDescending | |
| cub::DeviceSegmentedSort::SortPairs | hipcub::DeviceSegmentedSort::SortPairs | |
| cub::DeviceSegmentedSort::SortPairsDescending | hipcub::DeviceSegmentedSort::SortPairsDescending | |
| cub::DeviceSegmentedSort::Stabl | hipcub::DeviceSegmentedSort::Sta | |

| | | |
|---|---|---|
| eSortPairs | bleSortPairs | |
| cub::DeviceSegmentedSort::StablSortPairsDescending | hipcub::DeviceSegmentedSort::StableSortPairsDescending | |

### 5.8.4 Utilities

| CUDA | HIP | CUDA version* |
|---|---|---|
| cub::ArgIndexInputIterator | hipcub::ArgIndexInputIterator | |
| cub::CacheModifiedInputIterator | hipcub::CacheModifiedInputIterator | |
| cub::CacheModifiedOutputIterator | hipcub::CacheModifiedOutputIterator | |
| cub::ConstantInputIterator | hipcub::ConstantInputIterator | |
| cub::DiscardOutputIterator | hipcub::DiscardOutputIterator | |
| cub::TexObjInputIterator | hipcub::TexObjInputIterator | |
| cub::TexRefInputIterator | hipcub::TexRefInputIterator | |
| cub::TransformInputIterator | hipcub::TransformInputIterator | |
| cub::CacheLoadModifier | hipcub::CacheLoadModifier | |
| cub::CacheStoreModifier | hipcub::CacheStoreModifier | |
| cub::LoadDirectBlocked | hipcub::LoadDirectBlocked | |
| cub::LoadDirectBlockedVectorized | hipcub::LoadDirectBlockedVectorized | |
| cub::StoreDirectBlocked | hipcub::StoreDirectBlocked | |
| cub::StoreDirectBlockedVectorized | hipcub::StoreDirectBlockedVectorized | |
| cub::LoadDirectStriped | hipcub::LoadDirectStriped | |

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| | | |
|---|---|---|
| cub::StoreDirectStriped | hipcub::StoreDirectStriped | |
| cub::LoadDirectWarpStriped | hipcub::LoadDirectWarpStriped | |
| cub::StoreDirectWarpStriped | hipcub::StoreDirectWarpStriped | |
| cub::ThreadLoad | hipcub::ThreadLoad | |
| cub::ThreadStore | hipcub::ThreadStore | |
| cub::SHR_ADD | hipcub::SHR_ADD | |
| cub::BFE | hipcub::BFE | |
| cub::IADD3 | hipcub::IADD3 | |
| cub::PRMT | hipcub::PRMT | |
| cub::ThreadExit | | |
| cub::ThreadTrap | | |
| cub::RowMajorTid | hipcub::RowMajorTid | |
| cub::LaneId | hipcub::LaneId | |
| cub::WarpId | hipcub::WarpId | |
| cub::WarpMask | hipcub::WarpMask | |
| cub::LaneMaskLt | hipcub::LaneMaskLt | |
| cub::LaneMaskLe | hipcub::LaneMaskLe | |
| cub::LaneMaskGt | hipcub::LaneMaskGt | |
| cub::LaneMaskGe | hipcub::LaneMaskGe | |
| cub::CachingDeviceAllocator | hipcub::CachingDeviceAllocator | |
| cub::SwitchDevice | | |
| cub::KernelConfig | | |
| cub::ChainedPolicy | | |
| CubDebug | | |
| CubDebugExit | | |

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| | | |
|---|---|---|
| _CubLog | | |
| cub::Debug | hipcub::Debug | |
| cub::CurrentDevice | | |
| cub::DeviceCountUncached | | |
| cub::PtxVersion | | |
| cub::SmVersionUncached | | |
| cub::SmVersion | | |
| cub::SyncStream | | |
| cub::CUB_MAX | | |
| CUB_ALIGN | | |
| CUB_PREVENT_MACRO_SUBSTITUTION | | |
| CUB_MAX | | |
| CUB_MIN | | |
| CUB_QUOTIENT_FLOOR | | |
| CUB_QUOTIENT_CEILING | | |
| CUB_ROUND_UP_NEAREST | | |
| CUB_ROUND_DOWN_NEAREST | | |
| CUB_STATIC_ASSERT | | |
| CubVector | | |

### 5.8.5 GridModule

| CUDA | HIP | CUDA version* |
|---|---|---|
| cub::GridBarrier | hipcub::GridBarrier | |
| cub::GridBarrier::Sync | hipcub::GridBarrier::Sync | |
| cub::GridBarrier::SyncFlag | hipcub::GridBarrier::SyncFlag | |
| cub::GridBarrierLifetime | hipcub::GridBarrierLifetime | |
| cub::GridBarrierLifetime::Host | hipcub::GridBarrierLifetime::Hos | |

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| | | |
|---|---|---|
| Reset | tReset | |
| cub::GridBarrierLifetime::Setup | hipcub::GridBarrierLifetime::Setup | |
| cub::GridBarrierLifetime::sync_bytes | hipcub::GridBarrierLifetime::sync_bytes | |
| cub::GridEvenShare | hipcub::GridEvenShare | |
| cub::GridEvenShare::DispatchInit | hipcub::GridEvenShare::DispatchInit | |
| cub::GridEvenShare::BlockInit | hipcub::GridEvenShare::BlockInit | |
| cub::GridQueue | hipcub::GridQueue | |
| cub::GridQueue::AllocationSize | hipcub::GridQueue::AllocationSize | |
| cub::GridQueue::FillAndResetDrain | hipcub::GridQueue::FillAndResetDrain | |
| cub::GridQueue::ResetDrain | hipcub::GridQueue::ResetDrain | |
| cub::GridQueue::ResetFill | hipcub::GridQueue::ResetFill | |
| cub::GridQueue::FillSize | hipcub::GridQueue::FillSize | |
| cub::GridQueue::Drain | hipcub::GridQueue::Drain | |
| cub::GridQueue::Fill | hipcub::GridQueue::Fill | |
| cub::GridMappingStrategy | hipcub::GridMappingStrategy | |
| GRID_MAPPING_RAKE | GRID_MAPPING_RAKE | |
| GRID_MAPPING_STRIP_MINE | GRID_MAPPING_STRIP_MINE | |
| GRID_MAPPING_DYNAMIC | GRID_MAPPING_DYNAMIC | |

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# 5.9 thrust

## 5.9.1 Algorithms

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::copy | thrust::copy | |
| thrust::copy_n | thrust::copy_n | |
| thrust::swap_ranges | thrust::swap_ranges | |
| thrust::uninitialized_copy | thrust::uninitialized_copy | |
| thrust::uninitialized_copy_n | thrust::uninitialized_copy_n | |
| thrust::gather | thrust::gather | |
| thrust::gather_if | thrust::gather_if | |
| thrust::scatter | thrust::scatter | |
| thrust::scatter_if | thrust::scatter_if | |
| thrust::merge | thrust::merge | |
| thrust::merge_by_key | thrust::merge_by_key | |
| thrust::inclusive_scan | thrust::inclusive_scan | |
| thrust::exclusive_scan | thrust::exclusive_scan | |
| thrust::inclusive_scan_by_key | thrust::inclusive_scan_by_key | |
| thrust::exclusive_scan_by_key | thrust::exclusive_scan_by_key | |
| thrust::transform_inclusive_scan | thrust::transform_inclusive_scan | |
| thrust::transform_exclusive_scan | thrust::transform_exclusive_scan | |
| thrust::reduce | thrust::reduce | |
| thrust::reduce_by_key | thrust::reduce_by_key | |
| thrust::equal | thrust::equal | |
| thrust::count | thrust::count | |
| thrust::count_if | thrust::count_if | |
| thrust::min_element | thrust::min_element | |
| thrust::max_element | thrust::max_element | |

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| | | |
|---|---|---|
| thrust::minmax_element | thrust::minmax_element | |
| thrust::all_of | thrust::all_of | |
| thrust::any_of | thrust::any_of | |
| thrust::none_of | thrust::none_of | |
| thrust::is_partitioned | thrust::is_partitioned | |
| thrust::is_sorted | thrust::is_sorted | |
| thrust::is_sorted_until | thrust::is_sorted_until | |
| thrust::inner_product | thrust::inner_product | |
| thrust::transform_reduce | thrust::transform_reduce | |
| thrust::partition | thrust::partition | |
| thrust::partition_copy | thrust::partition_copy | |
| thrust::stable_partition | thrust::stable_partition | |
| thrust::stable_partition_copy | thrust::stable_partition_copy | |
| thrust::shuffle | thrust::shuffle | |
| thrust::shuffle_copy | thrust::shuffle_copy | |
| thrust::copy_if | thrust::copy_if | |
| thrust::remove | thrust::remove | |
| thrust::remove_copy | thrust::remove_copy | |
| thrust::remove_if | thrust::remove_if | |
| thrust::remove_copy_if | thrust::remove_copy_if | |
| thrust::unique | thrust::unique | |
| thrust::unique_copy | thrust::unique_copy | |
| thrust::unique_by_key | thrust::unique_by_key | |
| thrust::unique_by_key_copy | thrust::unique_by_key_copy | |
| thrust::unique_count | thrust::unique_count | |
| thrust::find | thrust::find | |
| thrust::find_if | thrust::find_if | |
| thrust::find_if_not | thrust::find_if_not | |

| | | |
|---|---|---|
| thrust::mismatch | thrust::mismatch | |
| thrust::partition_point | thrust::partition_point | |
| thrust::set_difference | thrust::set_difference | |
| thrust::set_intersection | thrust::set_intersection | |
| thrust::set_symmetric_difference | thrust::set_symmetric_difference | |
| thrust::set_union | thrust::set_union | |
| thrust::set_difference_by_key | thrust::set_difference_by_key | |
| thrust::set_symmetric_difference _by_key | thrust::set_symmetric_difference _by_key | |
| thrust::set_union_by_key | thrust::set_union_by_key | |
| thrust::sort | thrust::sort | |
| thrust::stable_sort | thrust::stable_sort | |
| thrust::sort_by_key | thrust::sort_by_key | |
| thrust::stable_sort_by_key | thrust::stable_sort_by_key | |
| thrust::adjacent_difference | thrust::adjacent_difference | |
| thrust::generate | thrust::generate | |
| thrust::generate_n | thrust::generate_n | |
| thrust::sequence | thrust::sequence | |
| thrust::tabulate | thrust::tabulate | |
| thrust::transform | thrust::transform | |
| thrust::transform_if | thrust::transform_if | |
| thrust::fill | thrust::fill | |
| thrust::fill_n | thrust::fill_n | |
| thrust::uninitialized_fill | thrust::uninitialized_fill | |
| thrust::uninitialized_fill_n | thrust::uninitialized_fill_n | |
| thrust::for_each | thrust::for_each | |
| thrust::for_each_n | thrust::for_each_n | |
| thrust::replace | thrust::replace | |

| | | |
|---|---|---|
| thrust::replace_if | thrust::replace_if | |
| thrust::replace_copy | thrust::replace_copy | |

### 5.9.2 Containers

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::host_vector | thrust::host_vector | |
| thrust::device_vector | thrust::device_vector | |

### 5.9.3 Parallel Execution Policies

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::host_execution_policy | thrust::host_execution_policy | |
| thrust::device_execution_policy | thrust::device_execution_policy | |
| thrust::host | thrust::host | |
| thrust::device | thrust::device | |

### 5.9.4 Function Objects

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::unary_function | thrust::unary_function | |
| thrust::binary_function | thrust::binary_function | |
| thrust::unary_negate | thrust::unary_negate | |
| thrust::binary_negate | thrust::binary_negate | |
| thrust::zip_function | thrust::zip_function | |
| thrust::not1 | thrust::not1 | |
| thrust::not2 | thrust::not2 | |
| thrust::make_zip_function | thrust::make_zip_function | |
| thrust::placeholders | thrust::placeholders | |
| thrust::plus | thrust::plus | |

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| | | |
|---|---|---|
| thrust::minus | thrust::minus | |
| thrust::multiplies | thrust::multiplies | |
| thrust::divides | thrust::divides | |
| thrust::modulus | thrust::modulus | |
| thrust::negate | thrust::negate | |
| thrust::square | thrust::square | |
| THRUST_UNARY_FUNCTOR_<br>VOID_SPECIALIZATION | THRUST_UNARY_FUNCTOR_<br>VOID_SPECIALIZATION | |
| THRUST_BINARY_FUNCTOR_<br>VOID_SPECIALIZATION | THRUST_BINARY_FUNCTOR_<br>VOID_SPECIALIZATION | |
| THRUST_BINARY_FUNCTOR_<br>VOID_SPECIALIZATION_OP | THRUST_BINARY_FUNCTOR_<br>VOID_SPECIALIZATION_OP | |
| thrust::bit_and | thrust::bit_and | |
| thrust::bit_or | thrust::bit_or | |
| thrust::bit_xor | thrust::bit_xor | |
| thrust::equal_to | thrust::equal_to | |
| thrust::not_equal_to | thrust::not_equal_to | |
| thrust::greater | thrust::greater | |
| thrust::less | thrust::less | |
| thrust::greater_equal | thrust::greater_equal | |
| thrust::less_equal | thrust::less_equal | |
| thrust::identity | thrust::identity | |
| thrust::maximum | thrust::maximum | |
| thrust::minimum | thrust::minimum | |
| thrust::project1st | thrust::project1st | |
| thrust::project2nd | thrust::project2nd | |
| thrust::logical_and | thrust::logical_and | |
| thrust::logical_or | thrust::logical_or | |

| | | |
|---|---|---|
| thrust::logical_not | thrust::logical_not | |

### 5.9.5 Iterators

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::input_device_iterator_tag | thrust::input_device_iterator_tag | |
| thrust::output_device_iterator_tag | thrust::output_device_iterator_tag | |
| thrust::forward_device_iterator_tag | thrust::forward_device_iterator_tag | |
| thrust::bidirectional_device_iterator_tag | thrust::bidirectional_device_iterator_tag | |
| thrust::random_access_device_iterator_tag | thrust::random_access_device_iterator_tag | |
| thrust::input_host_iterator_tag | thrust::input_host_iterator_tag | |
| thrust::output_host_iterator_tag | thrust::output_host_iterator_tag | |
| thrust::forward_host_iterator_tag | thrust::forward_host_iterator_tag | |
| thrust::bidirectional_host_iterator_tag | thrust::bidirectional_host_iterator_tag | |
| thrust::random_access_host_iterator_tag | thrust::random_access_host_iterator_tag | |
| thrust::constant_iterator | thrust::constant_iterator | |
| thrust::counting_iterator | thrust::counting_iterator | |
| thrust::discard_iterator | thrust::discard_iterator | |
| thrust::iterator_adaptor | thrust::iterator_adaptor | |
| thrust::iterator_facade | thrust::iterator_facade | |
| thrust::iterator_core_access | thrust::iterator_core_access | |
| thrust::permutation_iterator | thrust::permutation_iterator | |
| thrust::reverse_iterator | thrust::reverse_iterator | |

| | | |
|---|---|---|
| thrust::transform_input_output_iterator | thrust::transform_input_output_iterator | |
| thrust::transform_iterator | thrust::transform_iterator | |
| thrust::transform_output_iterator | thrust::transform_output_iterator | |
| thrust::zip_iterator | thrust::zip_iterator | |
| thrust::make_constant_iterator | thrust::make_constant_iterator | |
| thrust::make_counting_iterator | thrust::make_counting_iterator | |
| thrust::make_discard_iterator | thrust::make_discard_iterator | |
| thrust::make_permutation_iterator | thrust::make_permutation_iterator | |
| thrust::make_reverse_iterator | thrust::make_reverse_iterator | |
| thrust::make_transform_input_output_iterator | thrust::make_transform_input_output_iterator | |
| thrust::make_transform_iterator | thrust::make_transform_iterator | |
| thrust::make_transform_output_iterator | thrust::make_transform_output_iterator | |
| thrust::make_zip_iterator | thrust::make_zip_iterator | |
| thrust::advance | thrust::advance | |
| thrust::distance | thrust::distance | |

### 5.9.6 Memory Management

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::device_reference | thrust::device_reference | |
| thrust::device_ptr | thrust::device_ptr | |
| thrust::device_delete | thrust::device_delete | |
| thrust::device_free | thrust::device_free | |
| thrust::device_malloc | thrust::device_malloc | |

| | | |
|---|---|---|
| thrust::device_new | thrust::device_new | |
| thrust::operator<< | thrust::operator<< | |
| thrust::device_pointer_cast | thrust::device_pointer_cast | |
| thrust::swap | thrust::swap | |
| thrust::get_temporary_buffer | thrust::get_temporary_buffer | |
| thrust::return_temporary_buffer | thrust::return_temporary_buffer | |
| thrust::raw_pointer_cast | thrust::raw_pointer_cast | |
| thrust::raw_reference_cast | thrust::raw_reference_cast | |
| thrust::device_ptr_memory_resource | thrust::device_ptr_memory_resource | |
| thrust::device_allocator | thrust::device_allocator | |
| thrust::device_malloc_allocator | thrust::device_malloc_allocator | |
| thrust::device_new_allocator | thrust::device_new_allocator | |
| thrust::mr::allocator | thrust::mr::allocator | |
| thrust::mr::stateless_resource_allocator | thrust::mr::stateless_resource_allocator | |
| thrust::mr::polymorphic_allocator | thrust::mr::polymorphic_allocator | |
| thrust::mr::operator== | thrust::mr::operator== | |
| thrust::mr::operator!= | thrust::mr::operator!= | |
| thrust::mr::disjoint_unsynchronized_pool_resource | thrust::mr::disjoint_unsynchronized_pool_resource | |
| thrust::mr::disjoint_synchronized_pool_resource | thrust::mr::disjoint_synchronized_pool_resource | |
| thrust::mr::memory_resource thrust::mr::new_delete_resource | thrust::mr::memory_resource thrust::mr::new_delete_resource | |
| thrust::mr::unsynchronized_pool_resource | thrust::mr::unsynchronized_pool_resource | |
| thrust::mr::pool_options | thrust::mr::pool_options | |

| | | |
|---|---|---|
| thrust::mr::synchronized_pool_resource | thrust::mr::synchronized_pool_resource | |
| thrust::system::cpp::memory_resource | thrust::system::cpp::memory_resource | |
| thrust::system::cpp::universal_memory_resource | thrust::system::cpp::universal_memory_resource | |
| thrust::system::cpp::universal_host_pinned_memory_resource | thrust::system::cpp::universal_host_pinned_memory_resource | |
| thrust::system::omp::memory_resource | | |
| thrust::system::omp::universal_memory_resource | | |
| thrust::system::tbb::universal_host_pinned_memory_resource | | |
| thrust::universal_ptr | thrust::universal_ptr | |
| thrust::mr::tls_disjoint_pool | thrust::mr::tls_disjoint_pool | |
| thrust::mr::get_global_resource | thrust::mr::get_global_resource | |
| thrust::mr::tls_pool | thrust::mr::tls_pool | |

### 5.9.7 Complex Numbers

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::complex | thrust::complex | |
| thrust::complex::value_type | thrust::complex::value_type | |
| thrust::complex::complex | thrust::complex::complex | |
| thrust::complex::operator= | thrust::complex::operator= | |
| thrust::complex::operator+= | thrust::complex::operator+= | |
| thrust::complex::operator-= | thrust::complex::operator-= | |
| thrust::complex::operator*= | thrust::complex::operator*= | |
| thrust::complex::operator/= | thrust::complex::operator/= | |

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| | | |
|---|---|---|
| thrust::complex::real | thrust::complex::real | |
| thrust::complex::imag | thrust::complex::imag | |
| thrust::abs | thrust::abs | |
| thrust::arg | thrust::arg | |
| thrust::norm | thrust::norm | |
| thrust::conj | thrust::conj | |
| thrust::polar | thrust::polar | |
| thrust::proj | thrust::proj | |
| thrust::operator+ | thrust::operator+ | |
| thrust::operator- | thrust::operator- | |
| thrust::operator* | thrust::operator* | |
| thrust::operator/ | thrust::operator/ | |
| thrust::exp | thrust::exp | |
| thrust::log | thrust::log | |
| thrust::log10 | thrust::log10 | |
| thrust::pow | thrust::pow | |
| thrust::sqrt | thrust::sqrt | |
| thrust::cos | thrust::cos | |
| thrust::sin | thrust::sin | |
| thrust::tan | thrust::tan | |
| thrust::cosh | thrust::cosh | |
| thrust::sinh | thrust::sinh | |
| thrust::tanh | thrust::tanh | |
| thrust::acos | thrust::acos | |
| thrust::asin | thrust::asin | |
| thrust::atan | thrust::atan | |
| thrust::acosh | thrust::acosh | |
| thrust::asinh | thrust::asinh | |

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| | | |
|---|---|---|
| thrust::atanh | thrust::atanh | |
| thrust::operator<< | thrust::operator<< | |
| thrust::operator>> | thrust::operator>> | |
| thrust::operator== | thrust::operator== | |
| thrust::operator!= | thrust::operator!= | |

### 5.9.8 Random Number Generation

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::random | thrust::random | |
| thrust::random::ranlux24 | thrust::random::ranlux24 | |
| thrust::random::ranlux48 | thrust::random::ranlux48 | |
| thrust::random::taus88 | thrust::random::taus88 | |
| thrust::random::default_random_engine | thrust::random::default_random_engine | |
| thrust::random::minstd_rand0 | thrust::random::minstd_rand0 | |
| thrust::random::minstd_rand | thrust::random::minstd_rand | |
| thrust::random::ranlux24_base | thrust::random::ranlux24_base | |
| thrust::random::ranlux48_base | thrust::random::ranlux48_base | |
| thrust::random::normal_distribution | thrust::random::normal_distribution | |
| thrust::random::uniform_int_distribution | thrust::random::uniform_int_distribution | |
| thrust::random::uniform_real_distribution | thrust::random::uniform_real_distribution | |
| thrust::random::operator== | thrust::random::operator== | |
| thrust::random::operator!= | thrust::random::operator!= | |
| thrust::random::operator<< | thrust::random::operator<< | |
| thrust::random::operator>> | thrust::random::operator>> | |

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| | | |
|---|---|---|
| thrust::random::operator== | thrust::random::operator== | |
| thrust::random::operator!= | thrust::random::operator!= | |
| thrust::random::discard_block_engine | thrust::random::discard_block_engine | |
| thrust::random::xor_combine_engine | thrust::random::xor_combine_engine | |
| thrust::random::linear_congruential_engine | thrust::random::linear_congruential_engine | |
| thrust::random::linear_feedback_shift_engine | thrust::random::linear_feedback_shift_engine | |
| thrust::random::subtract_with_carry_engine | thrust::random::subtract_with_carry_engine | |

### 5.9.9 System

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::cpp | thrust::cpp | |
| thrust::system | thrust::system | |
| thrust::system::omp | | |
| thrust::system::cpp | thrust::system::cpp | |
| thrust::system::cuda | thrust::system::hip | |
| thrust::system::tbb | | |
| thrust::omp | | |
| thrust::tbb | | |
| thrust::system::cpp::pointer | thrust::system::cpp::pointer | |
| thrust::system::cpp::universal_pointer | thrust::system::cpp::universal_pointer | |
| thrust::system::cpp::reference | thrust::system::cpp::reference | |
| thrust::system::omp::pointer | | |

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| | | |
|---|---|---|
| thrust::system::omp::universal_pointer | | |
| thrust::system::omp::reference | | |
| thrust::system::tbb::pointer | | |
| thrust::system::tbb::universal_pointer | | |
| thrust::system::tbb::reference | | |
| thrust::system::errc | thrust::system::errc | |
| thrust::system::is_error_code_enum | thrust::system::is_error_code_enum | |
| thrust::system::is_error_condition_enum | thrust::system::is_error_condition_enum | |
| thrust::system::error_category | thrust::system::error_category | |
| thrust::system::error_code | thrust::system::error_code | |
| thrust::system::error_condition | thrust::system::error_condition | |
| thrust::system::system_error | thrust::system::system_error | |
| thrust::system::generic_category | thrust::system::generic_category | |
| thrust::system::make_error_code | thrust::system::make_error_code | |
| thrust::system::operator< | thrust::system::operator< | |
| thrust::system::operator<< | thrust::system::operator<< | |
| thrust::system::make_error_condition | thrust::system::make_error_condition | |
| thrust::system::operator== | thrust::system::operator== | |
| thrust::system::operator== | thrust::system::operator== | |
| thrust::system::operator!= | thrust::system::operator!= | |
| thrust::system::system_error::code | thrust::system::system_error::code | |
| thrust::system::system_error::wh | thrust::system::system_error::wh | |

| | | |
|---|---|---|
| at | at | |

### 5.9.10 Utility

| CUDA | HIP | CUDA version* |
|---|---|---|
| thrust::pair | thrust::pair | |
| thrust::tuple | thrust::tuple | |
| thrust::tuple_element | thrust::tuple_element | |
| thrust::tuple_size | thrust::tuple_size | |
| thrust::swap | thrust::swap | |
| thrust::make_pair | thrust::make_pair | |
| thrust::get | thrust::get | |
| thrust::make_tuple | thrust::make_tuple | |
| thrust::tie | thrust::tie | |
| thrust::proclaim_contiguous_iterator | thrust::proclaim_contiguous_iterator | |
| thrust::conjunction_value | thrust::conjunction_value | |
| thrust::disjunction_value | thrust::disjunction_value | |
| thrust::negation_value | thrust::negation_value | |
| thrust::remove_cvref | thrust::remove_cvref | |
| thrust::voider | thrust::voider | |
| thrust::integer_sequence | thrust::integer_sequence | |
| thrust::index_sequence | thrust::index_sequence | |
| thrust::make_integer_sequence | thrust::make_integer_sequence | |
| thrust::make_index_sequence | thrust::make_index_sequence | |
| thrust::make_reversed_integer_sequence | thrust::make_reversed_integer_sequence | |
| thrust::make_reversed_index_sequence | thrust::make_reversed_index_sequence | |

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| | | |
|---|---|---|
| thrust::integer_sequence_push_front | thrust::integer_sequence_push_front | |
| thrust::integer_sequence_push_back | thrust::integer_sequence_push_back | |
| thrust::is_contiguous_iterator | thrust::is_contiguous_iterator | |
| thrust::is_execution_policy | thrust::is_execution_policy | |
| thrust::is_operator_less_function_object | thrust::is_operator_less_function_object | |
| thrust::is_operator_greater_function_object | thrust::is_operator_greater_function_object | |
| thrust::is_operator_less_or_greater_function_object | thrust::is_operator_less_or_greater_function_object | |
| thrust::is_operator_plus_function_object | thrust::is_operator_plus_function_object | |
| thrust::conjunction | thrust::conjunction | |
| thrust::conjunction_v | thrust::conjunction_v | |
| thrust::disjunction | thrust::disjunction | |
| thrust::disjunction_v | thrust::disjunction_v | |
| thrust::negation | thrust::negation | |
| thrust::negation_v | thrust::negation_v | |
| thrust::remove_cvref_t | thrust::remove_cvref_t | |
| thrust::is_contiguous_iterator_v | thrust::is_contiguous_iterator_v | |
| thrust::is_execution_policy_v | thrust::is_execution_policy_v | |
| thrust::is_operator_less_function_object_v | thrust::is_operator_less_function_object_v | |
| thrust::is_operator_greater_function_object_v | thrust::is_operator_greater_function_object_v | |
| thrust::is_operator_less_or_greate | thrust::is_operator_less_or_greate | |

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| | | |
|---|---|---|
| r_function_object_v | r_function_object_v | |
| thrust::system::system_error::what | thrust::system::system_error::what | |
| thrust::is_operator_plus_function_object_v | thrust::is_operator_plus_function_object_v | |
| thrust::conjunction_value_v | thrust::conjunction_value_v | |
| thrust::disjunction_value_v | thrust::disjunction_value_v | |
| thrust::negation_value_v | thrust::negation_value_v | |
| THRUST_PROCLAIM_CONTIGUOUS_ITERATOR | THRUST_PROCLAIM_CONTIGUOUS_ITERATOR | |

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# 6 通讯组件兼容性列表

## 6.1 RCCL

### 6.1.1 通讯组创建及管理类函数

| NCCL | RCCL | NCCL version |
|---|---|---|
| ncclGetLastError | ncclGetLastError | |
| ncclGetVersion | ncclGetVersion | |
| ncclGetUniqueId | ncclGetUniqueId | |
| ncclCommInitRank | ncclCommInitRank | |
| ncclCommInitAll | ncclCommInitAll | |
| ncclCommInitRankConfig | | 2.14 |
| ncclCommFinalize | | 2.14 |
| ncclCommDestroy | ncclCommDestroy | |
| ncclCommAbort | ncclCommAbort | |
| ncclCommGetAsyncError | ncclCommGetAsyncError | |
| ncclCommCount | ncclCommCount | |
| ncclCommCuDevice | ncclCommCuDevice | |
| ncclCommUserRank | ncclCommUserRank | |

### 6.1.2 集合通讯函数

| NCCL | RCCL | NCCL version |
|---|---|---|
| ncclAllReduce | ncclAllReduce | |
| ncclBroadcast | ncclBroadcast | |
| ncclReduce | ncclReduce | |
| ncclAllGather | ncclAllGather | |
| ncclReduceScatter | ncclReduceScatter | |

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### 6.1.3 组调用函数

| NCCL | RCCL | NCCL version |
| --- | --- | --- |
| ncclGroupStart | ncclGroupStart | |
| ncclGroupEnd | ncclGroupEnd | |

### 6.1.4 点对点通讯函数

| NCCL | RCCL | NCCL version |
| --- | --- | --- |
| ncclSend | ncclSend | |
| ncclRecv | ncclRecv | |

### 6.1.5 数据类型

| NCCL | RCCL | NCCL version |
| --- | --- | --- |
| ncclComm_t | ncclComm_t | |
| ncclResult_t | ncclResult_t | |
| ncclDataType_t | ncclDataType_t | |
| ncclRedOp_t | ncclRedOp_t | |
| ncclScalarResidence_t | ncclScalarResidence_t | |
| ncclConfig_t | | 2.14 |

### 6.1.6 自定义规约操作符

| NCCL | RCCL | NCCL version |
| --- | --- | --- |
| ncclRedOpCreatePreMulSum | ncclRedOpCreatePreMulSum | |
| ncclRedOpDestroy | ncclRedOpDestroy | |

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